Valuation for Financial Engineering:

Part I: Foundations

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Comments welcomed!

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Preface for students

Some of you will savor the preface in order to forestall the inevitable labor of the seemingly unending chapters that follow. Others will skip the preface thinking that they should save time and avoid any material that won't appear on the midterm. I intend to surprise and please the former. In fact, the book reveals many unexpected surprises with only modest departures from and extensions of rigorous traditional finance theory.

But let's not get ahead of ourselves. *Financial engineering* is still a young field, lacking well-defined boundaries and consistent self-definition. Some authors define financial engineering as anything mathematical or computational in finance. Others restrict financial engineering to the mathematics and analysis of derivative securities and structured products. I prefer to adapt the traditional application of the term *engineering* to the field of finance, such as the one below from Wikipedia. The clarifications for finance are my own.

- **Engineering** – /ˌenjəˈniriNG/ - Engineering is the use of scientific principles to design and build machines, structures, and other things, including bridges, roads, vehicles, and buildings
- **Financial Decisions** – Valuation, trading, structuring (product design), risk assessment, hedging, risk management, investment, financing, lending, risk-based performance assessment.
- **Financial Engineering** – The branch of finance concerned with the design, building and use of financial decision-making tools and processes, financial computing processes and financial products.

What do financial engineers design and build? Here is a partial list:

- ✓ Financial decision frameworks, such as valuation, lending and trading
- ✓ Financial decision support, including data collection and analysis
- ✓ Design of securities, contracts and strategies used to raise capital and fund companies
- ✓ Other innovative financial product and strategy design
- ✓ Risk measurement and management strategies

To test the definition, does it correctly discriminate between activities that should be considered financial mathematics and those that should be considered financial engineering? Most financial economists would agree that financial engineering should include the study of financial derivative pricing and risk models. Even further, algorithmic trading models or any number of big or small data implementations that support derivatives pricing and trading activity should be included.

Second, financial risk models would reasonably be expected to fall within the scope of financial engineering. This can be attributed to the incredible efforts taken to implement accurate risk measurement and reporting when financial theory provides no easy answers. I recall one discussion in 2019 with an eminent financial economist who asked what the big deal was with risk management. *"Isn't it just sigma?"* A financial engineer would address the practical complications, such as variable volatilities, nonlinear relationships and unstable correlations within shaky volatility models --- maybe even using implied volatility --- unlikely to be content with a mere standard deviation calculation.

Third, while there is not as much literature in this area, many finance folks would probably agree that financial product design or security structuring falls solidly within the boundaries of financial engineering. After all, this is just another type of consumer or industrial product design, highly customized to particular clients and market environments. Financial products are the means by which we express trading opinions, design portfolios to meet investment objectives, and raise money in the most efficient manner.

Fourth, and perhaps most controversial, I submit that corporate finance can own significant real estate in the field of financial engineering. Though we still benefit from the celebrated theoretical work of Modigliani, Miller, Sharpe, Lintner, Mossin ,and Treynor (and their many worthy successors unnamed here), financial theory has

focused on highly idealized or stylized assumptions that were likely never meant to apply to all firms in practice at all times. Financial engineering applies financial theory to the vicissitudes of practice.

I realized this in my 25-year consulting practice with corporations. No company relies on the CAPM (Capital Asset Pricing Model) to determine discount rates for its expected cash flows. If they start with the CAPM discount rate, a hefty risk premium is usually added. See Jagannathan et al for a comprehensive survey.[1] Companies do value real options, but often lack the necessary engineering tools to apply financial theory rigorously to practice. They often end up mis-applying standardized option pricing models to unsuitable problems. Corporations make capital structure decisions, including security design and decisions about leverage levels, but using considerations far afield of those considered important in theoretical *classical* finance. Risk management decisions, which should be recast as *contingent capital structure and cash management decisions*, are roundly ignored or minimized in corporate finance textbooks --- which is ***shocking*** when you consider that so many risk management debacles have destroyed previously strong corporations. Brought down by contingent liabilities! Unreported by FAS 5![2]

Which brings me to the next point. With so much applied work to be done in corporate finance, who will actually do the work? Financial theorists would have to develop new theories for all types of corporations in many different circumstances. Financial engineers well-versed in the theory apply the tools of their trade to solve each problem that comes up. They pay particular attention to their employers' and clients' circumstances, objectives, constraints, and contemporaneous market conditions. Hopefully, you are reading this book in order to become one of those people.

So much for the *engineering* term. The single most important tool for any financial professional is *valuation*. The CAPM doesn't work for the kinds of complex cash flows we have today. We can't even define percentage returns or discount rates for large classes of securities, contract and positions. Valuations need to be both public and private, since our differences in valuation opinions drive trading. Capital constraints affect private valuations dramatically. Benchmarking is critical, but tools like P/E ratios are far too primitive. Valuation models developed in the 1960s and taught to MBAs ever since need vigorous review and revision, even though textbooks continue to recycle these models.

This book therefore aspires to compile, develop and explain a body of theoretical and practical valuation knowledge learned in a career in applied finance --- corporate, sovereign and fund consulting. With applied, bespoke valuation tools, corporations can:

- Value projects and their embedded options; make investment decisions
- Decide when to buy and sell assets
    - e.g. corporate assets differ from projects when they have limited liability
- Determine value-maximizing capital structure & risk management strategy
- Design securities
- Write optimal contracts

[1] Jagannathan, Ravi, Iwan Meier and Vefa Tarhan, "The Cross-Section of Hurdle Rates for Capital Budgeting: An Empirical Analysis of Survey Data"

[2] "[FAS 5] establishes standards of financial accounting and reporting for loss contingencies. It requires accrual by a charge to income (and disclosure) for an estimated loss from a loss contingency if two conditions are met: (a) information available prior to issuance of the financial statements indicates that it is probable that an asset had been impaired or a liability had been incurred at the date of the financial statements, and (b) the amount of loss can be reasonably estimated." - https://www.fasb.org/summary/stsum5.shtml Many derivative positions meet neither condition necessary for reporting under FAS 5, and therefore never reported.

Indeed, nearly every financial decision requires a valuation. Therefore, *Valuation for Financial Engineers* meets a well-defined need for financial problem-solvers to apply valuation theory rigorously in the face of specific corporate constraints and market circumstances.

It won't be that easy however. The book focuses on discrete-time modeling since our real world is not continuous. Closed-form solutions and easy formulae are rare. Nearly all risk-related and valuation problems in the real world require simulation to get a flexible solution. We will challenge long-held beliefs of financial theory, for example, the role of idiosyncratic risk in asset pricing. We provide cases, like real life, which should be expected to display a degree of complexity.

Finally, *Valuation for Financial Engineers* is not a glorified MBA textbook, though one trained as an MBA might see an occasional familiar equation. It is a toolkit for the modern financial engineer --- that applies advanced finance, undergraduate level mathematics, statistics, linear algebra and computational algorithms to practical problems in corporate and investment finance. It is particularly useful for applications where joint cash flow probabilities are known or can be simulated, and is not as useful for continuous time models or finite state environments.

Supplemental preface for faculty (Part I)

Most faculty don't read others' textbooks, but will perhaps suffer a preface. This preface summarizes the unusual and different approach we take to standard valuation topics. It does not follow the format of a valuation text or a corporate finance text. Generally, we approach the topic of valuation using a comprehensive simulation and benchmarking framework, where the benchmarks are based on traded assets or contract prices. For example, if one were to value an oil refinery, one would build a stochastic pro-forma, simulate cash flows, benchmark the cash flows to traded futures and option prices in energy, discount the cash flow components separately to reflect their different risks or benchmarks, and determine a final valuation. The value of an oil refinery should depend closely on energy futures prices, and options as well --- since a refinery in its essence comprises a collection of options to transform energy products from one form to another. In other words, the valuation of an oil refinery should take a similar approach to a complex derivative instrument, without the benefit and challenges of continuous-time financial mathematics.

Part I addresses valuation when discount rates are known. This assumption is surprisingly unrestrictive --- even real options can be modeled by estimating the effect of nonlinear risks on expected cash flows. Each chapter contains some surprises that I have not been able to find in other textbooks, but that I found extremely useful in practice. All the models herein are based on financial theory, but have not gotten much play outside academic papers. For example, the simple form valuations follow Lintner, Bogue and Roll, Brennan and Rubinstein's papers on cash flow valuation --- methodologies displaced by the detailed calibrations of the CAPM approximation undertaken by many valuation texts.

Here are some surprises you may expect: (yes I know this ruins the surprise, but in case you don't read the book…)

Chapter 1	• Characterizing valuation as a modeling process --- "always wrong, sometimes useful" • Introducing portfolio-based valuation as an important tool, especially in private valuations • Dealing with undefined percentage returns, e.g short positions of all types and long forward/futures or other leveraged positions • Introducing a single equation, the GVE, that unites all the valuation formulae in classical finance: Required return = Expected return, special cases resulting in the CAPM, Modigliani & Miller proposition 2, forward and option pricing formulae.

Chapter 2	• New and easier ways to develop annuity formulae using financial intuition rather than mathematical brute force; includes exponentially growing annuities • Focus on using Excel, as hand-held financial calculators have little use in modern finance • New types of annuities, including linear growth and mean reversion, foreshadowing the introduction of Arithmetic Random Walk (ARW), Geometric Random Walk (GRW) BM and Mean Reversion (MR) stochastic processes in Chapter 6
Chapter 3	• A grab-bag of complications to finding basic present values of cash flows: payment frequency, inflation and taxes • Teaches students how to handle numerous cash flow types and risks • The introduction of the zero-coupon bond pricing curve and FX forward pricing • Pricing event risk including default, in the style of Poisson processes • Floating rate bonds, including the valuation of individual FRN payments and cash flows that correlate to interest rates • Foreign currency-denominated cash flows
Chapter 4	• A comprehensive introduction to bonds traded in the U.S. and their informational content • Detailed pricing relationships in practice • Bootstrapping of the ZCB yield curve using nonlinear regression • Defaultable bonds, using the BIS framework (PD and LGD)
Chapter 5	• A comprehensive introduction to simulation techniques in finance, using Excel or Python • Variance reduction: Moment matching and antithetical variables • The control variate simulation technique (used later for advanced benchmarking) • Multivariate methods including the Cholesky decomposition and Gaussian copula transformations • Simulation and optimization with common random variables
Chapter 6	• Thorough discussion of introductory stochastic processes: Arithmetic & Geometric Random Walks and Mean Reversion in discrete time • Knowing which process to apply to any given financial variable of interest • Statistical properties, simulation and parameterization techniques

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Of course, I am sure there are still many errors, typos and unclear sentences, but with the help of future volunteers I hope to minimize those.

Chapter 1: Valuation Methods for Financial Engineers

INTRODUCTION

Problems in valuation extend far beyond the scope of this text. Most financial engineers will not be called upon to value paintings or used cars, so we may consider it safe to exclude these interesting valuations from consideration. However, financial engineers would clearly be expected to value financial instruments, like stocks and bonds, most of which would have no value were it not for the contractual or otherwise expected cash flows generated by those instruments. We may call these financial instruments *capital assets*. In this text, we define our coverage of assets to include both capital assets and physical assets whose values derive from the cash flows they produce, such as properties that produce rental income, not the intangible benefits they provide to any individual or institution. Indeed, the valuation of intangible benefits is a far more difficult problem we cannot presently solve. Of course, financial engineers might build artificial intelligence-based models to estimate the value of fine art or intangible benefits of yacht ownership. These are also valuation models, but they are also beyond the scope of this text.

If we limited our valuation scope to capital assets and physical assets that generate cash, it would not suffice. We must also include contracts in our valuation models, even though they may require little or no initial capital investment. Why? Because a contract generally becomes either an asset or a liability over time, even if it appeared to be of equal value to both parties, i.e. with a total value equal to zero, when it was originally agreed. For example, if you have a contract that allows you the exclusive right to sell hot dogs at Yankee Stadium in New York, the value of that contract might rise or fall significantly over time. The value would probably rise when stadium ticket sales increase, and fall when vegetarians dominate the stadium crowd, all other factors unchanged. Hopefully you could eventually sell or transfer the rights of the contract and realize a gain. Or pivot to tofu dogs instead of hot dogs! However, in either case, your contract may become a liability, in which case it would cost money to transfer or exit the contract.

Further, the valuation of corporate assets in many cases depends on what strategic or tactical options are available to manage the assets. Therefore, any comprehensive discussion of asset valuation should include corporate assets and liabilities, since these assets and liabilities are actively managed in order to maximize shareholder value. The values of these assets are determined in part by the choices made at corporate headquarters. These choices include the capital budgeting decision (when and how to invest), the capital structure decision (how to finance the company), the risk management decision (how to select and manage the company's risk profile), and the cash management decision (how to manage liquid assets). Each of these decisions depends on how the company values the alternative available asset and liability management strategies: ultimately they are all valuation problems. In fact, as we will see, every corporate decision of consequence can be thought of as a valuation problem.

Finally, options, forward and futures contracts can be used not only to value other assets, but are important in their own right, and should be integrated into the general theory of valuation. For this reason, we include the valuation of these *derivative contracts* as a special case of the general framework, without using the advanced mathematical tools typically needed for derivatives study.

Mathematically, the text focuses on assets or contracts that produce cash flows with known probability distributions, or at least cash flows that can be stochastically defined or simulated. This implies that the probability distributions of future cash flows will either be given, or can be estimated with actual or historical data. These probability distributions may be altered within defined bounds at the asset manager's discretion, in which case their values can be maximized. For example, a manufacturer may have the ability to change product design

to appeal more to customers in a profitable way. Or perhaps, a Chief Financial Officer can design a financing contract that transfers some risk profitably to lenders. In all cases, we normally determine the risk adjustments to expected cash flows based on market benchmarks, so that the adjusted *risk-neutral* cash flows can be discounted at the risk-free rate. The traditional CAPM method adjusts the discount rate instead of the cash flows. We will also show that the traditional CAPM method is a special case of a more general valuation framework.

Except for our discussion of stochastic interest rates, we generally assume the risk-free *term structure* is known. That is, the risk-free rate of interest used to discount cash payments paid at any time in the future is given. Since bond prices are readily available, this is not a limiting assumption in practice. In Part I, we assume the discount rates already include any risk adjustments, and we need only apply those discount rates to the expected cash flows in order to determine value. Part II offers several methods for valuing cash flows when risks evolve over time. This means that the cost of risk, or the *risk charges*, are based on careful benchmarking to traded assets. While some textbooks focus on a utility theory approach to determining the costs of risk, we prefer to estimate the cost of risk directly from comparables when available. The reason for this is simple --- no-one knows their own utility function, much less that of the market! For example, when one speaks of the expected return of the market portfolio, this includes a risk premium. While the risk premium cannot be determined with certainty, statistical methods can be used to estimate historical market risk premiums, which actuaries typically assume fall in the range of 6-8% per year. There is of course no way to know the market's choice of utility function and how it is parameterized.

After completing the text, financial engineers will have learned a single consistent framework within which capital assets, contracts, and corporate assets & liabilities, and physical assets defined by their cash flows can be valued. In each section, the relevant theories of valuation are presented first, followed by detailed chapters on applications. The theoretically inclined student may wish to focus on the valuation theory chapters, but the practical financial engineer will find significant value in the applications. There are also many opportunities in the text to develop or sharpen the student's Excel/VBA or Python programming skills using advanced financial applications.

1.1 ASSET TYPES

Assets may be tangible, such as a building or a ship. They may be intangible, such as a corporate investment opportunity, a celebrity's brand, or even a graduate education. Somewhere in between tangible and intangible assets lie capital assets, which are claims to cash flows. An example of a capital asset is a share of stock, which provides its owner with a percentage share of the appreciation and cash flows of an enterprise. The appreciation in price is called the *capital gain*, and corporate payouts to shareholders are called *dividends*. Another example of a capital asset is a bond. Bondholders receive interest payments called *coupons* over time in exchange for an initial cash investment, plus capital gains and scheduled *principal* payments. A third example of a capital asset is real estate, which is a physical asset as well. Owners of real estate assets receive capital gains and rental payments over time, while maintaining the properties by paying the needed operating expenses.

Can anything be a capital asset? Even owners of dairy cows receive cash from the sale of milk, and make payments for livestock feed, labor and facilities to produce the milk. One sh*udders* to think of the fatal consequences (to the cow) of a termination date, but this produces cash as well. Yikes. Even physical assets can often be thought of as capital assets if they can be used to produce cash flows. Coming full circle, buildings and ships generally have value because they produce cash flows for their owners, in the form of net rental fees and future cash proceeds from the sales of the assets. Even in the case of a private yacht used for personal cruises, the owner benefits from the use of the yacht while avoiding the payment of rental fees to

others. However, in this case, there are no doubt intangible benefits to owning a private yacht in addition to the measured cash savings, making it difficult to fully value the yacht in a market context.

Assets can be affected by their circumstances, particularly the legal environment. For example, in the U.S., most corporations enjoy government-granted *limited liability* which protects investors against claims that would exceed the value of the corporation's assets. This means investors would never be required to make additional payments to cover corporate losses. For example, a drug company could go out of business to avoid paying claims to consumers who were suing for damages caused by the drugs. A firm with limited liability protection may abandon its assets at any time, even if there are no lenders involved. However, the decision is irrevocable.

In *bankruptcy*, firms have the right to default on their debts by surrendering the assets of the firm to the lenders, or *debtholders*. Again, the decision is irrevocable. In an *involuntary bankruptcy*, the court forces a company into bankruptcy to satisfy the claims of its creditors. Tax incentives, such as the tax deductibility of interest paid on debt issued by corporations, may increase cash flows and therefore increase the value of the company. Therefore, the existence of limited liability, bankruptcy protection and tax subsidies all add value to a corporation's unenhanced assets. Shareholders in a limited liability company never have to pay a penny in damages to litigants, creditors or the government on behalf of the company in which they invested. They can never lose more than they invest. Therefore, the company may enjoy limited liability, and the individual shareholders may do likewise.

Contracts may also be thought of as *contingent assets*. Contracts have value because they commit two or more parties to actions that hopefully create value for all parties. Even if contracts have no value at inception, they may end up destroying value for one party over time, in which case the contract value to that party would be negative. For example, if a farmer contracts to sell wheat in the future for $4 per bushel, but the market price in the meantime increases to $5, he faces an opportunity loss of $1 per bushel, and the value of the contract is negative to him.[3]

In all of these cases, it will be necessary to value assets or contracts from time to time. In the case of futures contracts, i.e. exchange-traded standardized forward contracts, these revaluations occur continually. In the case of real estate, valuations occur mostly contemporaneously with financing, sales, and purchases. This text has to do with valuing financial assets and contracts, covering wide range of practical situations.

1.2 VALUATION MODELS

The *price* of an asset is the amount for which it is bought or sold. In a negotiation, there may be a price *offered* or *asked* by the seller, and another lower price *bid* by the buyer. The *transaction price* reflects the final agreed price of the asset at the time of sale. A market order to buy a share of stock on a stock exchange will be executed at the ask price, and a market order to sell will be executed at the bid. The dealer on the exchange looks to buy at the bid and sell at the offering or asking price.

However, the *value* of an asset might differ significantly from its price. For example, the *public value* of an asset likely differs from one party's *private valuation*. Indeed, one could argue that a public asset should only be bought by an individual if their private valuation meets or exceeds the public valuation. As another example, consider that the true value of a share of stock may differ from its price due to information unknown by the market. If a company's electric auto production is known privately by the CEO to be much lower than

[3] This type of contract is known as a forward contract, since the transaction occurs on a forward (future) date, and its terms, such as the future transaction price, are determined in advance.

market expectations, then the true value of the share of the company's stock is likely lower than the market price.

In sum, the value of an asset is not a simple calculation, and it is not necessarily a calculation at all. Models produce valuations based on assumptions. Assumptions simplify the complexities of reality and are used to manage the lack of complete information. When one prepares a valuation, assumptions are made, and therefore the results may always be questioned. The following quote from George Box and Norman Draper (1987) sums it up accurately:

> "Remember that all models are wrong; the practical question is how wrong do they have to be to not be useful."

Modelers should humbly keep this aphorism in mind when they prepare valuation analyses. For example, present value is not the same as value; it is but one model for valuation. There are many possible models of course, and any one of them might be considered suitable for a particular purpose at a point in time. Six of these models are listed in the table below:

Some useful valuation models

#	Name	Description
1	Liquidation Value	How much the highest available bidder will pay for an asset if sold in the near future
2	Replacement Cost	How much it would cost to replace or rebuild the asset
3	Comparable Value	The value of comparable assets, with valuation adjustments for imperfect comparables
4	Benchmark Valuation	Similar to comparables, but allowing a wide range of assets and contracts to be used to replicate the cash flows of the asset as well as possible, using securities and assets whose prices are known.
5	Discounted Cash Flow or Present Value	The present value of the net appreciation and cash flows that an asset creates incrementally; or, the amount of money one would have to set aside at the prescribed discount rate to produce a given pattern of future cash flows
6	Portfolio Valuation	The increase in the value of a portfolio as a result of the addition of the asset.

Liquidation Value is a popular and generally reasonable valuation model, especially for lenders, who would like to ensure they will be paid what they are owed even if the company fails and has to be liquidated. For others, liquidation puts a minimum guaranteed value on the asset, a very conservative standard. However, it does not reflect the full value of a company. For example, we sometimes refer to the value of a *going concern*, i.e. an asset in the context of an operating business. In *liquidation*, the forced sale of a going concern will likely reduce its ability to produce income. Also, the definition of available bidders can have a significant impact on asset value. In general, the more restricted the bidder pool, the lower the inferred liquidation value will be.

Replacement Cost is also practical in many cases. A designer home may have cost $2 million to build the first time, but it would cost much less to duplicate the second time. The owner may have "overpaid" for the asset because he or she placed a personal value on innovation, uniqueness, beauty or customization that went with the house. Not surprisingly, these owners often do not recoup the cost of designer homes when they are resold --- unless they can find buyers who value the unique characteristics of the home, and who do not have the ability to easily replicate it.

Comparable Value applies when assets can be directly compared, and the value is known for one of them . Two apartments in the same building, with the same features and similar views should be worth about the same amount. Therefore, if one apartment sells for $500,000, it can be used as a benchmark to estimate the value of a second apartment. But no valuation benchmark is perfect. One apartment may be in better physical condition than another, for example. Also, as time passes, the valuation benchmark becomes less relevant, since last year's sale price of one apartment may not reflect current price changes in the local real estate market.

Replacement Cost and Comparable Value are the most widely used valuation methods in accounting. Accounting valuation rules differ from financial valuation rules principally because accounting requires a degree of objectivity and verifiability --- accounting reports are for managerial, investor and public consumption. In contrast, financial valuation is expected to be subjective and private, with no need for consistency or verification, designed only to help managers make the best decisions possible. Accounting rules can help prevent owners of assets from abusing subjective aspects of their measurements to overstate or understate asset value. However, accounting procedures can but also lead to values which would be unrealistic from a financial valuation standpoint.

Benchmark Valuation is used when the cash flows of an asset can be approximated by other assets or contracts whose valuations are known. For example, when modeling Amazon stock, you might expect that some of the company's performance is related to the retail sector, while some is related to high-tech comparables. Benchmark valuation chooses the portfolio of retail and tech stocks that best matches Amazon's projected earnings in order to determine a valuation.

Discounted Cash Flow and *Net Present Value* models are used when we have a good sense of the expected cash flows that an asset will produce, and an appropriate discount rate reflecting the risk of those cash flows. In Part I of this text, we will take the discount rate as given, but in Part II, we will discuss models that can be used to determine appropriate discount rates considering risk adjustments.

Portfolio Valuation is used when new assets interact with existing assets in a portfolio. For example, an oil drilling company with 50 wells already in operation will value a new oil well differently than a new entrant into the industry. The established company will likely be better able to enjoy operating efficiencies and other *synergies* with the new acquisition. This implies the value of the new well will likely be higher for the established company than for the new entrant. To value the new well, the company would determine the value of the 51 wells together, and subtract the value of the 50 wells in order to determine a marginal private valuation for the new well. This methodology would allow the company to capture the value of synergies of the new well with their existing portfolio, and thereby come to a more accurate private valuation.

In traditional financial asset pricing models, portfolio valuation is unnecessary, since all parties are assumed to value an asset in the same way. In practice, private valuations can differ markedly from public valuations ---- both much higher and much lower. In this text, private valuations are considered as important as public valuations. This is because many investments are either unavailable or too specialized to be of interest to the general public, especially in the early stages. Venture Capital firms, for example, often invest heavily in a

small number of startup firms that lack access to the public markets. Accordingly, they value the portfolio not as well-diversified investors, but as investors with serious constraints on their capital and managerial resources.

Considering the goals of the text, we will focus mainly on *Benchmark Valuation*, *Discounted Cash Flow* and *Portfolio Valuation* methods. An asset for our purposes is anything that creates cash flows and capital appreciation or depreciation. The total return on an asset is defined as the incremental net benefit produced by the asset over a period of time. The cash flows can be positive or negative, depending on whether we will be receiving or making payments. The change in asset value, called the *capital gain*, records the appreciation or depreciation of the asset over time.

Summarizing this section, valuation methods 1, 2 and 3 are used by accountants to report information in an unbiased and consistent manner, one that is difficult to manipulate and hopefully will not mislead investors. Unfortunately, accounting rules can also sometimes lead to distortions that may mislead investors, whether intended or not. For example, book value (depreciated acquisition cost) is often used as a basis for valuation by accountants, when market value can be significantly higher or lower.

Financial valuation follows method 4, 5 or 6. Financial models are normally used by management and investors to make financial decisions based on economic criteria, and are expected to contain subjective assumptions of managers or investors. All six methods represent valuation models, and any one of them might be most appropriate for a given situation.

1.3 RATES AND RETURNS

The *interest rate* is the price of money. It is the cost of money to a borrower, and the benefit of money to a lender. The term *interest rate* can be interpreted as a contracted rate in a loan transaction, but in general, when traders speak about interest rates rising or falling, they usually are referring to risk-free government bond interest rates.

Investors compute returns as lenders do. If an investor purchases a security for $40 and sells it for $48, then she earns a 20% *return*. In this case, the *realized return* was 20%, but we don't know what her *expected return* was at the time of investment. Perhaps she expected a 10% return and was pleasantly surprised; if she expected 30%, then she may have been disappointed. Expectations always refer to the future, while realizations have already occurred in the past.

Expected return has two components. People buy shares because they expect them to appreciate in value, and sometimes because they pay dividends. The *capital gains rate* is the percentage increase in the price of an asset after dividends are paid, and the *dividend rate* is the annual percentage dividend paid. Capital gains and dividends can be expressed in percentages or dollars.

The term *required return* suggests the minimum amount one would need to earn to justify an investment. It would reflect my *opportunity cost*, or what one would give up in order to shift money from another investment to the current one. I might also expect a *risk premium* if the new investment were riskier than the old one. For example, if I use a risk-free asset as my starting point, my opportunity cost is the lost interest, and my risk premium is the total compensation I would receive for taking risk.

What is your required return on capital? If I borrow money from you paying 3% interest per year, assuming my default risk is negligible, would you be happy with that? The answer may differ for each person. For

someone earning 1% interest (i.e. a 1% return) on a savings account at a bank, with no better alternative, and who didn't likely need the cash for a while, a chance to earn 3% might look very attractive. For someone with low risk investments earning 8% on average, it might look unattractive to lend at 3%. Therefore, when we establish our required returns on investment, it is important to know our own *opportunity cost of capital*, i.e. what we are giving up on our current investments to invest in a new project.

The term *discount rate* is used specifically to denote the rate of return used to discount future expected cash flows to determine the current value. It is also the return on investment if we purchase the cash flows at that price, the cash flows are realized, and the asset is held until the last cash payment is made.

In some analyses, we will use *growth rates* to describe the rate of change in cash flows over time. The growth rate of cash flows should be distinguished from the rate of return on the asset composed of the cash flows. We use *default rates* to reflect the likelihood that the payer of the cash flows may fail due to bankruptcy or other reason. Normally, all rates are quoted on annual basis unless stated otherwise.

Example

Suppose you purchased a share of Walmart for $70 one year ago, with money from your bank account yielding 3% per year. You expected it would pay a $1.00 dividend and appreciate in value by $6.00. Then, it actually paid $0.50 in dividends and appreciated by $10 in value. Then we could say any and all of the following:

- The expected one-year total dollar return was $7.00 (10%) one year ago
- The expected one-year capital gain was $6.00 (8.57%) one year ago
- The expected dividend was $1.00 (1.43%) one year ago
- The realized one-year total dollar return was $10.50 over the last year(15%)
- The realized one-year capital gain was $10.00 over the last year (14.29%)
- The realized one-year dividend was $0.50 over the last year (0.71%)

These quantities are often expressed as percentages, when divided by $70.

Also, since you bought the share expecting a total return of 10%, you were pleasantly surprised. Your opportunity cost of funds was $2.10 in lost interest (3% of $70), so your implied incremental risk compensation was $4.90 ($7.00 – 2.10) or 7%.

This table summarizes the Walmart investment decision one year ago.

	Required total return (RTR)		Expected total return (ETR)	
Percentage	10%		10%	
Dollars	$7.00		$7.00	
	Time value	Risk compensation	Exp capital gain	Exp cash flow
Percentage	3%	7%	8.57%	1.43%
Dollars	$2.10	$4.90	$6.00	$1.00

What if you had the opposite opinion, that Walmart would fall 10% in value? In this case, you could sell Walmart stock directly if you owned it, and perhaps compute the expected benefit to your portfolio as a result. If you did not own Walmart stock, however, you could also *sell short*.

Selling short means that you would borrow the asset and sell it, with an agreement to repurchase the asset in the future and return it to its owner along with any dividends received. Investors purchase stocks generally believing their value will increase; they sell short when they think the value will fall.

In the case of buying Walmart, the maximum loss of $70 provided an easy way to convert dollar returns to percentage returns. In the case of selling Walmart short, however, the short seller would have to eventually buy back Walmart stock at whatever price it reached, facing a theoretically unlimited loss! Percentage return is therefore meaningless in the context of short sales.

1.4 WHEN RETURNS ARE UNDEFINED

The definition of percentage returns can be difficult. In the Walmart example of the last section, we discussed dollar returns, which were the amount of increase in value one would receive in an asset over a period of time plus any cash flows. We were able to divide these dollar returns by the price of Walmart stock at the beginning of the period to determine the percentage returns, since there was no possibility that additional cash would be required to fund the stock position. Said differently, because of limited liability, we can divide dollar returns by the share price to determine percentage returns because the share price represents the maximum investment we would have to make in one share of the stock. The share price also measures the maximum loss on investment. Much of traditional financial theory is based on percentage returns, but surprisingly, percentage returns often cannot be calculated, because either (a) the initial investment is insufficient to cover losses or (b) the maximum loss is unknown.

For example, consider once again a short position in Walmart stock. Suppose you sell short at $70, the share drops to $60, and you repurchase the share to earn $10. What was your percentage return? You might be tempted to say $10/70 \approx 14\%$, but you would be incorrect. Why? Because if the share had gone to $1070, you would have lost $1000 per share! Your loss could even have been higher than that. Since your maximum loss is undefined, your percentage return cannot be computed. We do not have any idea (yet) how to measure the amount of capital required by the short position.[4]

Similarly, suppose you enter a contract to buy a barrel of oil in one month's time for $50. The amount $50 is called the *forward price* since it specifies the price of a future transaction and not a current price. The value of the contract is zero at inception, since it costs nothing to establish the position. If the forward price of oil rises to $51, you could transfer your contract to someone else and take a $1 profit per barrel. You might be tempted to say you made a return of 1/50, or 2%. Unfortunately, you would be wrong, since it cost you nothing to enter the contract, and you could have lost $50 *or more*. If you had the misfortune to make this trade in April 2020, the futures price would have reached (negative!) -$37 per barrel, causing a loss of $87 per barrel in total! Your percentage return cannot be computed for the simple reason that division by zero (the investment amount) or the maximum loss (infinite) is undefined. Finally, if you had agreed to *sell* oil in one year's time, again, we would also be unable to compute the maximum loss, since there is no upper limit on the oil forward price. You could be forced to buy back the position at an incredible loss.

For these reasons, we use percentage returns only in the case of assets that are purchased and paid for in cash, and have limited liability protection. This means percentage returns are limited to stock, bond and

[4] For an interesting case study on the possible losses of short positions, you may like to read about the GameStop case in January 2021, where buyers of the stock caused prices to increase by 16 times their initial level within a month.

option purchases. These can be referred to as *long positions* that are fully funded. For short positions and forward contracts, we must use dollarized returns. In this text, we generally only use dollar returns, since dollar returns can be computed for all asset and contract positions.

Finally, a *very small* note on return vocabulary. A *basis point* is equivalent to 0.01%. Hence, 0.25% would be called either 25 basis points, 25 bp when written, or when spoken on a trading floor, 25 *"bips"*. In loan transactions, a *point* is used to refer 1% of a transaction's value. For example, a real estate mortgage broker might collect a transaction fee of two *points* on a $250,000 mortgage loan, or $5000 upfront.

1.5 THE MOST BEAUTIFUL EQUATION IN FINANCE

The most powerful valuation concept in finance can be captured in a single equation. This equation values cash flows, securities, derivatives, and corporate assets. In words, the equation states that the price of an asset must be set so that the required return on the asset will equal its expected return. If the expected return is a function of the asset price, we can then often solve the equation to determine the asset value.

How do we know this? A buyer must believe that the expected total return of the new investment (ETR) must equal or exceed his minimum required total return, including compensation for risk. Then ETR $\geq$ RTR. A seller believes the opposite, that ETR $\leq$ RTR. Since the buyer and seller transact at the same price in a single moment, unless one of them has superior private information, the only way this equation can be true is if ETR = RTR.

The General Valuation Equation (GVE)
(in dollars)

Required return = Expected return

Time value + Risk compensation = Expected capital gains + Expected dividends

The required return is what the asset buyer "requires" or must earn for giving up the time value of money, plus some compensation for risk as appropriate. In this chapter and the next, there is no risk, so the compensation for risk is necessarily zero. In later chapters, the risk compensation will be the cost per unit of risk multiplied by the risk measure. With no risk, the required return is the risk-free rate of interest. The required return can also be thought of as the opportunity cost of funds for similar assets.

The expected return of the asset is related to how the asset is managed and what the asset produces. Normally, assets produce some combination of capital gains, i.e. appreciation or depreciation, and cash flows.

To see how this works, we begin with the *discrete-time* case. This term implies that cash flows and valuations will occur on fixed dates, as opposed to the *continuous time* models, where cash flows, capital gains and valuations change continuously in time. In practice, most corporate valuations are based on discrete-time analysis, and most derivative contract valuations are based on continuous time analysis. Some applications require both, such as the valuation of a derivative contract at a number of set future dates. Examples of these applications will be covered in future chapters. We will use discrete time models for all valuations in this text, including derivatives.

Let's take the simplest possible discrete-time case, one for which you may already know the answer. Begin with an asset that pays a single risk-free cash flow of C in one year. Its value today will be denoted as V_0, and the risk-free interest rate r is a simple annual risk-free interest rate. How can we determine V_0?

The GVE must hold over the next year

Required return = Time value of money + Compensation for risk

Required return = rV_0 + 0

The compensation for time value can also be thought of as the interest an investor would have earned on the funds needed to purchase the asset, had he kept those funds in an account earning the risk-free rate of interest --- a risk-free opportunity cost of funds, i.e. rV_0. There is no compensation for risk, since the cash flow is assumed risk-free in this example.

Expected return = Expected capital gain + Expected cash flow

Expected return = (0 – V_0) + C

The expected capital gain is 0-V_0 since the asset holder gives up their asset for 0 at time 1, i.e. receives nothing more after the cash flow C is paid. After the cash flow payment, the asset is worthless. Alternatively, one may think of exchanging the asset V_0 for the final payment of C, in which case the capital gain would be C-V_0, and the additional "cash flow" would be 0. Both characterizations lead to the same result.

Now we may solve the GVE:

Required return = Expected return $\quad rV_0 = -V_0 + C \quad V_0 = C/(1+r)$

The present value of a single risk-free payment of C in one year is C/(1+r), as expected. While this result is obvious and well-known, the purpose of the derivation is to introduce the GVE and how to use it. In coming chapters, this knowledge will aid us in valuing a large class of simple and complex assets, including futures and options. The same equation that values a cash flow in one year can value the most complex derivative securities.

Example 1 (more difficult than it seems)

The U.S. government offers a Treasury bill (T-Bill) which pays $10,000 in one year. The risk-free discount rate is 5% per year. What is the value of the T-Bill today? Should I buy the T-Bill? If I can buy the same T-Bill from a dealer today for $9500, what is its expected return? Should I buy it from the dealer?

Solution 1

According to the net present value model, the value today is 10,000/1.05 = $9524, rounding to the nearest dollar. If I can buy the T-Bill for $9500, the expected return is 10,000/9500 – 1 = 5.26%, provided I hold it for the full year and do not sell it.

My private valuation will probably be lower than either $9524 or $9500, since any of the following factors might apply:

1. I can earn a higher return elsewhere taking little or no risk
2. I cannot take the risk of losing access to $9524 in the event of a cash shortfall in the next year

The T-Bill can be resold in the capital market between now and one year from now, which reduces the risk of factor #2, though there may be some transaction costs to be considered for selling the T-Bill prior to maturity,

and my return would not be 5.26%. In fact, if interest rates rise enough, the value could fall below $9524, in which case the realized return would be negative if I sold it.

If I am offered the T-Bill for $9500, the same broad concerns apply. However, I would be happy to buy the T-Bill for $9500 and sell it in the market for $9524 at the same time, provided the transaction costs did not exceed $24. If I could perform these transactions at the same time, it would be called a risk-free *arbitrage*, earning a risk-free profit with no risk. If I could borrow funds to buy the bonds, my costs would also be zero, and my gain would be limited only by the number of available bonds to purchase at $9500 each.

In sum, the net present value method is a useful model when a liquid public market exists for a security, but it is less useful as a private valuation model.

Example 2 (The Elon Musk problem)

Suppose Elon has $44 billion in assets in total. A bank offers Elon the following deal:

- $5.00 now
- 50% chance of receiving an additional $44 billion in one year
- 50% chance of paying $44 billion in one year

The risk-free rate of interest is assumed to be 10%, and the risk of the outcome is random, unaffected by the market. According to the usual finance textbook analysis, the present value is $5 plus the expected value of the outcome divided by 1+r, or 1.10. Since the expected value of the outcome is 0, the overall present value is $5, regardless of the discount rate chosen. Should Elon take the deal?

Solution 2

Common sense suggests that Elon should not take this deal, but the present value is positive, suggesting he should.

Let's try with the GVE. Suppose the risk measure is either the maximum loss ($44B) or the standard deviation of the payout (coincidentally also $44B in this case). An investment bank might put the cost of risk at 25% of the risk measure (standard deviation), or $11B.

From the GVE, we solve for V_0 without the $5 (as if the $5 mattered!)

Required return = Expected return

Time value + Risk compensation = Expected capital gains + Expected cash flow

$$rV_0 + k\sigma = (0 - V_0) + E[C_1]$$

$$0.10V_0 + 0.25(44B) = (0 - V_0) + 0.50\big(44B + (-44B)\big)$$

$$V_0 = -\frac{\$11B}{1.10} = -\$10B$$

In other words, under these assumptions, Elon should only accept the deal if he were paid $10B now. This makes more sense than a positive valuation of $5. Furthermore, the bank should not even offer this deal, because the bank itself would have to be paid $10B to take the deal! Interestingly, the GVE gives us two prices for the deal, one for the bid and one for the ask, at ±$10B.

The two examples above illustrate that a simple financial model blindly can lead to the wrong decision if it is applied improperly. However, if financial theory is applied correctly, it can also shed light on how to solve real-world valuation problems. In practice, financial engineers are charged with applying finance theory to solve unique problems in the real world. This text aims to expand their toolkit beyond the conventional basics taught in most finance textbooks.

In later chapters, we will show that the GVE can be used to value all of the following:

- Annuities: Sequences of fixed or growing future payments
- Perpetuities: Infinite sequence of fixed or growing future payments
- Stocks and bonds
- Structured debt products
- Risky cash flows
- Futures and forward contracts
- Financial options and real options
- Corporate assets and liabilities
- Financial institution assets and liabilities
- Privately held assets and portfolios

In addition, we will show that many of the famous equations in finance governing returns and prices are special cases of the GVE:

- Benchmarking models used for equity
- The Modigliani-Miller Proposition II
- The Capital Asset Pricing Model (CAPM)
- The option pricing differential equation of Black-Scholes-Merton (BSM)

Finally, we will show that the GVE can generalize traditional valuation models in many dimensions:

- Multiperiod/multi-asset valuation models allowing intertemporal correlations
- Models for assets with variable volatility
- Valuation models allowing for optimization
- Models of corporate decisions: capital budgeting, capital structure, and risk management

We must humbly keep in mind that as we study and build more advanced valuation models, they are still models, and therefore necessarily still wrong. We can only hope that the models we build are less wrong and more useful than those that have come before!

1.6 REFERENCES AND FURTHER READING

Box, G. E. P., & Draper, N. R. (1987). **Empirical model-building and response surfaces**, Wiley series in probability and mathematical statistics. Empirical model-building and response surfaces. Oxford, England: John Wiley & Sons.

1.7 INTERVIEW QUESTIONS

Answer the following questions using no more than 2-3 sentences each. Be concise, and prioritize your responses to the more complex questions.

1. What is the difference between valuation and a valuation model?
2. What valuation models are primarily used by accountants and why?
3. What valuation models are primarily used by financial engineers and why?
4. What is the difference between a financial asset and a contract?
5. How does the bankruptcy option add value for a corporation?
6. How does limited liability add value for a corporation?
7. What does a company lose in a liquidation?
8. How does a discount rate differ from a rate of return?
9. What is the benefit of a present value model?
10. In what situations are percentage returns undefined?
11. What is the difference between price and value in a stock? In a forward contract?
12. What is the relationship between expected returns and required returns in a competitive capital market? Why does this relationship hold?
13. Why do private valuations differ from public valuations?

1.8 NUMERICAL QUESTIONS

Assume the risk-free one-year interest rate is 10% for all these questions.

1. Suppose the assets owned by a company are currently worth $100, and the company has $120 in liabilities due next year. What can we say about the value of the company?
2. Suppose you are an excellent driver and a skilled insurance actuary. You have determined that your expected losses due to automobile accidents is $1000 per year (assume paid in one year), largely due to risks caused by other drivers, of course! However, the cost of insurance is $2000 per year, paid in advance. What is the present value of the one-year insurance contract to you? Should you buy auto insurance?
3. Suppose you go short GameStop at $40 per share, and the price rises to $80 one year later, at which point you repurchase the stock. What was your return, in dollars and percentage terms?
4. Suppose you bought Tesla at $400 and sold it at $600 per share in a one-year period. You thought it was going to go to $500, but you were pleasantly surprised. Tesla does not pay dividends, and the risk-free rate of interest is 10%. Please fill in the table below, based on information at the time you invested:

GVE elements	Expected value
Time value	
Risk compensation	
Capital gain	
Cash flow	

5. You believe Alibaba will pay a $2 dividend in the next year. Also, you expect the share price to rise to $128 in a year's time, with a standard deviation of $15. The cost of risk (as measured by the standard deviation) is 60%. According to the GVE, what is your Alibaba valuation today?
6. Continuing question 5, compute the derivative of V_0 with respect to the equation elements and indicate the sign of the derivative. Do these values make sense? Hint: Risk compensation is the price of risk (k) multiplied by the risk measure (σ).

	Formula	Sign (+ / -)
GVE		N/A
Solve for V_0		N/A
Derivative with respect to μ		
Derivative with respect to k		
Derivative with respect to σ		
Derivative with respect to r		

1.9 CHAPTER VOCABULARY

The list below provides a quick vocabulary reference that will be very useful to you while reading this text and in your study of finance. You may already be familiar with some of the terms.

Arbitrage	A trade that produces a risk-free profit, ideally at zero cost
Asking price	The best price at which an outside buyer could execute a share purchase on the stock exchange immediately. Also called "*offer price*", *offered price*, or simply *offer*.
Bankruptcy	The option to surrender a firm's assets to its creditors and exit the operation. May be involuntary, i.e. forced by the courts.
Basis point (b.p. or bip)	One one-hundredth of 1%, or 0.01%
Benchmark	Asset of comparable risk used to help price another asset
Bid price	The best price at which an outside seller could execute a share sale on the stock exchange immediately
Bond	A security that offers coupon payments and principal repayment in the future in exchange for cash today
Capital assets	Assets defined by the cash flows they produce
Capital budgeting	The corporate decision as to what investments to accept and reject
Capital gains	The increase in the price of an asset or contract
Capital structure	The corporate decision as to what securities and commitments it should issue in order to finance its operations
Comparable value	Estimation of an asset value using another asset that is similar
Contingent assets/liabilities	Contractual obligations that may become assets or liabilities in the future
Coupons	Payments made to bondholders other than principal
Debtholders	Investors who own bonds
Default rate	Percentage of an obligation expected to become unpaid or uncollectible within a period of time
Derivative contract	Forward, futures or option contract.
Discount rate	Interest rate used to find the present value of cash flows.
Discounted cash flow (DCF)	Conversion of future cash flows to present value equivalents for the purpose of comparing different cash flow levels and timing

Discrete cash flows	Cash flows paid at fixed dates (as opposed to continuous payments)
Dividends	Cash flows paid by corporations to their shareholders
Expected return	The expected cash flows and capital gains to be received from owning an asset. May be in percentage terms or currency.
Forward contract	Agreement to buy or sell something at a later date (the forward date) with terms specified today (such as price and quantity)
Future value	The value of an asset at some future date, if the payments are deposited or reinvested at the assumed return rates.
Futures contract	A forward contract that is standardized and traded on an exchange.
Hedge fund	Investment company that typically uses more complex strategies and analysis to provide superior returns to investors.
Limited liability	The right of a corporation to abandon its business, and disclaim any further obligations.
Liquidation value	The value of an asset if it is sold immediately.
Long and short	"Long" is a synonym for owning an asset or profiting if the asset rises in value. "Short" means one has sold short, agreed to sell in the future, or profits if the asset falls in value.
Offered price	See *asking price*.
Opportunity cost of capital	When taking cash from one investment to another, the opportunity cost is the yield on the prior asset
Options	A wide-ranging term. Generally a contract that allows the owner to make a decision in her own best interest at the expense of the writer of the contract.
Physical assets	Assets valued for their actual use, compared to financial assets, which are valued for their cash flow benefits.
Portfolio valuation method	The value of an asset is the value of one's portfolio with the asset minus the current value of one's portfolio.
Present value	The conversion of a cash flow in the future to its value today.
Principal	The base amount of a bond debt, used to compute the coupon payments and the final payment of a bond.
Rate of growth	In cash flows, the expected percentage change in the cash flows over time.
Rate of return	The rate of increase in the value of an investment, whether through capital gains or cash flows.
Real options	Compared to exchange-traded options, real options give the owner a choice to take an action in the future that may add value for the owner.
Realized return	Past returns.
Replacement value	What it would cost to replace an asset.
Required return	The return an investor needs to earn to hold an asset.
Risk charges	Usually measured in currency terms. Negative adjustments to expected cash flows to obtain risk-neutral equivalents.
Risk-neutral	Risk cash flows minus risk charges, which can be discounted at the risk-free rate of interest.
Security	A legal and negotiable (tradeable, transferable) instrument binding the issuer to certain obligations, like a share of stock or a bond.
Selling short	Borrowing an asset to sell and later repurchase in the hopes of declining asset prices.

Stochastic	Random
Stock	Percentage share in the residual value of an enterprise.
Synergy	1+1=3. The value of two things together exceeds the sum of their separate values.
Tax deductibility	The ability to reduce taxes by reducing taxable income.
Term structure	The risk-free interest rate for a cash payment at every date in the future.
Undefined return	Percentage returns that cannot be determined because (a) the initial investment does not represent the full cost or (b) the maximum loss may be unlimited
Venture capital	An investment firm that specializes in funding startup companies that typically have no access to capital markets.

1.10 PRESENT VALUE IN PRACTICE: USING MICROSOFT EXCEL

There are many basic Excel tutorials online (such as YouTube) available free of charge, so we will focus only on selected financial functions here. This is a list of functions directly relevant to the materials covered in the first two chapters. Arguments in square brackets are optional. Parenthetical values refer to the equivalent notation used in this chapter.

Be sure to follow these tips:

- The number of payments and payment size correspond to the correct period length. For example, all parameters should be monthly in a problem with monthly payments.
- When entering the PV and PMT (payment values) as arguments to a function call, be sure to enter the PV as a negative number if the payments are positive.
- As you begin to type the formula, you will see the list of arguments appear in the correct order, allowing you to speed up your formula entry.

Excel Function	Arguments	Purpose and Use
=PV(rate,nper,pmt,[fv],[type])	rate=periodic interest rate (r) nper=number of periods (n) pmt=payment (C) fv=future value of annuity after all payments are made type=1 if first payment is immediate, 0 if in one year	Finds the present value of a flat annuity (PVA)
=NPER(rate,pmt,pv,[fv],[type])	pv=present value	Finds the value of n that makes the present value of the flat annuity equal to pv
=RATE(nper,pmt,pv,[fv], [type],[guess])	guess=initial value guess to speed up calculation	Finds the periodic interest rate that makes the present value of the payments equal to pv.
=NPV(rate,value1,[value2],…)	value1=first cash flow in one period (C_1)	Finds the present value of a range of equally spaced cash flows at the given rate.

=IRR(values,[guess])	values=set of cash flows, usually a negative cash flow (investment) followed by positive cash flows	Finds the rate that makes a sequence of cash flows have pv=0. *May not be unique if there are more than one sign changes in the cash flow.
=XNPV(rate,values,dates)	dates=Excel-format dates corresponding to the cash flows	Finds the present value of a set of cash payments made on specific dates.
=XIRR(values,dates,[guess])		Finds the discount rate that makes the present value of a set of cash flows on given dates equal to zero.

MINIPROJECT 1: LEARNING EXCEL WITH LOAN MATH

The purpose of this project is to become very familiar with Excel (formulas and graphs for now) and to learn something about loan economics and mathematics. This project continues into Miniproject #2, at the end of the next chapter.

All financial engineers must become expert in Excel. You may feel that Python or C++ suffices for your analytical needs, and that these programs are quicker and more powerful than Excel. If you believe this, you would be correct. But that is not the point. It is fine to adopt this position if you only communicate with other quants, who are happy to receive C++ code or swap Jupyter notebooks. They can easily understand whatever analysis you are offering. But you may be surprised at the capabilities of Excel (matrix math, statistics, optimization and simulation), or to learn that advanced applications available in VBA, C++ or Python can be incorporated into Excel when additional firepower is needed. You can make Excel your Graphical User Interface (GUI) for extremely advanced and fast applications. Finally, advanced proficiency in Excel should be listed among the computing skills on your resume!

Considering that your future income and promotions in the financial field depend on more than quant skills, you need to communicate effectively with all of the following people, most of whom will never read your code:

- Managers (who determine your bonus)
- Non-technical colleagues (who may become your managers at your current or future company)
- Clients (who may hire you someday)
- Fellow conference attendees (who want to be able to talk about your work)
- Outside publications (to publish your work)
- Regulators (who need to understand your work)
- Ratings agencies (who will use your work to rate fixed income instruments)
- Future employers (you never know!)

These people will generally lack your programming skill will need to be impressed with your work. They can understand your spreadsheets and presentations better than they can understand code.

Anyway, I digress. Here is the project. In Excel.

Loan project

Design an Excel spreadsheet for your client taking out a loan from either Chase or Wells Fargo. Assume the loan maturity is 2 years. The payments are monthly. The Chase loan has a simple annual rate of 10% and no fees. The Wells Fargo loan charges only 7% simple annual interest, but also charges 2% in loan fees (2 points or $1000) which are financed into the loan at inception. The borrower needs $5000. Allow for the two loans to be evaluated simultaneously in the same spreadsheet to help the borrower decide which loan to take.

A. For each month from 1-24, compute the key elements of the amortization table, which shows the balance of the loan every period under the assumptions that:
   1. The loan payments are made exactly as scheduled until the maturity date
   2. There is no risk of default
   3. The loan is not *prepaid*, meaning the borrower does not pay off the balance of the loan prior to maturity.

Loan Amortization Term	**GVE Equivalent**
Beginning principal balance (start with i=0)	V_i
Interest	rV_i
Payment	C
Payment – interest (this is the reduction in principal)	$C - rV_i$
New principal amount	V_{i+1}

If you have done the amortization table correctly, the final balance of the loan will be exactly zero on the maturity date.

B. Produce a graph for the user showing the breakdown of each payment into principal and interest on each date (stacked bar). Produce a separate graph showing what the payoff balance would be as a function of the payoff date (line chart).
C. Provide a validation formula in your spreadsheet for determining the payoff balance on each date that does not use the amortization table.
D. Compute the present value of all the payments of each loan as a function of the early payoff date, discounting both loan payment streams at the Chase loan rate. Make a graph of these and state any conclusions.
E. Design your spreadsheet so that the user can change all the terms of the two loans (except the 2-year maturity), and make a clean comparison between the loans. Also provide a recommendation of which loan should be chosen, based on your calculations.

Your final report should be a 1-2 page maximum pdf showing graphs, analysis and conclusions.

Make sure your report looks attractive and easy for your client to read!

- Label your charts and column headings
- Emphasize text where appropriate with **bold text** or larger font sizes.
- Exclude extraneous information.
- Show your dollar values to two decimal places without dollar signs, and your interest rate percentages down to the basis point.

Chapter 2: Annuities and Perpetuities, Basic and Complex

INTRODUCTION

Cash flows come in many forms. Very often, they follow standard patterns. Given a discount rate, we could use the NPV function in Excel or any similar spreadsheet program to find the present value of any given set of payments easily, regardless of the pattern. However, sometimes we need to be able to value frequently occurring cash flow patterns using closed-form analytic formulae. For example, a financial engineer might need to develop an application involving millions of present value calculations, such as determining the value of an option to convert a corporate bond to shares of stock. In this case, she would need an advanced valuation method, but the relatively simple present value computations would be a critical component of the analysis.

For this reason, Chapter 2 focuses on several different variants of perpetuities and annuities found in advanced modeling exercises, with the objective of helping the student derive closed-form analytic present value formulae now and more efficient valuation algorithms in the future. In all cases, as in all the Part I chapters, we assume the discount rate is known and constant. The GVE can value these complex cash flow patterns with surprising ease. After completing this chapter, students will understand how to value several unusual types of growing annuities and perpetuities. There are some references to advanced math for the curious student, but advanced math is not required. Most everything can be completed with college-level algebra and basic calculus.

2.1 ANNUITIES AND PERPETUITIES

Simple *annuities* are defined as sequences of equally spaced fixed payments over time, such as a fixed loan payment of $1000 per month for 24 months. By default, the first payment of an annuity is assumed to occur one period in the future. A fixed rate loan is a good example of an annuity, since the borrower receives the loan proceeds in the beginning, at time 0, and makes regular constant payments every period {1..n} until the loan is repaid. Another example of an annuity is an employment contract, wherein you might receive equal payments every month for the duration of your contract.

Annuities can also be scheduled to grow. Growing annuities have built-in increases or decreases over time, usually expressed as the percentage change (g) in the payment from one period to the next period. For example, an employment agreement will often contain guaranteed percentage increases, perhaps as compensation for expected inflation, or to reward employees for staying with the organization.

A *deferred annuity* is one that starts at a future date more than one period in the future. Each period an annuity is deferred, its present value is discounted one period. Conversely, if an annuity is advanced so the first payment is now instead of one period from now, the present value is multiplied by (1+r). When the first payment occurs immediately, this is sometimes called an *annuity due*, or an *annuity in arrears*.

In some instances, annuities may grow by a fixed dollar amount each period instead of by a percentage. In other cases, particularly when annuity payments are linked to commodity prices or interest rates, we may find that the underlying index decays exponentially to a higher or lower long run mean level. This is called *mean reversion*. If cash flows are linked to these indices, they should be discounted using formulae that account for mean reversion. Annuities whose payments grow at fixed percentage rates, fixed dollar rates, and mean-reverting rates occur frequently in advanced financial models.

Perpetuities are perpetual annuities, which are more common than you might think. For example, the owner of a preferred share of stock issued by a corporation would receive a contracted dividend each year with no fixed

maturity date. In practice, preferred shareholders may have the option to convert their shares to common (regular) shares of the company, so the preferred shares might not last forever in their original form. These are called *convertible preferred* shares. The conversion option will be valued in a later chapter, when we discuss the valuation of options. As another example of a perpetuity, if the owner of a bank account paying a fixed interest rate withdrew the interest she earned every year, she would also create a perpetuity for herself. Again, the payments might not last forever, but there is no predetermined ending date, since the balance never would never fall below the initial deposit.

Finally, in this chapter, we also often value perpetuities as a first step in order to find closed-form mathematical expressions for the annuity values. All of the mentioned annuities and perpetuities are valued in this chapter, using the GVE to do so efficiently. You may have previously learned other methods to value these instruments based on mathematical relationships, such as those provided in tables of integrals or transforms. In this chapter we focus on the GVE to obtain the values. This also sets the stage for more advanced valuations later in the text using the GVE. As a bonus, we can use the GVE technology to compute simpler analytic expressions for traditional bond risk measures like duration and convexity.[5]

Optional advanced comment!

Students who are very mathematically inclined may notice that the discrete present value function maps to the one-sided z-transform, with z = (1+r). This means that for most of the examples in Chapter 2, one could simply look up or adapt the present value formulae from a one-sided z-transform table. However, even advanced students are encouraged to study the methods presented in this chapter, since these methods are not only more intuitive, but they also solve present value problems that do not appear in these tables. It is also interesting to note that the GVE itself may be considered a difference equation, and the methods presented in this chapter are also used to solve difference equations in other mathematical applications. Indeed, the continuous complement of difference equations, i.e. differential equations, was proposed by Black-Scholes (1973) and Merton (1974) to value options and other derivative securities. It makes sense that discrete cash flows could be valued using difference equations.

2.2 NOTATION

In order to proceed with the valuation of many different types of annuities, it is convenient to define notation that is consistently applied throughout the chapter. In valuing cashflows occurring at discrete intervals, or in discrete time, we will generally use time 0 as the present, and the subscripts "i" as a time index. **Unless otherwise specified, the time step between cash flows is one period, the first cash flow occurs in one period, and the discount rate is the simple periodic rate.** We will address exceptions and variations to these cash flows in Chapter 3.

Definition of terms

n = date of final payment

r = simple discount rate per period

C = amount of a single cash flow, or bond coupon

C_i = cash flow on a particular date i

[5] Duration is proportional to the value of a linear growth annuity, and convexity is proportional to the value of a quadratic growth annuity, both of which are valued in this chapter.

F = bond face value or principal amount

g = percentage growth rate in cash flow over one period, i.e. $C_2 = C_1(1+g)$

b = fixed dollar increase in a cash payment per period, i.e. $C_2 = C_1 + b$

μ = long-run mean (asymptotic) cash payment

κ = speed of reversion of cash flow to long-run mean per period; the percentage movement of a cash flow towards its long-run mean in one period

V_i = valuation of a financial instrument as of date i, so V_0 = value today

$V_1 - V_0$ = the capital gain on an asset over one period, after cash flows at time 1 have occurred

d = 1/(1+r), an analytic convenience that makes the formulae appear simpler and make them easier to remember

$d_n = 1/(1+r)^n$, the discount factor or present value of $1 paid at time n, using discount rate r.

r* = (r-g)/(1+g), the discount rate applied to a flat annuity in order to value a growing annuity

$d_n^* = 1/(1+r^*)^n$, substituting r* for r in the discount factor.

2.3 VALUING A SINGLE PAYMENT AT ANY FUTURE DATE

In Chapter 1, we valued a single risk-free cash flow in one year. We now begin with a two-year time period, and use mathematical induction to derive the general case. To value a single payment C_2 in two years, it is useful to work *backwards*. Using this method, we first value the asset at time 1, and then value the asset at time 0 knowing the value at time 1. Following this logic, at time 1, the GVE holds:

$$rV_1 = (0 - V_1) + C_2$$

and moving back to time zero the GVE must also hold, substituting the now known value of V_1:

$$rV_0 = (V_1 - V_0) + 0$$

Note how the capital gain is treated differently at times 0 and 1. At time 1, we receive the general capital gain on the asset (V_1-V_0) and no cash flow. At time 2, we are giving up the asset and receiving a cash flow. Solving the first equation, we compute $V_1 = C/(1+r)$. Substituting this into the second equation, we arrive at the valuation of the cash flow in the second period

$$V_0 = C_2/(1+r)^2$$

Using this backward recursion technique, it follows that the value of a risk-free payment of C at any time i is given by

$$V_0 = C_i/(1+r)^i$$

Furthermore, the present value of a sequence of cash flows would be valued as the sum of the present values of the individual cash flows. The general formula for valuing a possibly infinite sequence of cash flows is then

$$V_0 = \sum_{i=1}^{\infty} \frac{C_i}{(1+r)^i}$$

This is the most widely used discounting equation in finance. In practice, cash flows are often variable and complex, and are not valued on handheld calculators, but by machines using spreadsheets and code. The valuation shortcuts proposed in this chapter only apply to special cases restricting the values of C_i to follow recognizable patterns.

Even though this equation is the most general we will see in this chapter, it should be noted that discrete cash flow valuation inherits the properties of a linear function. This implies that cash flow present values are additive, and that if all cash flows are multiplied by the same constant, the overall valuation is multiplied by that constant as well. This latter property is also called *homogeneity (of degree 1)*. In later chapters, cash flows may be risky and correlated over time, in which case the present value operator will not be additive.

Exercise: You are expecting a legal settlement of $2,000,000 to be paid in 3 years. If the discount rate is 5%, what is the value of the claim today? If someone offers you $1.5 million for your settlement, paid today, should you take it?

Answer: The PV is $1.73 mm. If your true discount rate is 5%, you should reject the offer. If the cash allows you to do something else which is worth more than $0.23 mm to you now, such as buying a house for your family, or paying for cancer treatments, perhaps you should take the offer. Remember, present value is only one model for value.

2.4 VALUING A SIMPLE PERPETUITY, OR A RECURRING PERPETUAL CASH FLOW

What if an asset makes a risk-free payment of C every year forever, starting one year from today? This is called a *perpetuity*. There are several ways to value perpetuities; we will demonstrate both the traditional method and the GVE method.

The traditional method recognizes that because of value-additivity of risk-free cash flows, we already have an equation for perpetuity valuation:

$$V_0 = \sum_{i=1}^{\infty} \frac{C}{(1+r)^i}$$

Note that the constant C has replaced the variable C_i. To simplify this, let $d = 1/(1+r)$ for convenience. Then we have

$$V_0 = Cd \times [1 + d + d^2 + d^3 + \cdots]$$

The term in brackets is an infinite geometric series, which converges as long as d<1, i.e. r>0. Therefore

$$[1 + d + d^2 + d^3 + \cdots] = \frac{1}{1-d}$$

$$V_0 = \frac{Cd}{1-d} = \frac{C}{r} \; for \; r > 0$$

Now let's solve using the GVE. An asset that pays C in perpetuity today must be worth the same amount next year, after the first payment is made --- the future cash flow diagram stays the same. We can write the cash flow diagrams for the perpetuity in a clear way below to support the derivation. At time 0 and 1, we can say that the value of the perpetuity at those times is given by the present value of the corresponding cash flow sequence:

Cash flow diagrams

$$V_0 = PV\{C, C, C, C, \dots\}$$

$$V_1 = PV\{C, C, C, C, \dots\}$$

The future cash flows as of time 1 reflect that the first payment at time 1 has already been made, leaving us in the *exact same position* as we started at time 0. Since the two sequences of future cash flows are perpetual and identical, their values must be equal. Therefore $V_1 = V_0$ and the capital gain on V is always 0. Then, using the GVE, we get

Time value (rV_0) + Risk compensation (0) = Capital gain (0) + Cash flow (C)

$$rV_0 + 0 = 0 + C$$

$$V_0 = C/r$$

Isn't the GVE method simpler and more intuitive?

Exercise: If you expect a real estate investment to produce $100,000 per year net return forever, and your discount rate for real estate is 4% per year, what is the private value of your investment? If the market value is $3,000,000, should you sell the property?

Answer: The present value of the perpetuity is $2,500,000. You might sell the property for $3,000,000 if you could earn the same 4% return, i.e. $120,000 per year, in another investment with similar risk.

2.5 VALUING A DEFERRED PERPETUITY

Suppose you are expecting to inherit a sequence of annual payments of amount C, but starting in n+1 years. What is the valuation of this stream today if the annual discount rate is r?

Let's begin with the simple perpetuity, where $V_0 = PV\{C, C, C, \dots\} = C/r$. Suppose we defer the perpetuity by one year, then we have $V_0(deferred\ one\ period) = PV\{0, C, C, C, \dots\}$. Since all the payments are made one year later, the present value of each payment can be divided by (1+r). Factoring out the (1+r) term using the homogeneity property, we have

$$V_0(deferred\ one\ period) = PV\{0, C, C, C, \dots\} = V_0/(1+r) = \frac{C}{r}d$$

Similarly, if the first payment occurs in n+1 periods, there must be a sequence of n zeroes preceding the cash flows in the deferred perpetuity. Therefore,

$$V_0(deferred\ n\ periods) = \frac{V_0}{(1+r)^n} = \frac{\frac{C}{r}}{(1+r)^n} = \frac{C}{r} d_n$$

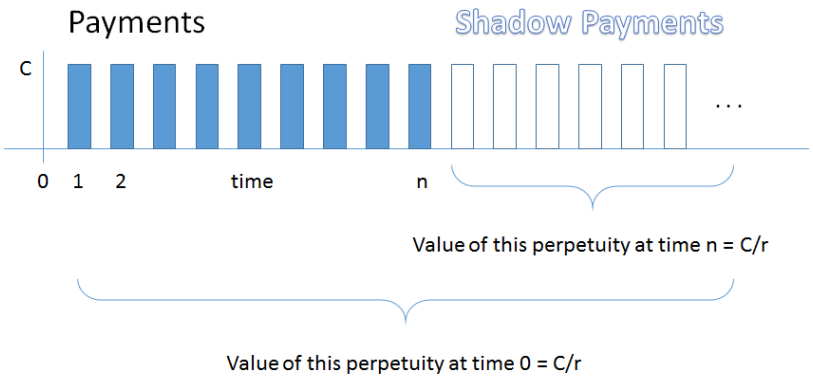
The term d_i is defined as the present value discount factor for a payment of \$1 at time i, used in order to clarify the interpretation of the equations.

$$d_i = \frac{1}{(1+r)^i}$$

This equation works for any sequence of cash flows, not only perpetuities with constant payments. Symmetrically, an accelerated perpetuity is worth (1+r) times its value for each period it is accelerated. Of course, in this case, one needs to worry about how to handle payments that occurred in the past! Yet, a simple conclusion is that for any model that assumes payments in arrears, if we are able to negotiate payments starting immediately, the value in arrears is multiplied by (1+r) to find the new value.

2.6 VALUING A SIMPLE ANNUITY

To value an annuity, we can use the traditional approach, express the annuity's value as the sum of its payments, and use the known formula for the sum of a finite geometric series. Perhaps the most popular approach is to observe that an annuity, a sequence of n payments of C, starting in one year for n years, can be seen as the difference between two perpetuities, as shown in the figure below.



The present value of all the payments, including the shadow payments, is simply C/r. However, the shadow payments have a present value themselves of C/r at time n, one period before the first payment at n+1. The value at time n can be discounted to time zero as if it were a single cash flow:

$$PV\ of\ shadow\ payments\ (the\ deferred\ perpetuity) = \frac{C}{r} d_n$$

Therefore, the value of a regular annuity is the difference in value between the perpetuity and the deferred perpetuity.

$$V_0 = \frac{C}{r}(1 - d_n) \quad \text{for r>0}$$

$$V_0 = nC \quad for\ r = 0$$

When r→0, you can use L'Hôpital's rule to derive the limit, or simply realize that with no discounting, one can simply add up the payments!

Now let's try the GVE. The value of the annuity can be computed at time 0 and at time 1 for the purposes of determining the capital gain $V_1 - V_0$, which is simply the negative of the present value of the last payment, since the PV of all the payments 1..n-1 cancel in the subtraction.

$$V_0 = \Sigma_{i=1}^{n} C d_i$$

$$V_1 = \Sigma_{i=1}^{n-1} C d_i$$

$$V_1 - V_0 = -C d_n$$

The required return is the risk-free rate, and the cash flow is C, therefore according to the GVE:

$$rV_0 = -C d_n + C$$

$$V_0 = \frac{C}{r}(1 - d_n) \quad \text{for r>0}$$

Exercise: Your client has been in an automobile accident and was awarded $50,000 per year for 25 years as a settlement, starting in one year. If his discount rate is 8%, what is the present value of the settlement? Assuming the insurance company is willing to double the annual payment as long as the present value stays the same, for how many years can he receive $100,000 per year?

Answer: The present value is about $534K. If the payment is doubled, it would last for only 7.24 years. Hint: If you had trouble with the second part, solve the annuity equation for n. If you get stuck, consider taking natural logs.

2.7 VALUING A SIMPLIFIED BOND

In its simplest form, a bond normally offers investors a combination of fixed regularly spaced coupons (C), together with a final repayment of principal, face value (F) on the final coupon payment date. For example, if a corporation were to issue a bond for $1000 with annual payments of 5% (C=50) for 10 years (n=10), it would pay $50 at the end of each year, return the last coupon plus principal, or $1050, in 10 years. As such, the valuation of a bond is quite straightforward, since it is the sum of an annuity together with a final payment.

$$V_0 = \frac{C}{r}(1 - d_n) + F d_n = \frac{C}{r} + d_n \left(F - \frac{C}{r}\right)$$

This is called a simplified bond valuation function because it assumes that the payments are equally spaced, and starting in one period. Also, the discount rate is assumed constant. Traded bonds can of course be bought on any day between coupon payment dates, so we must take account of accrued interest to determine the value. Also, the time between coupons can vary by a few days.

We may use the simplified bond function to value bonds that make semiannual, quarterly or monthly payments as well. In practice, the convention for bond yields is to divide the simple annual yield by the number of compounding periods per year to determine the periodic discount rate. Chapter 3 considers the many complications involved with valuing bonds in practice.

Exercise

What is the value of the bond described above (C=50, n=10, F=1000) if the discount rate is 4, 5, or 6%? What do we learn from this calculation?

Solution

$1081.11, $1000.00, $926.40

The exercise shows us three facts:

1. Par bonds, for which the face value and the present value are the same or $V_0 = F$, have a periodic discount rate equal to the periodic coupon rate (see the second bond in the exercise)
2. Higher discount rates are associated with lower bond prices
3. Bond prices are convex in the discount rate: They decrease in value at a decreasing rate as interest rates rise. The drop from 4-5% is $81.11, and the drop from 5-6% is $73.60.

Zero-coupon bonds (ZCBs)

A special case of a bond that pays no coupons is known as a *zero-coupon bond* or ZCB. Two examples of these are U.S. government-issued T-Bills and STRIPS, which are discussed in Chapter 3. These securities are useful to us if we want to value a stream of cash flows with different discount rates for different maturities. Of course, if the discount rate r(t) is known at time t, the value of a ZCB is its discounted face value (F), or $V_0=F/(1+r(t))^t$.

2.8 VALUING AN EXPONENTIAL GROWTH PERPETUITY

Suppose the first cash payment is C_1, and future payments grow at a percentage rate of g, so $C_2=(1+g)C_1$ and so on. The cash flow diagrams are shown below:

$$V_0 = PV\{C_1, C_1(1+g), C_1(1+g)^2, \dots\}$$

$$V_1 = PV\{C_1(1+g), C_1(1+g)^2, C_1(1+g)^3 \ \dots\}$$

Even if we don't know the value V_0 at time zero, we can see that at time 1, the cash flows have all been multiplied by (1+g), again using homogeneity, it must be the case that $V_1 = V_0(1+g)$, so

$$V_1 - V_0 = gV_0$$

From the GVE we know

Time value (rV_0) + Risk compensation (0) = Capital gain (0) + Cash flow (C)

$$rV_0 + 0 = gV_0 + C_1$$

therefore

$$V_0 = \frac{C_1}{r-g}$$

This equation only holds if g<r. If g≥r then the value is infinite.

It is convenient to define $C_0 = C_1/(1+g)$, and $r^* = (r-g)/(1+g)$. With these substitutions, the value of the growing perpetuity is expressed as if it were a constant perpetuity of C_0 starting one period from now with a first payment of $C_0(1+g)$, discounted at a rate of r*

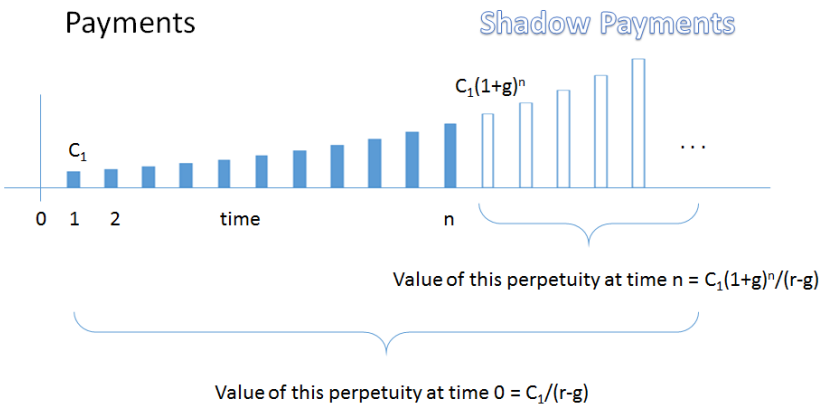
$$V_0 = \frac{C_1}{r-g} = \frac{C_0}{r^*}$$

Important and useful fact:

If every term in a sequence of payments C_i (from 1 to n) is multiplied by $(1+g)^i$, it has the same present value effect as changing the discount rate to r* from r, and reducing the first payment to $C_0 = C_1/(1+g)$. This applies for any value of g<r, including negative values. For example, if the first payment is $100, the discount rate is 5%, but the probability of payments stopping forever is 2% per year every year, then g=-0.02 and r* = (0.05+0.02)/(1-0.02) = 7.1429%. One can use the same formula without the growth adjustment, substituting r* for r and C_0 for C_1. That is, $V_0 = 100/(0.05-(-0.02)) = (100/0.98)/0.071429 = \$1428.57$.

2.9 VALUING AN EXPONENTIAL GROWTH ANNUITY

Now suppose an annuity makes an initial payment of C_1, but the payment grows at a rate of g per year for n payments in total. Then using the same observation we made for an annuity, we can draw a chart for a growing annuity:



The value of the payments is the total value of payments and shadow payments, minus the present value of the shadow payments:

$$V_0 = \frac{C_1}{(r-g)}\left[1 - \frac{(1+g)^n}{(1+r)^n}\right]$$

To simplify the computation, apply the *important and useful fact* from the last section: we made for a growing perpetuity

$$r^* = \frac{r-g}{1+g} \qquad C_0 = \frac{C_1}{(1+g)}$$

and then value the asset using the regular annuity formula using r*:

$$V_0 = \frac{C_0}{r^*}(1 - d_n^*)$$

This is particularly useful when using financial calculators or spreadsheets to compute the value of a growing annuity, or any time the function is not provided. It is interesting to note that the formulae for a growing annuity are valid *even if* g > r. If g = r, then growth exactly cancels out discounting, so the value of the annuity is $V_0 = nC_0$, the same as an annuity when r = 0.

Example

	A	B	C	D	E	F	G	H
1								
2		No growth		Growth				
3	1	100		98.00				
4	2	100		96.04				
5	3	100		94.12		EXCEL ANNUITY CALCULATION		
6	4	100		92.24			r	5.00%
7	5	100		90.39			g	-2.00%
8	6	100		88.58			r*	7.14%
9	7	100		86.81				
10	8	100		85.08				
11	9	100		83.37				
12	10	100		81.71				
13								
14		$772.17		$697.74				
15		=NPV(0.05,B3:B12)		=NPV(0.05,D3:D12)				
16								
17		$697.74						
18		=NPV(r*,B3:B12)						

The no-growth annuity is valued in column B using the 5% discount rate and the NPV function in Excel at $772.17.

The growing annuity (g = -0.02) is valued in column D using the 5% discount rate, with growth starting at time 0, at $697.74.

The growing annuity can also be valued as a flat annuity with

PMT = 98/(1 – 0.02) = 100 and r* = 7.14%.

The present value is the same, at $697.74.

Exercise

Your client has been in an automobile accident and was awarded $50,000 next year **growing at 2% per year** for a total of 25 annual payments as a settlement. The 2% growth rate is meant to cover expected increases in lost wages. If his discount rate is 8%, what is the present value of the settlement? Assuming the insurance company is willing to double the monthly payment for the same present value, for how many years can he receive $100,000 per year?

Solution

$634K, 8.37 years.

2.10 ADVANCED ANNUITIES: LINEAR GROWTH

In advanced modelling applications, one encounters annuities of various types that do not fit the constant or exponential growth patterns discussed in the last sections. The two most common of these alternative models are *linear growth* and *mean reversion*. These three annuity types tend to occur frequently because they are the discrete equivalents of geometric random walks (exponential growth), arithmetic random walks (linear growth), and Ornstein-Uhlenbeck mean-reverting stochastic processes, which are all used to describe future cash flows of different types.

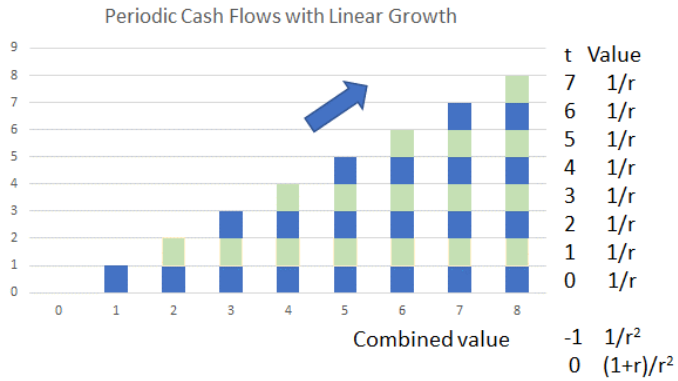
Definition of linear growth

In a linear growth payment pattern, the equally spaced payments increase by a constant amount each period. For example, a broker who introduces a client to a bank might receive a step-down compensation scheme that defers part of his commission. The broker's payment might be $100 in year 0, $80 in year 1, $60 in year 2, and so on until zero is reached. The broker might like to know the present value of his payments, or might not care much. The bank definitely wants to know the present value of all of the expected commissions paid in the program, in order to judge the value of the program.

To see how to value linear-growth annuities, let's begin with a perpetuity, as we did when we first studied flat annuities. Suppose a payment stream begins with $1 in one year, $2 in two years, $3 in three years and so on indefinitely, as in the cash flow diagram below:

$$V_0 = PV\{1, 2, 3, \dots\}$$

Any other linear growth pattern can be expressed as a linear function of a constant payment pattern plus a multiple of these payments. For example, the present value of the sequence PV{3,5,7,…} can be expressed as PV{1,1,1,…} + 2×PV{1,2,3,…] due to the linearity property of present value.



Note: "t" represents the valuation date for the $1 perpetuity that begins at t+1

To value this particular annuity, we could "cut" the cash flows horizontally, as shown with the alternate color shading. Then, we can see that the linear growth perpetuity can be valued as the sum of an infinite number of perpetuities. The first perpetuity, the bottom block in each column shaded darker, pays $1 per year forever, and is worth 1/r at time 0 --- this is recorded in the right column under "value". The second perpetuity, the second-from-the-bottom line shaded lighter, pays $1 per year forever starting at time 2, so it is worth 1/r **at time 1**. The present value of the perpetuity of perpetuities in the right column, i.e. all the cash flows, is therefore

$$V_{-1} = (1/r)/r \; = \; 1/r^2$$

at time -1. The present value at time 0 is *(1+r)* times higher, or $V_0 = (1 + r)/r^2$.

Alternatively, we can do this the easy way! After the first payment, we can see that the remaining payments are all exactly one dollar higher than they were on the preceding payment date, hence the value of the capital gain $V_1 - V_0 = 1/r$.

$$V_0 = PV\{1, 2, 3, \dots\}$$

$$V_1 = PV\{2, 3, 4, \dots\}$$

$$V_1 - V_0 = PV\{1, 1, 1, \dots\} = 1/r$$

Substituting into the GVE, we have

$$rV_0 = (V_1 - V_0) + C_1 = \frac{1}{r} + 1 = \frac{1+r}{r}$$

Present value of a linear growth perpetuity {1,2,3, …}

$$V_0 = (1+r)/r^2$$

Exercise: A new B2B (business to business) marketing program expects to bring in one major client per year for the next 15 years. If each client is worth $50,000 per year to the company in perpetuity, and the corporate discount rate is 12%, what is the value of the marketing program? Answer: About $3.2 mm

Valuing a linear growth annuity

For the annuity, we follow the same procedure with a finite number (n) of payments

$$V_0 = PV\{1,2,3, \dots, n\}$$

$$V_1 = PV\{2,3, \dots, n, 0\}$$

$$V_1 - V_0 = PV\{1,1,1, \dots 1, -n\} = \frac{1}{r}(1 - d_{n-1}) - nd_n$$

$$rV_0 = \frac{1}{r}(1 - d_{n-1}) - nd_n + 1$$

$$Using\ d_n(1+r) = d_{n-1}, we\ have$$

$$V_0 = \frac{(1+r)}{r^2}[1 - d_n] - \frac{n}{r}d_n$$

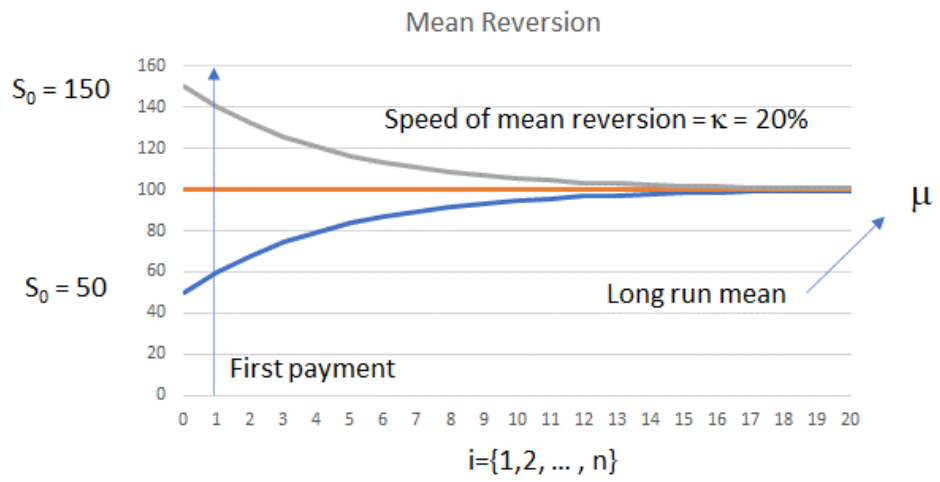
2.11 ADVANCED ANNUITIES: MEAN REVERSION

Mean-reversion occurs frequently in the modeling of interest rates, foreign exchange rates and commodity prices. For example, when oil prices are high, demand falls while supply increases, forcing future prices downward. When oil prices are low, demand for oil increases and supply decreases, causing prices to rise in the future. Cash flows linked to interest rates or commodity prices tend to mean-revert over time.

If oil follows a simple deterministic mean reverting process, then the expected spot price (S) moves in the direction of the long-run mean price mu (μ, pronounced "mew") at a speed of kappa (κ), as described by the following equation:

$$S_{t+1} = S_t + \kappa(\mu - S_t)$$

For different possible starting values of S_0, we might see either an upward sloping or downward sloping exponential decay towards the long run mean, as in the figure below.



To find the present value of the perpetuity of payments equal to S_i every period we may again use the GVE:

$$V_0 = PV\{S_1, S_2, S_3, \dots\}$$

$$V_1 = PV\{S_2, S_3, S_4, \dots\}$$

$$V_1 - V_0 = PV\{S_2 - S_1, S_3 - S_2, S_4 - S_3, \dots\}$$

$$= PV\{\kappa(\mu - S_1), \kappa(\mu - S_2), \kappa(\mu - S_3), \dots\}$$

$$= PV\{\kappa\mu, \kappa\mu, \kappa\mu, \dots\} - \kappa PV\{S_1, S_2, S_3, \dots\}$$

$$V_1 - V_0 = \frac{\kappa\mu}{r} - \kappa V_0$$

Now substituting into the GVE

$$rV_0 = \frac{\kappa\mu}{r} - \kappa V_0 + S_1$$

$$V_0 = \frac{\frac{\kappa\mu}{r} + S_1}{r + \kappa}$$

Present value of the mean reverting perpetuity (after some algebra)

$$V_0 = \frac{\mu}{r} + \frac{S_0 - \mu}{r^*}$$

$$r^* = \frac{r + \kappa}{1 - \kappa}$$

Alternatively, we can see now from the last shown value of V_0 that the value of the mean reverting perpetuity is the sum of two perpetuities. The first is the present value of the flat long-run mean component, and the second is a growing perpetuity starting at $C_0 = S_0 - \mu$, but growing at a rate of $-\kappa$. Notice that the sign of the annuity changes correctly according to whether $S_0 > \mu$ or $S_0 < \mu$.

Seeing this fact, we can easily value the mean-reverting annuity by taking the sum of a flat annuity and a growing annuity:

Present value of a mean-reverting annuity

$$V_0 = \frac{\mu}{r}(1 - d_n) + \frac{S_0 - \mu}{r^*}(1 - d_n^*) \qquad r^* = \frac{r + \kappa}{1 - \kappa}$$

Exercise: Farmer John believes that corn prices, currently $3.50 per bushel (bu) will rise to a long run mean of $5/bu over time, at a rate of 10% per year. The first harvest is one year away, and Farmer John expects to operate the farm for 20 years. If the discount rate is 5%, and he produces 1,000,000 bu of corn per year, what is the present value of his revenues? Check your answer by computing all the cash flows in Excel. Answer: $53.7 mm

2.12 CHALLENGE PROBLEMS

You should be able to complete problems 1-4 without much difficulty. In all cases, be aware that simpler solutions may be available using valuation results you already know.

1. Derive the valuation formula for quadratic perpetuities and annuities using the GVE. In this case,

$$V_0 = PV\{1,4,9,16, \dots, i^2, \dots\}$$

HINT: $(i+1)^2 = i^2 + 2i + 1$

2. Using the GVE, find the value of a growing perpetuity that pays the value of the Fibonacci series, i.e.

$$V_0 = PV\{1,1,2,3,5,8,13, \dots\}$$

HINT: Assume $r > \varphi - 1$ where $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$, the Golden Ratio, to ensure it converges.

3. Find the present value of the Fibonacci annuity for any number of payments n.

HINT: The nth term in the Fibonacci sequence is also given by the formula $F_n = \frac{1}{\sqrt{5}}\left[\varphi^n - \left(\frac{-1}{\varphi}\right)^n\right]$

4. The following bond risk measures are related to the first and second derivatives of the simple bond price with respect to the yield to maturity (r) per period. Use the GVE to find closed-form expressions for Macauley's duration (weighted average life), modified duration and bond convexity. The term V_0 represents the value of the underlying bond.

$$Macauley's\ Duration = \frac{\sum_{i=1}^{n} iC_i d_i}{V_0}$$

$$Modified\ duration = Macauley's\ duration * d_1$$

$$Convexity = \frac{d_2}{V_0}\Sigma_{i=1}^{n}(i^2 + i)C_i d_i$$

HINT: Use the equations provided in the summary valuation table at the end of this chapter.

5. VERY DIFFICULT SUPER CHALLENGE (generalizing #1)

Find the present value of the perpetuity for a constant discount rate r

$$V_0 = PV\{1^m, 2^m, 3^m, \dots\} \; where \; m \; is \; a \; positive \; integer$$

HINTS: Consult a detailed one-sided z-transform table, let z = 1+r. The solution requires polylogarithms and Euler's number triangle (yikes!).

2.13 CHAPTER SUMMARY AND PV REFERENCE

This chapter was necessarily a technical one, because it builds the foundation for applying the GVE to more advanced problems and applications we will encounter in future chapters. Also, the present value table following will prove an invaluable reference for advanced analysis and coding. The table summarizes the discrete-time cash flow types we have valued so far, using the notation provided in Section 2.2.

Key ideas:

a. Perpetuities can be valued for their own sake, or as a step to valuing annuities.
b. It is surprising how many different cash flow sequences can be valued using the GVE, even the Fibonacci sequence.
c. If we multiply every cash flow i in *any* sequence $\{C_i\}$ by $(1+g)^i$, the resulting present value will be as if we substituted r* for r, with r* = (r-g)/(1+g).
d. An annuity can be expressed as the difference between a perpetuity and a deferred perpetuity, which may be sufficient to value the annuity.
e. A complex annuity can also often be valued as a combination of different types of simpler annuities.
f. Any deferred payment sequence that is deferred by j periods can be valued by taking the present value of the original sequence and multiplying by d_j.
g. Linearity in the PV function allows us to solve even more complex valuation problems.

Present Values of Complex Annuities and Perpetuities (first payment in one period)

Selected Cash Flow Patterns	**Present Value at Discount Rate r**
Single payment of C at time n	$V_0 = Cd_n$
Payments C_i at different times i	$V_0 = \Sigma_{i=1}^{\infty} C_i d_i$
Perpetuity of C per period forever	$V_0 = \frac{C}{r} \; for \; r > 0$
Annuity payment of C for n periods (first payment in one period)	$V_0 = \frac{C}{r}(1 - d_n) \; for \; r \neq 0$ $V_0 = nC \; if \; r = 0$
Growing perpetuity	$V_0 = \frac{C_1}{r - g} \; for \; r > g$
Growing annuity	$V_0 = \frac{C_1}{(r-g)}\left[1 - \frac{(1+g)^n}{(1+r)^n}\right] for \; r \neq g$

	$V_0 = nC_1 \ \ if\ r = g$
Growing annuity converted to a flat annuity (using the regular annuity formula)	$V_0 = \frac{C_0}{r^*}(1 - d_n^*) \ \ for\ any\ r^* \neq 0$ $r^* = \frac{r - g}{1 + g}$ $C_0 = \frac{C_1}{1 + g}$ $V_0 = nC_0 \ \ if\ r^* = 0$
Linear growth perpetuity $V_0 = PV\{1,2,3, \dots\}$	$V_0 = \frac{1 + r}{r^2} \equiv LGP\ for\ r > 0$
Linear growth annuity $V_0 = PV\{1,2,3, \dots, n\}$	$V_0 = \frac{(1 + r) - d_n(1 + r + nr)}{r^2} \equiv LGA(n)\ for\ r \neq 0$ $V_0 = \frac{1}{2}\, n(n + 1) \ \ for\ r = 0$
Quadratic growth perpetuity $V_0 = PV\{1,4,9,16, \dots\}$	$V_0 = \frac{(r + 1)(r + 2)}{r^3} \ for\ r > 0$
Quadratic growth annuity $V_0 = PV\{1,4,9,16, \dots, n^2\}$	$V_0 = \frac{(1 + r) - d_n(1 + r(n + 1)^2) + 2rLGA(n)}{r^2} \ for\ r \neq 0$ $V_0 = \frac{1}{6}\, n(n + 1)(n + 2) \ \ for\ r = 0$
Mean-reverting growth perpetuity	$V_0 = \frac{\mu}{r} + \frac{S_0 - \mu}{r^*}$ $r^* = \frac{r + \kappa}{1 - \kappa}$
Mean-reverting growth annuity	$V_0 = \frac{\mu}{r}(1 - d_n) + \frac{S_0 - \mu}{r^*}(1 - d_n^*)$ $r^* = \frac{r + \kappa}{1 - \kappa}$
Fibonacci perpetuity $V_0 = PV\{1,1,2,3,5,8,13, \dots\}$	$V_0 = \frac{r + 1}{r^2 + r - 1} \ \ for\ 1 + r > \varphi$

2.15 PRESENT VALUE EXERCISES

Note: For many of these problems, it is helpful to think in terms of periods of time other than years. For example, if a problem has a simple annual interest rate of 6% but payments are made monthly, compute the simple monthly rate of 0.5%, and count months instead of years. These problems are meant to be solved by hand, but may be solved or checked in Excel as well, depending on your instructor's preferences.

1. A government bond pays $100 in coupons per year starting in one year for 8 years, and $1000 principal along with the last coupon. If the discount rate is 4% per year, what is the value of the bond today?
2. A homeowner wishes to borrow $500,000, but must repay interest and principal with monthly flat payments over 30 years. If the simple annual interest rate is 6%, i.e. the monthly rate is 0.5%, what is the monthly payment? What is the remaining balance (amount owed) after five years of payments?
3. Suppose a bank wanted to offer the borrower in #2 an option to have the payments grow at 1% a month for the full 30 year period. What would the initial payment be in this case, in order to have the same present value? Can you see any advantages or disadvantages to the bank with this option?
4. A retiree saved $250,000 and wants to withdraw $5000 per month for expenses, starting next month. How many months (n) will the money last if the annual discount rate is 5%? What is the general formula to solve for n as a function of r? *Hint:* You may need to take a natural log.
5. How does your answer to #4 change if the retiree increase the withdrawal by 0.5% per month?
6. Annuity Plan A has an initial annual payment that starts at $10,000 and grows at 5% per year. Plan B has a flat payment of $12,500 for the 10 years.
    a. Which 10 year annuity is better, if the annual discount rate r = 8%?
    b. At what discount rate would the annuity owner be indifferent between the two plans?
7. An alternating annuity pays $1 in one year, –$2 in two years, $4 in three years, –$8 in four years, which each subsequent payment doubling and changing sign for 20 years. If the risk-free discount rate is 5%, what is the value of this alternating annuity? (*Hint:* Check your result with Excel using a high value of r.)
8. A company plans to pay a dividend of $0.50 per share next year rising by $0.25 per year until it reaches $2.50, at which point it will remain flat forever. If the discount rate is 9% per year, what is the present value of the dividends, i.e. the value of a share of stock?
9. The price of oil is currently $50 per barrel (bbl), but the long-run mean is believed to be $75, and the annual speed of mean reversion is 15%. An oil drilling company wishes to value a prospective new property. The financial details are below:
    a. Production levels will start at 500,000 bbl for the first 10 years, and will drop to 250,000 bbl for the next 10 years, at which time the oil will be exhausted.
    b. The fixed costs of producing the oil will be $5,000,000 per year.
    c. The variable costs of producing oil will be $10 per barrel
    d. The discount rate is 9% per year, and the first cash flow is in one year.

    What is the value of the oil well today?

2.16 EXCEL EXERCISE

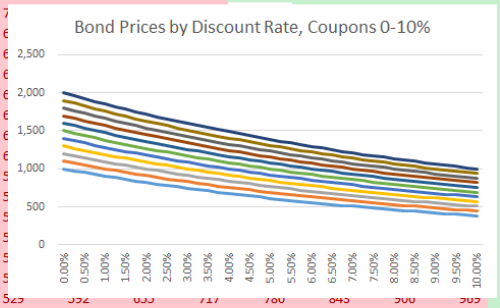
Consider a bond that makes annual payments. For example, a 6% coupon bond with face value $1000 pays $60 per year in payments. Create a "bond table" that would provide a reference to a bond trader. The trader chooses a bond term in one cell of the spreadsheet. The table shows coupon rates (0%-10% in 1% increments) across the columns and discount rates (0-10% in 25 bp increments) across the rows, and in each cell of the report, the bond price is quoted. Using your table, answer the following questions.

A. Graph the bond prices as a function of discount rates. How would you describe the relationship between discount rates and bond prices?
B. On your table, use conditional formatting to identify bonds that trade at a premium or discount to par value of $1000.

Your solution should look like this:

Face value	1000
Bond term (years)	10

	Coupon rates										
	0.00%	1.00%	2.00%	3.00%	4.00%	5.00%	6.00%	7.00%	8.00%	9.00%	10.00%
Discount rates 0.00%	1,000	1,100	1,200	1,300	1,400	1,500	1,600	1,700	1,800	1,900	2,000
0.25%	975	1,074	1,173	1,271	1,370	1,469	1,567	1,666	1,764	1,863	1,962
0.50%	951	1,049	1,146	1,243	1,341	1,438	1,535	1,632	1,730	1,827	1,924
0.75%	928	1,024	1,120	1,216	1,312	1,408	1,504	1,600	1,696	1,792	1,888
1.00%	905	1,000	1,095	1,189	1,284	1,379	1,474	1,568	1,663	1,758	1,852
1.25%	883	977	1,070	1,164	1,257	1,350	1,444	1,537	1,631	1,724	1,818
1.50%	862	954	1,046	1,138	1,231	1,323	1,415	1,507	1,599	1,692	1,784
1.75%	841	932	1,023	1,114	1,205	1,296	1,387	1,478	1,569	1,660	1,751
2.00%	820	910	1,000	1,090	1,180	1,269	1,359	1,449	1,539	1,629	1,719
2.25%	801	889	978	1,066	1,155	1,244	1,332	1,421	1,510	1,598	1,687
2.50%	781	869	956	1,044	1,131	1,219	1,306	1,394	1,481	1,569	1,656
2.75%	762	849	935	1,022	1,108	1,194	1,281	1,367	1,454	1,540	1,626
3.00%	744	829	915	1,000	1,085	1,171	1,256	1,341	1,427	1,512	1,597
3.25%	726	810	895	979	1,063	1,147	1,232	1,316	1,400	1,484	1,569
3.50%	709	792	875	958	1,042	1,125	1,208	1,291	1,374	1,457	1,541
3.75%	692	774	856	938	1,021	1,103	1,185	1,267	1,349	1,431	1,513
4.00%	676	757	838	919	1,000	1,081	1,162	1,243	1,324	1,406	1,487
4.25%	660	740	820	900	980	1,060	1,140	1,220	1,300	1,381	1,461
4.50%	644	723	802	881	960	1,040	1,119	1,198	1,277	1,356	1,435
4.75%	629	707	785	863	941	1,020	1,098	1,176	1,254	1,332	1,410
5.00%	614	691	768	846	923	1,000	1,077	1,154	1,232	1,309	1,386
5.25%	599	676	752	828	905	981	1,057	1,134	1,210	1,286	1,362
5.50%	585	661	736	812	887	962	1,038	1,113	1,188	1,264	1,339
5.75%	572	646	721	795	870	944	1,019	1,093	1,168	1,242	1,317
6.00%	558	632	7								1,294
6.25%	545	618	6								1,273
6.50%	533	605	6								1,252
6.75%	520	591	6								1,231
7.00%	508	579	6								1,211
7.25%	497	566	6								1,191
7.50%	485	554	6								1,172
7.75%	474	542	6								1,153
8.00%	463	530	5								1,134
8.25%	453	519	5								1,116
8.50%	442	508	5								1,098
8.75%	432	497	5								1,081
9.00%	422	487	5								1,064
9.25%	413	476	5								1,048
9.50%	404	466	5								1,031
9.75%	394	457	519	581	643	705	767	829	891	953	1,016
10.00%	386	447	508	570	631	693	754	816	877	939	1,000



Excel skills learned: PV function, absolute and relative addressing, formatting and conditional formatting, graphing, what-if tables.

MINIPROJECT 2: THE OTHER SIDE OF THE LOAN STORY

Let's consider Wells Fargo's view of the loans offered in Minicase #1. There are several differences between the way Wells Fargo looks at a loan when compared to borrowers. First, their cost of funding the loan is much lower than the rate, let's say 2% per year. Second, customer loans under $5000 are too expensive to collect, so the bank does not bother. However, borrowers default at a certain rate, let's say 6% per year, or 0.5% per month. In default, we assume we receive nothing for these loans. Also, people prepay their loans. If the prepayment rate is 1% per month, this means in addition to the scheduled payments being made, people are making additional payments (more than the required payment) equal to the monthly prepayment rate times the previous balance. All other parameters are as in Minicase #1.

You may now focus your spreadsheet analysis on 1000 identical loans, but apply the default and prepayment rules at the loan portfolio level rather than considering each loan separately.

Develop the following reports for the bank:

A. The projected pattern of aggregate interest and principal over time (Stacked bar graph).

B. Current valuation of the loan portfolio to the bank, as a function of the contract interest rate and the points charged. (Hint: This is a two-way what-if table in Excel)
C. Current valuation of the loan portfolio to the bank, as a function of the default rate and the prepayment rate.
D. Three-dimensional plots of the analyses in B. (Hint: These are called *surface charts* in Excel.)
E. Your analysis of the sensitivities. Do they make sense?
F. Starting from your baseline valuation, suppose you are tasked with increasing the loan portfolio value by 5%, holding the number of loans and the principal balance constant. How much do you have to raise lending rates, holding everything else constant, in order to achieve this goal? (Hint: Use Excel *solver*.*) You may assume that this will not affect your loan applications, though in reality it might!

* This is a free add-in, but it must be installed from the *File/Options/Add-ins* menu. Also, please install the Analysis ToolPak at this time for use in later chapters.

Chapter 3: Complications and resolutions

INTRODUCTION

In Chapter 2, we valued some simplified cash flow patterns commonly seen in advanced financial engineering practice, but we made several simplifying assumptions.

- We assumed that the discount rate was constant, interest was compounded once per period, the first cash flow occurred in one period, and that the remaining cash flows were known and riskless.

We need to adjust for all types of payment frequencies and discounting or yield conventions.

- Cash flow valuations in practice are affected by inflation and also by taxes; hence we must be able to adjust our valuations to properly reflect these adjustments.
- Events may cause our cash flows to suddenly change or even cease, for example, a default by the payer, the death of the receiver, or a contract provision triggering a windfall or an early termination.
- We would like to be able to value cash flows linked to fluctuating interest rates or inflation rates.
- We should know how to perform valuations in alternative currencies.

These are the subjects of Chapter 3. The major element missing from the list of concerns above is the proper treatment of risk. This will be forthcoming in Part II. For now, we will make some adjustments for expected losses due to risk, but address the pricing of risk in later chapters. In some special cases, however, we are fortunate enough to know a *risk-neutral* equivalent price, which has a risk adjustment embedded, so that the risk-neutral price can be discounted at the riskless rate to find its present value. We can use risk-neutral pricing to determine the value of the individual payments in floating rate bonds, and the value of foreign-currency denominated cash flows. Risk-neutral pricing will become increasingly important in later chapters.

To help you understand all the rate adjustments in this chapter, we offer you the following lexicon as a reference:

RATE	DEFINITION
Average tax rate	Total taxes divided by total income, expressed as a percentage
Effective annual rate	The rate of interest earned or paid for a year considering the effect of periodic compounding and earning interest on interest
Forward rates	The year-by-year implied future returns embedded in a bond price.
Foreign exchange forward rate	The pre-agreed exchange rate for a contracted currency exchange in the future
Inflation rate	The percentage increase in the value of a fixed basket of goods over a period of time, typically one year. Inflation also reflects the decline in purchasing power per year of a fixed amount of money.
Marginal tax rate	The amount of tax paid on an incremental dollar of income, expressed as a percentage
Nominal rate	The discount rate used to value actual dollars paid (ignoring inflation)
Periodic rate	The simple annual rate divided by the number of periods a year
Real rate	The discount rate used to value inflation-adjusted cash flows, or the rate of growth in spending power of savings

Simple annual rate	The discount rate used to discount annual cash flows, i.e. as if interest were compounded once a year
Spot rate	The annualized rate of return on a bond bought today and held to maturity.

3.1 CHANGING THE FREQUENCY OF PAYMENTS

I early inquired the rate of interest on invested money, and worried my child's brain into an understanding of the virtues and excellencies of that remarkable invention of man, compound interest.

- Jack London, 1906

In Chapter 2, we assumed all payments were annual, and were discounted at the simple annual interest rate. Suppose we have payments every month, like consumer loans? Or every quarter (3 months), like some corporate bonds? Or every half-year, like some corporate bonds and all U.S. government bonds? How are valuations affected? In practice, as a matter of convention, we usually simply divide the quoted simple annual rate by the number of periods per year. Therefore, if "r" is the quoted simple annual rate, a monthly time step would imply a simple monthly interest rate of r/12. A 6% interest rate on a loan with monthly payments therefore has a monthly interest rate of 0.50% or 50 basis points per month (bppm).

At first, this concept may puzzle some students. If we invest $1,000,000 now and reinvest the 0.5% per month for 12 months at the same rate, the future value in one year would be $\$1{,}000{,}000 \times (1.005)^{12} = \$1{,}061{,}678$, which *effectively* amounts to 6.17% over the year, well over 6% per year thanks to the effect of compound interest. That is because interest is being paid on the interest. This seems like the more honest calculation than simple interest, since it considers the return using actual compounding. Multiplying and dividing an interest rate by a constant seems intuitively incorrect, even though it is simple, convenient and a widely used convention. It is also not necessarily the case in practice that interest can be reinvested at the same return that the principal earns.[6]

It is important to realize that in almost every capital market, there are different legacy quoting conventions for reporting prices and rates of return, such as the simple annual rate, but usually some adaptation needs to be made by investors in order to compare effective rates of return between investments. To help investors or borrowers distinguish between the quoted rate and the actual earned return, we often call the compounded return the *effective annual yield* or *effective annual rate*.

Quick quiz 1: If the simple annual rate on a loan is 8%, and interest is compounded quarterly, what is the effective annual rate? (Ans: $8.24\% = (1+0.08/4)^4 - 1$)

Quick quiz 2: If the effective annual rate is 12.55%, and the payment frequency is quarterly, what was the simple annual rate? (Ans: 12%)

Quick quiz 3: What assumptions are made in calculating the effective annual rate? (Ans: The instrument is held to maturity, and interest payments are reinvested at the same yield as the principal.)

There is another way to remember this easily. For example, when you consider the yield to maturity of a bond, also known as the *bond equivalent yield (BEY)*, the reported yield is not the same as the effective yield. However, the semi-annual yield *(BEY/2)* equals the effective yield on the bond per six-month period on the

[6] This risk is also known as *reinvestment risk*.

assumption that the bond is held to maturity and the coupons are reinvested at the same rate as the principal. It is only when we annualize the returns that we have to compound returns on assets with different payment frequencies.

To summarize in an equation, if r is the simple annual interest rate, EAY is the effective annual yield, and k is the number of compounding periods per year:

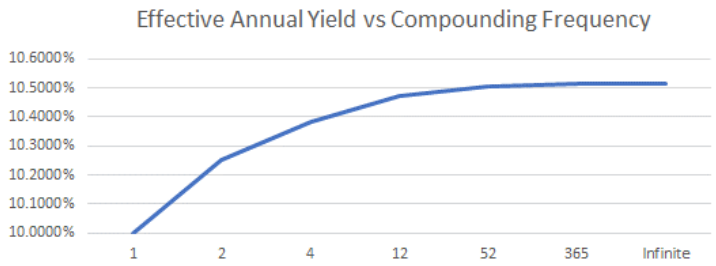
$$1 + EAY = (1 + r/k)^k$$

The effective annual rate is rarely computed in practice when simple rates can be compared among similar instruments. However, it is a necessary tool in careful return analysis, particularly when comparing the returns of assets with different payment frequencies.

Continuous compounding

As we saw in the last section, increasing the compounding frequency to 2, 4 or 12 times a year increases the effective annual return. What would happen if we increased to 52, 365, or even an infinite number of times per year?

Frequency	EAY
1	10.0000%
2	10.2500%
4	10.3813%
12	10.4713%
52	10.5065%
365	10.5156%
Infinite	10.5171%



This exhibit shows the effective annual yield for a 10% simple annual rate at different compounding frequencies on a nonlinear scale. The maximum rate that can ever be attained is 10.5171%. As the number of compounding periods tends to infinity, the discrete model is said to become *continuous*. We will refer to the 10% figure as the continuously discounted rate, and continue to refer to 10.5171% as the effective annual yield in this case.

In general, we may continue to interpret r as the simple annual interest rate, but $e^r - 1$ is the effective annual rate when the number of compounding periods is infinite. This can be derived by taking the limiting value of a $1 investment in one year, and compounding n times a year. As n goes to infinity, the future value of $1 is given by[7]:

$$\lim_{k\to\infty}\left(1+\frac{r}{k}\right)^k = e^r$$

This means that the present value of any payment C_t at time t, discounted continuously at the simple annual rate of r is given by

[7] This can be verified by applying L'Hôpital's rule in computing the limit of the natural log.

$$PV = C_t e^{-rt}$$

In many applications, especially when we study fixed income and derivatives, you may see all discount rates converted to their continuously compounded equivalents. Like the EAY, this has the benefit of eliminating differences in yield calculations caused by cash flows being discounted at different periodic rates.

Fortunately, there is an easy conversion between effective annual yield and the continuously compounded simple interest rate:

$$r = ln(1 + EAY)$$

Exercise

If an investment offers 3% simple return per half-year, what is the equivalent continuously compounded discount rate?

Solution

(Ans: The value of $1 + EAY$ is 1.03^2. The continuous yield is $5.91\% = \ln(1.03^2)$)

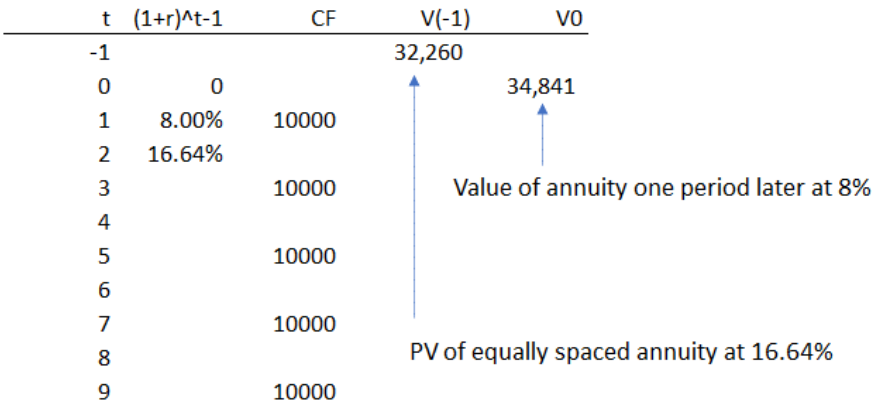
In practice, continuously compounded rates are rare in corporate finance, but are used extensively in valuations involving fixed income and exclusively when valuing derivatives. For this reason, the financial engineering student must be very comfortable with continuous discounting formulae and procedures. Also, for analytical purposes, it is often most efficient to use continuous discounting for discrete cash flows.

Payment frequency and discounting frequency

Payment frequency refers to the number of payments per period. *Discounting frequency* refers to the number of periods used per period to discount cash flows. When you encounter problems with different payment frequency and discounting frequency, it is best to begin by computing the discount rate corresponding to the payment frequency. For example, the interest rate on consumer loans is quoted on an annual basis, but payments are made on a monthly basis. In this case, we divide the simple annual rate by 12 in order to get the monthly rate, understanding that this is the standard rate convention --- ignoring the temptation to "de-compound" the return.

Now suppose we have a biennial payment (once every two years) of $10,000 starting in one year, repeating every second year for 5 payments. If the simple annual discount rate is 8%, what is the present value? We'd like to use the annuity formula to find the present value, but there are two problems. First, the quoted interest rate is annual, while payments occur every two years. Second, the first payment is in one year, which is in one-half of a two-year time period. To solve this problem, we first compute the effective two-year discount rate, which is $1.08^2 - 1$ or 16.64%. Then we find the PV of the annuity, or $32,260. But this is the present value *a year ago,* one period (one half of a two year period) before the first payment in one year. Hence we must multiply by 1.08 to find the present value today, or $34,841.

The Excel presentation summarizes this calculation:



This section discussed how to compute discount rates for different compounding periods and starting dates. Because consumer loans make monthly payments, corporate loans make quarterly or semiannual payments and government loans make semiannual payments, financial analysts frequently need to be able to convert yields from one compounding frequency to another. Also, for investment purposes, analysts need to convert simple rates of return into effective rates in order to compare different investments. As noted, this means that we are assuming the payments are reinvested at the same rate as the principal, and that the bonds are held to maturity.

In sum, the usual convention is to take a simple annual rate and divide by the number of compounding periods per year, ignoring the effects of compound interest. Then, an effective annual yield may be determined considering the effect of compounding. Finally, continuously compounded rates are sometimes used as a uniform discounting standard that allows analysts to compare yields at different compounding frequencies on an equivalent basis, even when payments are not continuous.

3.2 INFLATION

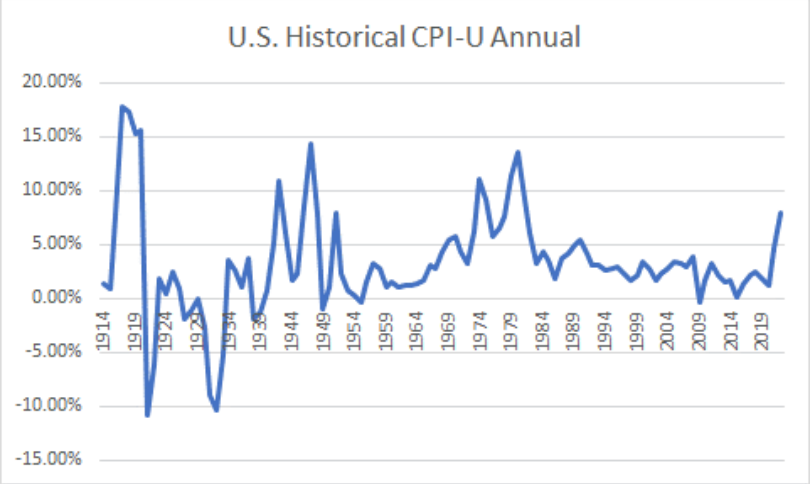
"I was reading in the paper today that Congress wants to replace the dollar bill with a coin. They've already done it. It's called a nickel."
— Jay Leno

"*The arithmetic makes it plain that inflation is a far more devastating tax than anything that has been enacted by our legislature. The inflation tax has a fantastic ability to simply consume capital. It makes no difference to a widow with her saving in a 5 percent passbook account whether she pays 100 percent income tax on her interest income during a period of zero inflation, or pays no income taxes during years of 5 percent inflation. Either way, she is 'taxed' in a manner that leaves her no real income whatsoever. Any money she spends comes right out of capital. She would find outrageous a 120 percent income tax, but doesn't seem to notice that 5 percent inflation is the economic equivalent."*
— Warren Buffett

Until now, we have ignored inflation. In reality, the purchasing power of a currency generally declines over time, so that the value of $1 today does not buy as many goods in 10 years. The rate of decline in purchasing power is also the rate of inflation of the currency. For example, if a can of beans cost $1 ten years ago, and it costs $1.50 today to buy the same can of beans, then the inflation rate, as measured by beans would be $(1.50/1.00)^{1/10} - 1 = 4.14\%$ per year. It is also possible that a currency may *deflate*, such as in the U.S. Great

Depression of 1929-33, indicating a negative rate of inflation. In this unusual situation, the spending power of a currency increases over time.

In the 2010's in the U.S., the rate of inflation averaged just under 2% per year. Inflation is commonly measured by the CPI-U (Consumer Price Index for All Urban Consumers), which is computed by the U.S. Bureau of Labor Statistics. The CPI-U is computed as a price index over time, although inflation is normally reported as the year-on-year change in the CPI-U. In 2022 for example, the CPI-U inflation rate was 8%. The chart below shows a historical perspective. In the 1970s and early 1980s, the U.S. experienced its highest modern inflation rates, both high in level and in volatility. Since then, rates fell and become more stable, at least until COVID arrived in 2020.



Source: Minneapolis Federal Reserve Bank

The CPI-U includes consumer costs in the following categories: Food & beverages, housing, apparel, transportation, medical care, recreation, education, communication, and other goods & services. The CPI-U includes taxes such as sales taxes that are associated with consumer expenses, but excludes other taxes. It excludes all investment related items.

Of course, the CPI-U will not in general be able to match anyone's actual increase in expenses, since some will necessarily spend relatively more on medical care vs education and vice versa. Nevertheless, it is considered close enough for most consumers. In fact, many consumer payments, such as cost-of-living adjustments to wages (so-called *COLA* or inflation-protection clauses), and pension benefits are linked to the CPI. If your wages are *indexed* to the CPI, this means your actual cash income will rise in proportion with the CPI. If you make $50,000 in 2017, your base wage does not increase, you have a COLA clause in your wage contract, and the CPI inflation rate ends up being 2%, you will nominally earn $51,000 in 2018, but your spending power will be $50,000 in 2017 dollars.

As an index choice, the CPI does not work in all situations. There are custom regional indices available for example, which you might use to negotiate a COLA contract specific to Syracuse or Seattle. There are also indexes related to producer prices (PPI), since manufacturing companies purchase baskets of industrial commodities that differ significantly from consumer baskets. Finally, some analysts prefer to use the GDP

(gross domestic product) deflator, which necessarily considers all goods and services produced in an economy as the "basket", and measures the rate of price increase of that basket. The GDP deflator better reflects the ability of USD to purchase a pro-rated basket of total U.S. production of goods and services, as compared to the previous year.

Can inflation be a good thing?

We've been considering the effect of inflation on wage-earners, who would be hurt by inflation if they received fixed wage payments while inflation was present. However, if the same wage-earner were making fixed payments on a mortgage while his salary was indexed to inflation, he would be pleased to see his purchasing power increasing on a relative basis rather than decreasing.

Nominal and real cash flows and rates

Whatever inflation index you use going forward, we will need to differentiate between *nominal* cash flows and *real* cash flows. Real cash flows have been adjusted for inflation so that the purchasing power of a real dollar in one year is the same as another year. For example, if your nominal income went from $50,000 in 2022 to $51,000 with 2% inflation, your real income in terms of constant spending power stayed the same at $50,000 in 2022 dollars.

The valuations we have done so far have all been based on nominal cash flows, indeed in practice, almost all valuations are like this, since cash payments are normally expressed without any inflation adjustment. Income and expenses are estimated in nominal dollars, and are discounted at nominal interest rates. Whenever we use "r" in this text, it represents the nominal rate of interest.

Real cash flows are those that have been adjusted for spending power. With inflation present, the spending value of a nominal dollar will depend on the inflation rate. Real dollars need to be discounted at a different rate of interest. We will use the Greek letter rho "ρ" to denote the real rate of interest. Also, "i" is the inflation rate.

The famous **Fisher Equation** links one period inflation to one period spending power. Interest raise the dollar's spending power in the future (multiplying by (1 + r)), while inflation reduces it (dividing by (1 + i)). The resulting ratio expresses your real increase in purchasing power (1 + ρ).

$$\textbf{\$1 for 1 yr} \rightarrow \textbf{(1+r)/(1+i)} \equiv \textbf{(1+}\rho\textbf{)}$$

$$or\ \rho = (r - i)/(1+i)$$

With continuously discounted inflation and interest rates, we have exactly.

$$e^r = e^{\rho}e^i \quad or \quad r = \rho + i$$

This is easily computed by taking natural logs of both sides. For example, if the interest rate is 5%, but the inflation rate is 2%, a risk-free investment increases in spending power by about 3% per year. One nominal dollar now has the same real value in today's dollars, unless we define it otherwise using a historical benchmark, such as "1960 dollars". After a year, a nominal dollar is worth $1.05, but its spending power is about $1.03 in today's dollars.

To value real annuities, therefore, we must adjust the growth rate to reflect a currency's decline in spending power. However, we have already solved this problem analytically in Chapter 2. Let the cash flow growth rate $g = -i$ to convert nominal cash flows to real cash flows. Then apply a real discount rate to the real cash flows.

As an example, let's consider a COLA wage annuity that pays an inflation-linked cash flow, with C_0 being the base income, $C_1 = C_0(1+i)$ being the first nominal payment in one year, $C_2 = C_0(1+i)^2$, and so on.

We can easily value this as a growing annuity with g = i

$$V_0 = \frac{C_0(1+i)}{(r-i)}\left[1 - \frac{(1+i)^n}{(1+r)^n}\right]$$

But if we let $r^* = (r - i)/(1+i)$, we have the regular annuity formula

$$V_0 = \frac{C_0}{r^*}\left[1 - \frac{1}{(1+r^*)^n}\right]$$

Which is exactly the simplification we learned in Chapter 2 for valuing exponentially growing annuities. Of course r* is actually ρ in this case. If ρ=0, we can simply add up the real cash flows to determine the value

$$V_0 = nC_0 \qquad if\ r^* = 0$$

This relationship is particularly important to understand when inflation rates are unknown. In this case, it would be very difficult to forecast nominal dollars and know what rate to use to discount them. We could however assume that the real payment was the same in every year, and that the cash flows were a constant annuity, to be discounted at the rate of ρ.

Example

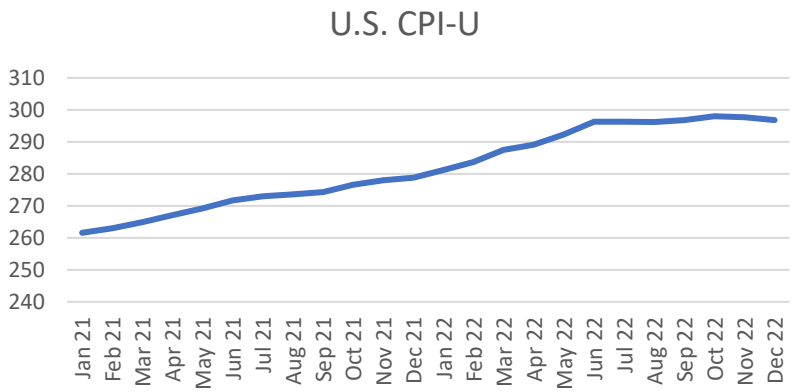
Paul earned $100,000 in the past year. He just signed a 3-year employment contract with a COLA clause. His base salary will be adjusted upward in the future according to the CPI, currently 2% per year. If the nominal rate of interest is 4.5%, what is the present value of the contract revenues?

Solution

$286K; find the present value of three years of constant base pay at the real rate of interest.

Recent CPI-U values (through 2022):

The graph below shows the historical values of CPI-U for 2021 and 2022. This was a time of high inflation concerns, considering the COVID expenses, other high government expenditures and the Ukrainian War. The top graph shows the price of the CPI-U basket leveling off, suggesting that inflation is approaching zero. However, since inflation is usually computed as a percentage change from the previous year, the accompanying table below shows that the reported CPI inflation rates appear to be falling.



Mo/Yr	2022	2021	%Δ
Jan	281.148	261.582	7.48%
Feb	283.716	263.014	7.87%
Mar	287.504	264.877	8.54%
Apr	289.109	267.054	8.26%
May	292.296	269.195	8.58%
Jun	296.311	271.696	9.06%
Jul	296.276	273.003	8.52%
Aug	296.171	273.567	8.26%
Sep	296.808	274.31	8.20%
Oct	298.012	276.589	7.75%
Nov	297.711	277.948	7.11%
Dec	296.797	278.802	6.45%

www.rateinflation.com

Inflation rate calculation

Dec 2022 CPI-U: 296.797
Dec 2021 CPI-U: 278.802
Inflation rate: 296.797/278.802 – 1 = 6.45%

If you are fortunate enough to have a cost-of-living-adjustment (COLA) clause in your wage contract, this means that your future wage will be determined by the formula:

Future wage = Base wage × Future CPI index level / Starting CPI level

Of course, actual inflation is unknown ahead of time, but COLA clauses protect people from unforeseen increases in the inflation rate as defined by the CPI.

In sum, the most important things to know about inflation are:

- ✓ The CPI-U is the most popular inflation index used in the U.S., but it is not suited to all purposes. It would be impossible to create a single index to measure everyone's change in purchasing power. Regional CPI indices (for employers), the PPI (for manufacturers) and the GDP deflator (for investors) are the most popular alternatives.

- ✓ *Nominal* cash flows are actual dollar cash flows, while *real* cash flows are adjusted for inflation. Real cash flows are expressed in a previous year's dollar value, e.g. 1960 dollars.
- ✓ Discount nominal or real cash flows at their respective interest rates r and ρ, never mix them.
- ✓ The growing annuity tools can be used to value real and nominal cash flows.
- ✓ $\rho = (r - i)/(1+i)$, nearly equal to $r - i$ for small i. This is exact with continuous discounting: $\rho = (r - i)$
- ✓ In today's dollars, the value of nominal cash flows discounted at nominal rates equals the value of real cash flows discounted at real rates.

The last point may be the most important one. It may seem difficult to value a stream of cash flows linked to an unpredictable future CPI index level. However, if the cash flows are inflation-adjusted, we can discount the future payments expressed in today's dollars at the real rate of interest. As we will see in Chapter 4, the real rate of interest may be estimated using the yield to maturity of inflation protected bonds (TIPS).

3.3 TAXES

One year while I was at his Princeton home preparing his return, Mrs. Einstein asked me to stay for lunch. During the course of the meal, the professor turned to me and with his inimitable chuckle said: "The hardest thing in the world to understand is income taxes." I replied: "There is one thing more difficult, and that is your theory of relativity." "Oh, no," he replied, "that is easy." To which Mrs. Einstein commented, "Yes, for you."

LEO MATTERSDORF New York City

In real life, taxes are complicated. Nevertheless, taxes can play a pivotal role in valuations and in investment selections. Governments use tax rules to encourage people to invest in government-favored assets, increasing the value of some assets relative to others. Individuals seek alternative investments to reclassify or offset income in order to reduce their taxes. A precise understanding of the taxes in any activity will be important for you to determine accurate valuations and appropriate investment strategies. However, your private valuations will depend on your tax situation. It will be impossible to cover all the complications of taxes here. Simulation methods covered in Chapters 5 and 6 can deal fairly easily with complex tax schemes, even when it might be difficult to develop analytic pricing formulas.

Let's begin with some terminology. *Nominal tax rates* refer to the stated tax rate for a given situation. *Effective tax rates* refer to the rate actually paid. Taxes may be levied on capital gains and cash flows, usually both, albeit at different marginal tax rates. The percentage tax paid on an incremental dollar of income is called the *marginal tax rate*. Total taxes divided by the investment amount is called the *average tax rate*. Taxes may be assessed on capital gains, but often, tax is only due in the tax period the capital gain is *realized*, which occurs by definition when the asset is sold or as required by the applicable tax code. For example, in mark-to-market accounting, taxes are computed based on the change in market value of an asset regardless of whether it was sold or not.

While it is impossible to list all the complications surrounding real life taxes, here is a list of important things to know:

Realistic assumptions:

- Capital gains are generally taxed when they are realized, on the sale of an asset
    - Capital losses are deducted from taxes, but must also be realized before they are taxed

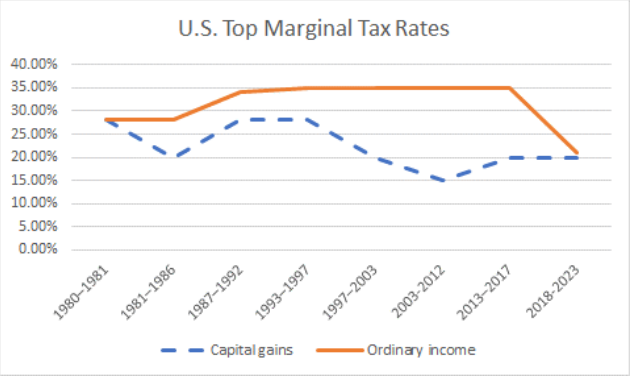
    - There are generally limits to allowable deductions
    - Losses may be allowed or required to be *carried forward*, i.e. recognized in future periods
- In some cases, accrued but unrealized capital gains may be taxable (e.g. mark-to-market accounting)
- Capital gains tax rates differ from ordinary income (cash) tax rates
- Tax rates are progressive, meaning that people with higher incomes have higher marginal tax rates
- Tax rates differ for different individuals and corporation types
- Tax rates change over time

For the purposes of this chapter, we will make the simplifying assumptions as follows:

- Capital gains and losses are taxed at the time they occur (i.e. they are all treated as realized)
- Tax rates are constant, and do not change based on income or ownership
- Loss deductions are unlimited

For the moment, we will allow capital gains tax rate (τ_{CG}) to differ from the ordinary income tax rate (τ_{OI}). A rule of thumb in finance states that you may discount pretax cash flows at pretax interest rates, or after tax (AT) cash flows at after tax interest rates. These calculations should yield the same present value, and in many cases, this equivalence defines the *effective* after-tax discount rate. When capital gains (CG) and ordinary income (OI) are taxed at the same rate τ, the after-tax of return $r_{AT} = r(1 - \tau)$, if r is the pretax rate of return.

Historically the two tax rates have generally differed in the U.S. since 1922, with the top income tax rate exceeding the capital gains rate. However, they are equal in 2018-23. Here are the two rates from 1980-2023:



Source: Wolters Kluwer CCH

Example

A par bond is one whose value is the same as its principal, often the case on the issuance date. Consider a par bond with annual coupons, which costs $1000 and pays annual coupons at a rate of 5% ($50 per year) until maturity. At maturity, the $1000 principal is repaid along with the final coupon. If the tax rate $\tau = 40\%$, then the bondholder only keeps $30 from the coupon payment, which is $(1 - \tau)$ or 60% of the $50, since $20 is paid in taxes. There is no capital gain to tax, since the bond is redeemed for $1000 at maturity.

The pretax present value is $1000 if the interest rate is also 5%. The after-tax cash flows resemble a pretax bond paying $30 per year and $1000 at maturity, so the realized rate of return, $r_{AT} = 3\%$. Indeed, in this case the equation holds --- $3\% = (1 - 0.40) \times 5\%$.

The effect on the GVE of taxes is simple in theory. If τ is the tax rate, and capital gains and cash flows are taxed at the same rate, and our assumptions from above hold, then after-tax required and expected returns are $(1 - \tau)$ times the pretax returns in dollars:

Ignoring risk, when the capital gains rate is the same as the ordinary income rate, the GVE states,

$$\text{Pretax required return} \times (1 - \tau) = \text{Pretax capital gain} \times (1 - \tau) + \text{Pretax cash flows} \times (1 - \tau)$$

Since all the components are multiplied by $(1 - \tau)$, we can see that the valuation equation is not affected by the taxes under our assumptions. If there is only one tax rate, then $(1 - \tau)$ cancels in the GVE. One may either express the GVE in pretax or post-tax terms, and arrive at the same valuation.

Different tax rates can also be incorporated into the GVE. Again ignoring risk,

$$\text{Pretax required return} \times (1 - \tau_e) = \text{Pretax capital gain} \times (1 - \tau_{CG}) + \text{Pretax cash flow} \times (1 - \tau_{OI})$$

This equation assumes that all capital gains are taxed at the same constant rate τ_{CG}, all cash income (or losses) are taxed at a potentially different constant rate τ_{OI}. The after-tax required return is determined from the pretax return and the effective tax rate is τ_e. Implicitly, this representation of the valuation assumes any losses are fully deductible from taxes paid --- negative taxes would reduce an investor's tax burden or even lead to a refund. For example, if one had to defer capital losses, then "$(1 - \tau_{CG})$ Capital gains" could be replaced with "$(1 - \tau_{CG})$ Max(Capital gains, 0)".

Here is a numerical example for a 10-year annuity of \$50 per year, and r=5%, and both tax rates equal to 40%:

	ACTUALS			TAX			
Time	CF	V	CG	CF	CG	Total	AT
0		386.09					
1	50.00	355.39	(30.70)	20.00	(12.28)	7.72	42.28
2	50.00	323.16	(32.23)	20.00	(12.89)	7.11	42.89
3	50.00	289.32	(33.84)	20.00	(13.54)	6.46	43.54
4	50.00	253.78	(35.53)	20.00	(14.21)	5.79	44.21
5	50.00	216.47	(37.31)	20.00	(14.92)	5.08	44.92
6	50.00	177.30	(39.18)	20.00	(15.67)	4.33	45.67
7	50.00	136.16	(41.14)	20.00	(16.45)	3.55	46.45
8	50.00	92.97	(43.19)	20.00	(17.28)	2.72	47.28
9	50.00	47.62	(45.35)	20.00	(18.14)	1.86	48.14
10	50.00	0.00	(47.62)	20.00	(19.05)	0.95	49.05

In each time period, the annuity value is determined as the PV of the remaining payments. The capital gain (CG) is the change in value, which is negative for its life. The tax is 40% of the payment minus a credit for the capital loss. The after-tax cash flow (AT) is the pretax cash flow (payment) minus taxes. In this case, as expected, the present value of the after-tax cash flows at r_{AT} = 3% is \$386.09, exactly as in the pretax case.

What if $\tau_{CG} \neq \tau_{OI}$? To value an asset in this situation, the problem is that we don't know the value of r_{AT} from the outset. To solve the after-tax problem, we therefore take the following steps:

- ✓ Compute the value of the pretax instrument V_0 at pretax discount rate r and set this equal to the after-tax value
- ✓ Find the discount rate r_{AT} that makes the present value of the after tax cash flows equal to this value.
- ✓ The effective tax rate τ_e satisfies $r_{AT} = (1 - \tau_e)r$.

Example

A risk-free security sale is expected to generate $20 in dividends, $30 in capital gains, and a $1000 return of principal in one year. The security's value today is $1000, implying a 5% risk-free discount rate. The tax rate for ordinary income is 40%, and the capital gains rate is 15%. What is the effective tax rate and the after-tax rate of return on the security?

Solution

The pretax valuation is $1000, as is the post-tax value. The after-tax cash flows are $37.50 (= 0.6(20) + 0.85(30)) in one year. This suggests $r_{AT} = 3.75\%$ so $(1-\tau_e) = \frac{3.75}{5} = 0.75$, and the effective overall tax rate is 25%.

Finally, we may consider valuation with taxes to be something like a private valuation, in that investors with different tax situations will value the same set of pretax cash flows or investment returns differently. This may make certain investments, such as nontaxable municipal bonds, attractive for some investors and unattractive for others. In particular, nontaxable bonds will be preferred by investors in the highest tax brackets compared to investors in the lowest tax brackets.

In sum, it is difficult to make generalizations about valuations with taxes unless the capital gains and ordinary income tax rates are the same. In practice, it may be best to simulate after-tax outcomes of different investment programs to rank possible investments according to the tax circumstances for a particular investor. This implies that most tax-based valuations can be considered private valuations rather than public valuations of securities or contracts.

3.4 EVENT RISKS

Cash flows or capital gains may be subject to shocks at random times. If that is the case, we need to know how to value them. In this section, we demonstrate the application of the GVE to event risks when the probability of the occurrence of the event is constant in each period.

A life insurance policy

Consider the example of a single premium life insurance policy:

- V_0 = Value of the insurance policy now.
  The policy is purchased for V_0, and there are no recurring payments.
- C = Payoff at the time of death. The policy has zero residual value after the payment is made
- π = Probability of death each year, assumed constant.

For now, let's assume there is no risk premium. Then according to the GVE we have

$$\text{Required return} = rV_0$$

$$\text{Expected return} = \text{expected capital gain} + \text{expected cash flow} = -\pi V_0 + \pi C$$

The capital gain is $-\pi V_0$ because in any given year the policy is active, the possibility of terminating the policy is π and the capital loss is V_0. Similarly, the probability in any year of getting the cash flow is π so the expected cash flow is πC.

Setting these returns equal and solving for V_0

$$V_0 = \frac{\pi C}{r + \pi}$$

For example, suppose an insurance policy pays C = $100,000 at the time of death, and for a 30-year old, the conditional likelihood of death is 1% per year, at least for a while. The risk-free rate is assumed constant at 5%. Then the value of the insurance policy is $16,667 regardless of when it is bought, provided the probability of death remains constant. The resulting formula may be interpreted as a kind of perpetuity: every year, the expected payment is πC. We cannot discount at r since the policy will not last forever, but increasing the discount rate to $r+\pi$ automatically adjusts for the expected time to collect on the policy.

Now we may turn our attention to the risk premium. Under the assumptions made above, the insurance company would not make a return on the policy and could not recover any of the underwriting or management costs. If we assume the insurance company must earn a fraction k of the policy's value every year (let's say 0.5%), then the buyer must pay that total risk premium as follows:

$$\text{Required return} = rV_0 - kV_0 = (r - k)V_0$$

We could also add kV_0 to the right-hand side of the GVE, reflecting the fact that the insurance company earns part of the capital gain and cash flow of the investment. Whether we adjust the left or the right side of the GVE, to solve for V_0, we adjust the discount rate downward to 4.5%, and the cost of the policy rises to $100{,}000(0.01)/(0.045+0.01) = \$18{,}182$.

The increase in cost should not surprise anyone. In practice, sales, origination, administrative and risk and regulatory charges can be a significant part of the overall cost of any insurance policy.

Auto insurance

Consider the example of a single premium auto insurance policy:

- V_0 = Value of the insurance policy now.
    - The policy is purchased for V_0, and there are no recurring payments.
- C = Payoff in case of an accident. The policy remains in force after the accident and covers the policyholder for future accidents, until death.
- δ = Probability of death in any year, assumed constant.
- π = Probability of accident each year, assumed constant.
- k = Premium paid to the insurance company, as a percentage of the policy value

We assume for simplicity that the two events, death and accidents, do not interact with each other. We assume the occurrence of death or a single accident causes a payment at the end of the period. Then according to the GVE we have

$$\text{Required return} = rV_0 - kV_0$$

$$\text{Expected return} = \text{expected capital gain} + \text{expected cash flow} = -\delta V_0 + \pi C$$

The capital gain is $-\delta V_0$ because in any given year the policy is active, the possibility of terminating the policy is π and the capital loss is V_0. Similarly, the probability in any year of getting the cash flow is π so the expected cash flow is πC. The single premium value is therefore

$$V_0 = \frac{\pi C}{(r-k)+\delta}$$

If the automobile policy that will pay C=$15,000 for each year you are in an accident, the probability of an accident every year is π=10%, the risk-free rate is 5%, the death probability is 1% per year and the risk premium s 0.5%, then the value of the policy is $27,273.

Notice in this example, we can separate the probability of the cash event (in the numerator) from the probability of the termination event (in the denominator). The adjustment to the discount rate appears in the denominator due to the assumption of proportional fees & costs.

3.5 VARIABLE INTEREST RATES

In previous chapters, we assumed discount rates were constant over time. Of course, it is obvious from business news reports that interest rates are changing all the time. Even at a point in time, longer term discount rates will differ from shorter-term rates closer to the present. The graph of yield as a function of time or term is known as the *term structure* of interest rates. How can we determine the present value of cash flows defined by variable discount rates?

For expositional convenience, let's work with one-year intervals and focus on zero-coupon bonds. Using a randomly chosen historical date, one can see the term structure of zero-coupon bonds by deducing the rates from government bond prices, which is explained in detail in Chapter 4. For now, let's examine the rates and see what information they contain.

We use the following notation:

Notation	Meaning
r_{0i}	Annualized spot rate for a risk-free zero-coupon bond maturing at time i, and bought at the present time 0. This value is known when the bond is purchased.
d_i	The present value of a riskless $1 paid at time i, either $(1+r_{0i})^{-i}$ for discrete compounding or $\exp[-r_{0i}i]$ for continuous compounding
r_{ij}	With j > i > 0, this means an annualized future spot rate for a zero-coupon bond bought at time i that matures at time j. *This value is unknown at time 0.*
$E[r_{ij}]$	The expected value of r_{ij} based on information at time 0. This may be assumed or based on a model.
f_{ij}	The forward rate of interest from i to j, determined at time 0. This is the annualized rate of return locked in from time i to time j achieved by buying the bond with maturity j now, and earning r_{oi} until the forward bond starts.

To motivate our discussion, suppose we want to know the one year spot rate one year from now. We currently may know the one year spot rate and the two year spot rate, but we do not know what the one year spot rate will be in one year. When the dates are one period apart, we could say (discrete followed by continuous):

$$d_2 = \frac{d_1}{1+r_{12}} \qquad or \qquad d_2 = d_1 e^{-r_{12}}$$

Then, we would realize that r_{12} is unknown to us, usually modeled as a random variable. Is there a constant I can substitute for r_{12} to make this equation true? $E[r_{12}]$ is a tempting answer, but we don't really know the expectation either, unless we assume a form of the random variable. But even if one could assume a probability distribution, there is no guarantee that the value of the expected value would make this equation true.

What if we define a constant, the forward rate, that forces the equation to be true? Then it would be as if we bought a two-year ZCB, earned r_{01} in the first year (a known rate) and a locked-in forward rate "f_{12}" from year 1-2. Then the defined forward value would be (discrete followed by continuous, dates one period apart)

$$f_{12} = \frac{d_1}{d_2} - 1 \qquad or \qquad f_{12} = \ln(\frac{d_1}{d_2})$$

When the dates are not one period apart, and the distance is $t_2 - t_1 > 0$,

$$f_{12} = \frac{d_1}{d_2}^{1/(t_2 - t_1)} - 1 \qquad or \qquad f_{12} = \ln(\frac{d_1}{d_2})/(t_2 - t_1)$$

To summarize, the forward rate is the single fixed rate in year 1-2 that gives us the same return as if we had bought a 1 year ZCB and reinvested the proceeds to receive exactly $1 at time 2. It is not the same as r_{12} (a random variable) or even its expectation. It is the price of money from year 1-2 implied by bond spot prices that replaces the value of the unknown future cost. An investor would be equally happy with accepting an implied fixed return of f_{12} in exchange for the unknown rate r_{12}, as long as they accept the ZCB bond prices. Therefore, the random variable r_{12} (the unknown cost of money from time 1 to 2) may be replaced with f_{12}, which is treated as a constant and discounted at a risk-free rate.

The term f_{12} in this context is called the *risk-neutral* equivalent of the unknown r_{12}. In this context, it can also be called the forward rate, or the forward price of money from time 1 to 2, implying that it is a rate determined now for future time periods.

Risk-neutral values are one of the single most important concepts in finance, since they can be used to replace risky future cash flows having unknowable discount rates with risk-neutral quantities discountable at risk-free rates. Without risk-neutral equivalents, we need to build models to determine the proper costs of risk and make the necessary adjustments. With risk-neutral values, no further risk adjustments need be made.

To illustrate the concept in practice, the table below shows the spot rates of ZCBs at a point in time as a function of different maturities measured in years. The rates are called spot rates because the bonds are purchased and paid for on a spot basis. Therefore, one could invest for 1 year and earn 1.22%, or 2 years and earn 1.40% per year.

Year				Simple
Start	End	Spot	Forw	Avg
0	1	1.22	1.22	1.22
1	2	1.40	1.58	1.40
2	3	1.54	1.83	1.54
3	4	1.67	2.04	1.67
4	5	1.78	2.24	1.78
5	6	1.89	2.41	1.89
6	7	1.98	2.57	1.98
7	8	2.07	2.71	2.08
8	9	2.16	2.84	2.16
9	10	2.24	2.96	2.24

One might also ask, what return is earned from year 1 to year 2? Can we determine the implied return from the given data? You might say, if I make 1.22% for one year (0-1) and 1.40% on average for 2 years (0-2), it is as if I am earning 1.58% in the second year, so that the average rate would be 1.40%. If you said this, you would be approximately correct. If you were discounting continuously, you would be exactly correct! The exact discrete formula would suggest comparing two strategies, specifically a two-year investment, or a one-year investment being guaranteed a given *forward rate* in year 1 for 1 year. These strategies should yield the same result. Therefore,

$$(1 + r_{02})^2 = (1 + r_{01})(1 + f_{12}) \qquad discrete\ formula$$

$$e^{2r_{02}} = e^{r_{01}} e^{f_{12}} \rightarrow 2r_{02} = r_{01} + f_{12} \qquad continuous\ formula$$

The term f is called a forward rate because it implicitly allows us to compute the return as if we had a contract to purchase the 1-year bond in 1 year with the proceeds of a 1-year bond invested in one year. The forward rate curve, we can see, rises faster than the spot rate. Since the spot rate is approximately the average of the forward rates, this makes sense. For the average spot rate to rise over time, the forward price must be increasing faster.

VALUING RISK-FREE CASH FLOWS WITH VARIABLE INTEREST RATES

Suppose we have a sequence of known cash flows of $100 each in years 1-10, continuing the example above. The present value of the cash flows is quite simple in the case, since we can discount each cash flow separately at the corresponding spot rate of interest. The result is $900.77, which is shown in the Excel calculation below, along with a computation of the *Internal Rate of Return* (IRR),a single average compounded yield corresponding to the pricing of the cash flows.

Date	Spot	PV(100)	CF
0			-900.77
1	1.22%	98.79	100
2	1.40%	97.26	100
3	1.54%	95.51	100
4	1.67%	93.60	100
5	1.78%	91.55	100
6	1.89%	89.40	100
7	1.98%	87.16	100
8	2.07%	84.85	100
9	2.16%	82.51	100
10	2.24%	80.14	100
	Total	900.77	
	IRR		1.95%

Another application associated with variable interest rates has to do with cash flows that change as a function of the interest rates. In particular, a *floating rate bond or note* is one whose coupons vary based on a pre-specified interest rate index. How should we value floating rate bonds?

First, we should know that the interest rate on a floating rate bond is normally determined at the beginning of the payment period, not the end. This convention of declaring the interest rate at the beginning of the effective period is known as quoting interest *in arrears*. For example, a floating rate bond with annual payments would be structured so that the first payment equals the one-year spot rate now at time 0 (r_{01}) multiplied by the principal balance (F). Using the GVE, we can write it this way:

Valuation of a one-period FRN

$$r_{01}V_0 = (F - V_0) + r_{01}F$$

This implies that the floating rate bond is priced at par, or $V_0 = F$. If we had a two-year floating rate bond, the price at time 1 and 0 would be determined by

Recursive valuation of a two-period FRN

$$At\ time\ 1: \quad r_{12}V_1 = (F - V_1) + r_{12}F \ \ or \ \ V_1 = F$$

$$At\ time\ 0: \quad r_{01}V_0 = (V_1 - V_0) + r_{01}F \ \ or \ \ V_0 = F$$

implying that after each floating coupon is paid, the value of the bond resets to the principal value, *regardless of the value of the new spot rate*. This is important especially since r_{12} is not known at time 0. The value of floating rate bonds constructed in this matter is straightforward --- they always stay at par value if they are risk-free, at the moment the last payment is made. Interest may accrue, i.e. become owed but not included in the bond price, but the bond price resets to par after the floating interest rate is paid.

What if we needed to value a single cash flow that was a function of the future spot interest rate, and not part of a floating rate bond? In this case, the unknown r_1 would not cancel as neatly as with the FRN. This would occur for example if you were trying to value a company whose earnings fluctuated with interest rates. In this case, we would need to find a number that represented the market value of the variable payment. Fortunately, we already know the forward rate serves that role, and that information is embedded in the spot yields. Therefore, if the actual cash flow equals $r_{t-1,t}\,C(t)$ at time t, the present value is given by

$$PV = \frac{f_{t-1,t}\, C(t)}{(1 + r_{0t})^t}$$

The forward rate f in this formula represents the forward rate one year prior for a payment on the same payment date. Continuing the previous example, suppose we want to value each FRN bond payment separately. We substitute the forward price for the unknown future spot price and discount it at the risk-free spot rate. Here are the details:

FRN Payments - Individual Present Value:

F=100		Forw	PV using
	Spot	Equiv	Spot
1	1.22%	1.22	1.21
2	1.40%	1.58	1.54
3	1.54%	1.83	1.75
4	1.67%	2.04	1.91
5	1.78%	2.24	2.05
6	1.89%	2.41	2.16
7	1.98%	2.57	2.24
8	2.07%	2.71	2.30
9	2.16%	2.84	2.34
10	2.24%	102.96	82.51
		Total	100.00

Of course, the total value is *F = 100*, as expected. This demonstration shows us how to value any risk-sensitive cash flow at any time to a risk-free forward equivalent, and determine its value by discounting at the risk-free spot rate.

Example

A company has determined that its cash flows for the next three years are sensitive to interest rates. The estimated cash flow each year is estimated to be $100 MM times the one-year spot rate in the previous year. What is the present value of the first 3 cash flows using the same term structure as the one shown in this section?

Answer: $4.5 mm.

3.6 FOREIGN EXCHANGE

The U.S. Federal Reserve Bank reports spot exchange rates for the major currencies on every trading day. "Spot rates" refer to the exchange rate on a given day, while "Forward rates" refer to exchange rates for future transactions whose terms are agreed today. In particular, the exchange rate to be applied in the future is determined today. According to Investopedia, the interbank foreign exchange market trades on average about $5 trillion *per day* in 2023, making it the largest capital market in the world.

Most currencies are quoted in terms of the number of units of foreign currency received for each $1 USD. This reporting standard is called a *direct quote*. There are a few exceptions, as noted with asterisks (*) in the following table (from 12/22/2017):

COUNTRY	CURRENCY	CODE	12/22/2017
*AUSTRALIA	DOLLAR	AUD	0.7713
BRAZIL	REAL	BRL	3.3209
CANADA	DOLLAR	CAD	1.2748
CHINA, P.R.	YUAN	CNY	6.575
DENMARK	KRONE	DKK	6.2866
*EMU MEMBERS	EURO	EUR	1.1839
HONG KONG	DOLLAR	HKD	7.8137
INDIA	RUPEE	INR	64.02
JAPAN	YEN	JPY	113.3
MALAYSIA	RINGGIT	MYR	4.077
MEXICO	PESO	MXP	19.5905
*NEW ZEALAND	DOLLAR	NZD	0.7015
NORWAY	KRONE	NOK	8.3387
SINGAPORE	DOLLAR	SGD	1.3441
SOUTH AFRICA	RAND	ZAR	12.605
SOUTH KOREA	WON	KRW	1076.87
SRI LANKA	RUPEE	LKR	152.85
SWEDEN	KRONA	SEK	8.358
SWITZERLAND	FRANC	CHF	0.9898
TAIWAN	DOLLAR	TWD	29.89
THAILAND	BAHT	THB	32.73
*UNITED KINGDOM	POUND	GBP	1.3379
VENEZUELA	BOLIVAR	VEF	9.975

*indirect quotes: USD per currency unit

Near the end of 2017, the spot exchange rate for Chinese Yuan (CNY) was 6.50 Yuan per $1 U.S. dollar (USD).



We can see that the spot CNY exchange rate grew from about 6 to about 7 in three years 2014-2017, implying an annual growth rate of about 5%. A positive growth rate indicates that the CNY was weakening relative to the dollar, since more units of the Chinese currency were required to purchase 1 USD as time passed. We might like to know how we should expect to see the exchange rate move in the future.

There are many theories relating to the determination of an exchange rate between two currencies, but the main drivers are relative interest rates, inflation, current account (trade) deficits, public debt levels, political

stability and economic performance[8]. The best available indicator of expected future exchange rates is the *forward* exchange rate, ignoring any risk premium. A forward contract may specify for example that $1 could be exchanged for CNY 7 in 5 years time. This would indicate expected future weakening of the currency.

There is a strict relationship between spot rates, forward rates and interest rates governing relative prices known as IRP or **Interest Rate Parity**. The table below demonstrates IRP by showing two ways to invest 1 USD to produce CNY in the future. It can either be invested in the U.S. and converted to CNY using a forward contract, or it can be converted now at the spot exchange rate and invested in a Chinese bond.

Interest rate parity illustration for direct quotation currencies

Time	Now	In t years	In CNY
USD	$1	$(1+r_{USD})^t$	$F(t)(1+r_{USD})^t$
CNY	¥1 (X_{CNY})	$X_{CNY}(1+r_{CNY})^t$	$X_{CNY}(1+r_{CNY})^t$

The final column entries are guaranteed to be equal by the threat of arbitrage; any discrepancy can be made profitable by borrowing in one currency to invest in the other, until the discrepancy disappears. This implies that F(t) forward prices at time t are completely determined by the spot exchange rate (X) and the ZCB interest rates of the two currencies.

The interest rate parity equation is a restatement of the equality of the last two table entries:

$$\frac{F(t)}{X_{CNY}} = \frac{(1+r_{CNY})^t}{(1+r_{USD})^t} \equiv (1+\delta)^t$$

If the Chinese interest rate exceeds the US interest rate, then the USD will become more expensive to purchase over time at the rate of δ, the CNY currency depreciation rate.

In sum, the purpose of this section is to allow you to model foreign-currency denominated cash flows. Rather than trying to estimate the expected future exchange rate, one can convert future foreign payments into domestic payments at the forward rate and then discount them at the risk-free rate. Alternatively, foreign risk-free payments can be discounted at the foreign interest rate and converted to domestic currency using the spot exchange rate.

[8] There are also instances where the spot exchange rate is fixed or pegged. One famous example is Argentina's pegging of the peso to the USD from 1991 to 2002.

3.7 SUMMARY

This chapter may look like a seemingly unconnected list of useful cash flow tricks, but it is more than that. Our objective by the end of Part I is to be able to simulate and value cash flows of any type. The purpose of this chapter was to introduce you to the many alternatives. For example, your cash flows might be subject to timing irregularities, different compounding frequencies, and inflation adjustments. They might be subject to taxes or event risks.

In other cases, we might have unknown variables affecting our valuations, such as changing interest rates and inflation rates. While we have not yet studied models of stochastic rates, we did learn how to recover forward interest rates and forward foreign exchange rates from underlying ZCB values and spot exchange rates. As it turns out these forward rates take the place of the unknown variable rates in the future for valuation purposes. When these forward rates are included in cash flow determination, the cash flows can be discounted at risk-free rates.

Chapter 3 was your introduction to risk-neutral valuation, a topic that will remain important throughout this text.

3.8 REFERENCES AND FURTHER READING

3.9 EXERCISES FOR THE CALCULATOR

1. Which has a higher effective annual yield, 10% interest compounded quarterly or 9.8% compounded continuously?

2. *(Harder than it looks!)* Consumer home loans (mortgages) usually have monthly payments. Some companies are advertising the possibility of making 26 biweekly payments instead of 12 monthly payments. If the simple annual rate for a normal mortgage is 4.5%, and we have a $500,000 loan for 15 years, what are the payments on the regular mortgage and on the biweekly mortgage? What is the sum of all the payments in both cases? Which do you think is better? *Note: The biweekly mortgage payment is computed as half of the regular monthly payment, meaning that the mortgagor would make 13 monthly payments a year.*

3. Rank the following mortgage offers in terms of which is best for the consumer, using a 30 year loan of $225,000. Be sure to list your assumptions and explain your reasoning.

    a. A 3.5% mortgage with 3 points in closing costs
    b. A 3.75% mortgage with 1.5 points in closing costs
    c. A 4.00% mortgage with no closing costs

    Note: A loan with 3 points in closing costs implies that the consumer must pay a fee of 3% of the value of the mortgage as compensation to the bank or broker. The fee is normally financed, meaning that the payment is computed using the loan amount plus the closing costs as the principal

4. If the nominal interest rate is 5% and the inflation rate is 2% (simple annual rates), then what is the real interest rate? What is the answer if the two rates are continuously compounded rates?

5. Your employer gives you a choice of two compensation plans. In Plan A, your base wage will be increased now by 10% and stay flat for the next 10 years. In Plan B, your current wage will be indexed to inflation. The nominal interest rate is 5%, and the expected inflation rate is 2%. What is the present value of each plan? What other factors might cause you to choose Plan A over Plan B?

6. Your pension plan offers you one of two options:

    a. A fixed annuity of $5000 per month
    b. An inflation-indexed annuity starting at a $4500 base level today, but with the first payment in one month increasing at the rate of inflation.

    If the interest rate is 5% and the inflation rate is 2%, what is the value of each plan, assuming you are going to live 20 years longer?

    If you have a 0.25% chance of dying per month, what are the new plan values? HINT: Divide all annual rates by 12 to determine monthly rates.

7. A stock is expected to pay 2% of its value in dividends per year, and grow at a net rate (after dividends) of 7% per year. If the capital gains tax rate is 15% and the ordinary income tax rate is 40%, then what is the expected after-tax return on the stock? Note: Dividends are taxed as ordinary income.

8. A bank's energy loan portfolio is estimated to have an average LGD of 70% given that oil prices have halved in the last few years. If the default probabilities are 8% per year, the principal is $100,000,000, and the scheduled interest payments are fixed at 6% per year for 12 years, what is the value of the energy loan portfolio today? How does your answer change if oil prices fall further? Assume the risk-free rate is 4%. *Note:* LGD (loss given default) is the percentage loss of each promised payment in the event of default.

9. If you bought a single payment life insurance policy for $150,000 today which paid $2,000,000 on death, the risk-free interest rate is 4%, and there is no risk premium, what is the implied annual probability of death?

3.10 EXERCISES TO BE DONE IN EXCEL

A. Revise the Excel tax computation in the chapter to value a retirement annuity instead of a bond. The annuity pays $10,000 per year for 20 years, and the cost of the annuity is determined by discounting the payments at the contract rate of 3% per year.

    1. What is the valuation and effective after tax rate for the annuity if the capital gains rate is 15%, the ordinary income rate is 40%, and the market risk-free discount rate is 4%? Assume the annuity is purchased from pretax funds. What is the value from the annuity provider's point of view?

    2. What is the answer to part 1 if the annuity is purchased from after tax funds?

NOTE: Annuity taxation differs from bonds. This quote is from Kiplinger in the context of retirement annuities: "If you buy the annuity with pretax money, then the entire balance will be taxable. If you use after-tax funds, however, then you'll be taxed only on the earnings. If you cash out a deferred annuity in a lump sum, then you'll have to pay income taxes on all of the earnings higher than your original investment. Jul 10, 2009".

MINIPROJECT 3: BOOTSTRAPPING THE U.S. TREASURY ZCB YIELD CURVE FROM BOND PRICES

Ask your instructor to choose a more recent date or download U.S. Treasury Notes and Bond quotations as of 2/8/23 from the Wall Street Journal. You are about to perform one of the most important functions of an investment bank, which is to know the exact ZCB yield curve at any point in time. The risk-free ZCB curve is used to determine forward interest rates, foreign exchange forward rates and derivative prices. Any cash flow receipts or commitments of the bank must be continually revalued as the term structure changes. Therefore, you should consider this problem seriously.

The details of bootstrapping are found in section 4.4. The idea is to imagine that we are trying to find the continuous discount factor $d(t)=\exp(-r(t)t)$ for all dates t from 0 to 30 years. The term d(t) is a complex and unparameterized function of time. We will compare the pricing of alternative d(t) parameter values to find the parameter set that finds the closest fit between our model prices and the actual market prices of all the bonds.

You report should include the following:

1. Your completed Excel spreadsheet
2. A plot of r(t)
3. A plot of market price vs theoretical price

Chapter 4: Bonds

INTRODUCTION

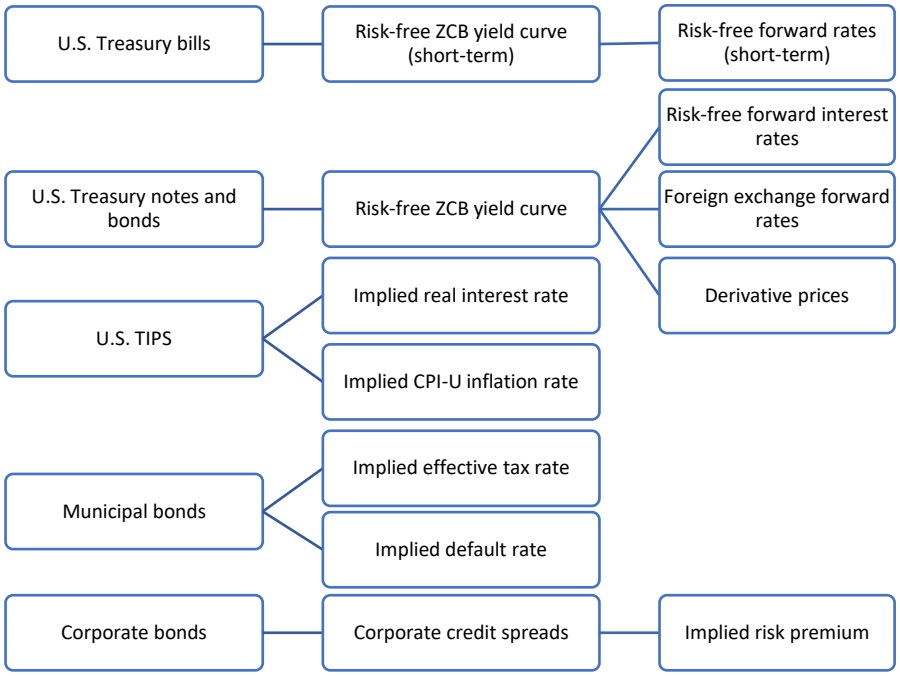
From one perspective, any financial engineer expecting to work in a capital markets function must learn the operational details of bonds and bond valuation. Once an analyst knows the yield to maturity, the coupon accrual conventions, the quoting conventions, applicable tax treatment, inflation rate, default rates, recovery rates and everything else that affects the value of the bond, he or she may proceed to value the bond with some confidence.

Although valuation is important, market price is usually the better measure of value than the result of a theoretical model. Hence, together with a model, bond prices can also be used to estimate the *implied* model parameters that might be unknown or difficult to estimate. In Chapter 3, for example, we learned how to discover bond forward rates and currency forward prices from the observed ZCB rates. In this chapter, we will extend this logic to explore the determination of the 30-year implied ZCB yield curve, real rates of interest, inflation rates, tax rates, default rates and estimates of loss given default (LGD) for corporate bonds.

Ultimately, we are focused on valuation of all cash flows, and therefore the information we collect from bond prices informs our valuations. Therefore, our purpose on the chapter is threefold:

- Understand and build models to value fixed income assets
- Discover implied valuation model parameters from fixed income asset and contract prices
- Use this information to value other cash flows

The chart below summarizes much of the information that can be extracted from bond prices.

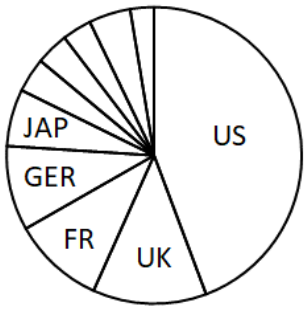


4.1 GLOBAL AND U.S. GOVERNMENT DEBT

The top 10 countries in the world had about $70 trillion in external debt at the end of 2022. The largest debtor in the world was the U.S. at that time, with about $31 trillion outstanding.

External Debt - Top 10 Countries - 2022

	USD trillions	%
United States	31	44%
United Kingdom	8.73	12%
France	7.04	10%
Germany	6.46	9%
Japan	4.36	6%
China	2.64	4%
Italy	2.51	4%
Spain	2.26	3%
Canada	3.2	5%
Australia	1.83	3%
Total	70.03	100%



Source: Wikipedia: List_of_countries_by_external_debt

The U.S. $31 trillion figure includes marketable debt and non-marketable debt. Marketable debt is in the form of tradeable U.S. government bonds, which collectively represent the largest fixed income market in the world. Marketable U.S. government bonds were worth about $24 trillion in 2022, with the remaining $7 trillion of debt being composed of "Federal employee & veteran benefits payable" and "other."

The federal debt does not include the following (2022 figures for 1 & 2 from SIFMA)[9]:

1. State and local (municipal) debt, $4 trillion
2. U.S. government agency debt (e.g. FHLB, FNMA, FHLMC, FFCB, …), $1.9 trillion
3. Unfunded liabilities of entitlement programs (e.g. Social Security and Medicare), $7.5 trillion?

Most U.S. government debt is sold in the form of tradeable securities. However, there are exceptions. Savings Bonds, Series EE and Series I, are held electronically and owned by individuals, unable to be traded. U.S. Treasury securities do not trade on any exchange. They are initially sold at auction to bond dealers, but are then traded *over-the-counter*, that is, privately between counterparties. TreasuryDirect.gov acts as a counterparty for retail and other investors, and is the exclusive manager of U.S. Savings Bonds.

4.2 U.S. GOVERNMENT SECURITIES

Marketable securities include Treasury Bills (T-Bills), STRIPs, T-Notes, T-Bonds, Treasury Inflation Protected Securities (TIPS), and Floating Rate Notes (FRNs). These instruments are considered risk-free in the sense that the U.S. government is highly unlikely to default on the debt. This is because the debt is denominated in dollars, and the issuance of dollars is controlled by the U.S. government. Hence, while investors are guaranteed to receive dollars, they are not guaranteed what those dollars will be worth. Also, while the payments are considered risk-free, bond prices will change over time as interest rates change, causing losses for investors who sell those bonds

[9] Entitlement program shortfalls are normally reported as annual projections, subject to actuarial assumptions. Present value figures are hard to find. However, cumulative 10-year projected deficits for the two programs are estimated at $2.5 trillion and $5 trillion.

prior to maturity. Therefore, the term risk-free should be considered to apply only to the level of the promised payments --- not to the market value of bonds.

A brief history

U.S. debt dates back to the Revolutionary War in 1776, but at that time, the debt was privately held by individuals and foreign governments. T-notes were introduced in 1812 and were originally issued as a means to finance the war with Britain. T-notes offer semiannual fixed payments and a range of 2-10 years to maturity from the issuance date. Principal is repaid on the maturity date. T-bonds are identical in structure to T-notes, except that the maturity at issuance runs from 10 to 30 years.

T-Bills were first introduced in 1929, in order to finance U.S. government programs during the Great Depression. The auction strategy for selling T-bills allowed more flexibility and higher frequency of issuance than existed for bonds and notes. The denomination of each T-Bill is $10,000, payable as a single principal payment due within one year of issuance. T-Bills are classified as short-term ZCBs (zero coupon bonds) because they are sold at a discount to par and do not make any payments prior to maturity. The short-term designation refers to the maturity occurring with one year of issuance.

The U.S. Treasury STRIPS (Separate Trading of Registered Interest and Principal of Securities) program was first introduced in February 1985. The program allows investors to hold and trade individual interest and principal payments of eligible Treasury notes and bonds as separate securities. In other words, STRIPS convert coupon bonds into portfolios of ZCBs. Since STRIPS maturities range from 0-30 years, they are an excellent reference point to estimate the risk-free discount rate for any given date within that range. STRIPS quotes were once widely reported but are no longer readily available. For this reason, we do not focus much on STRIPs in this text. Also, most analysts prefer to use T-note and T-Bond prices to determine the ZCB yield curve, given that the bond market is much deeper and is more actively traded, providing more reliable pricing.

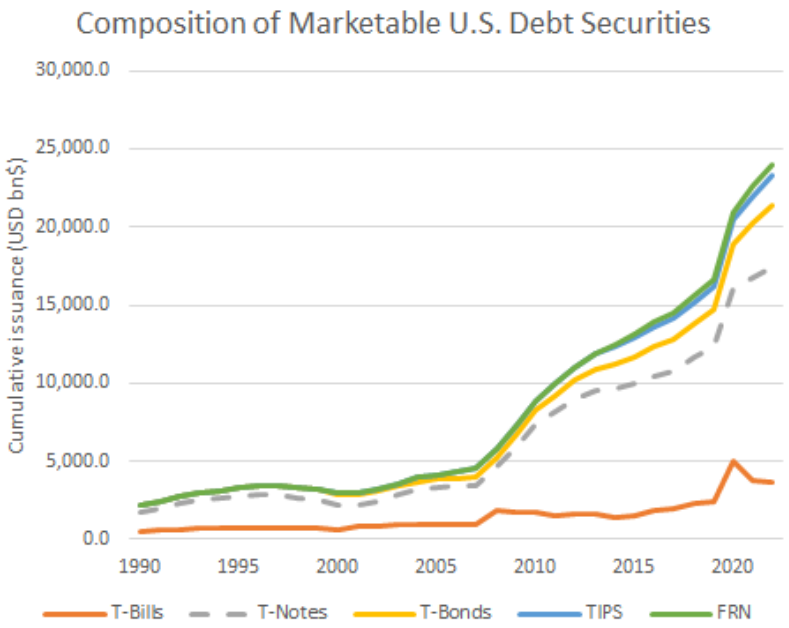
TIPS were created in 1997, and then expanded from a 20-year maximum maturity to 30 years in 2009. These securities provide investors with interest and principal payments that adjust according to the CPI-U inflation rate, defined later in the chapter. Floating rate notes (FRNs) were introduced in 2013. The floating rate payment is based on the most recent 13-week T-Bill auction high rate, and FRNs are issued at 2-year maturities.

Taxation of U.S. Government bonds from the Raymond James website

“Interest income from Treasury securities is subject to federal income tax but exempt from state and local taxes. Income from Treasury bills is paid at maturity and, thus, tax-reportable in the year in which it is received. Although not paid until maturity, income from zero-coupon STRIPS is taxable in the year in which it accrues. Increases in TIPS’ principal value as a result of inflation adjustments are taxed as capital gains in the year they occur, even though an investor does not collect these gains until TIPS are sold or mature. This is known as a ‘phantom income’ tax. Conversely, decreases in the principal amount due to deflation can be used to offset taxable interest income from other investments.”

Market size for U.S. Treasury issues

The sizes of the five marketable security components of the government debt are shown in the table below, from 1990-2022. The stacked line chart shows the cumulative amount of financing, with the bottom line representing T-Bills outstanding, with each successive category being added to the previous sum.



U.S. Marketable Debt as of Dec 31, 2022*	
Bills	3,697
Notes	13,752
Bonds	3,960
TIPS	1,908
FRN	617
Total	23,934

*USD Billions

Source: SIFMA data, author's graph

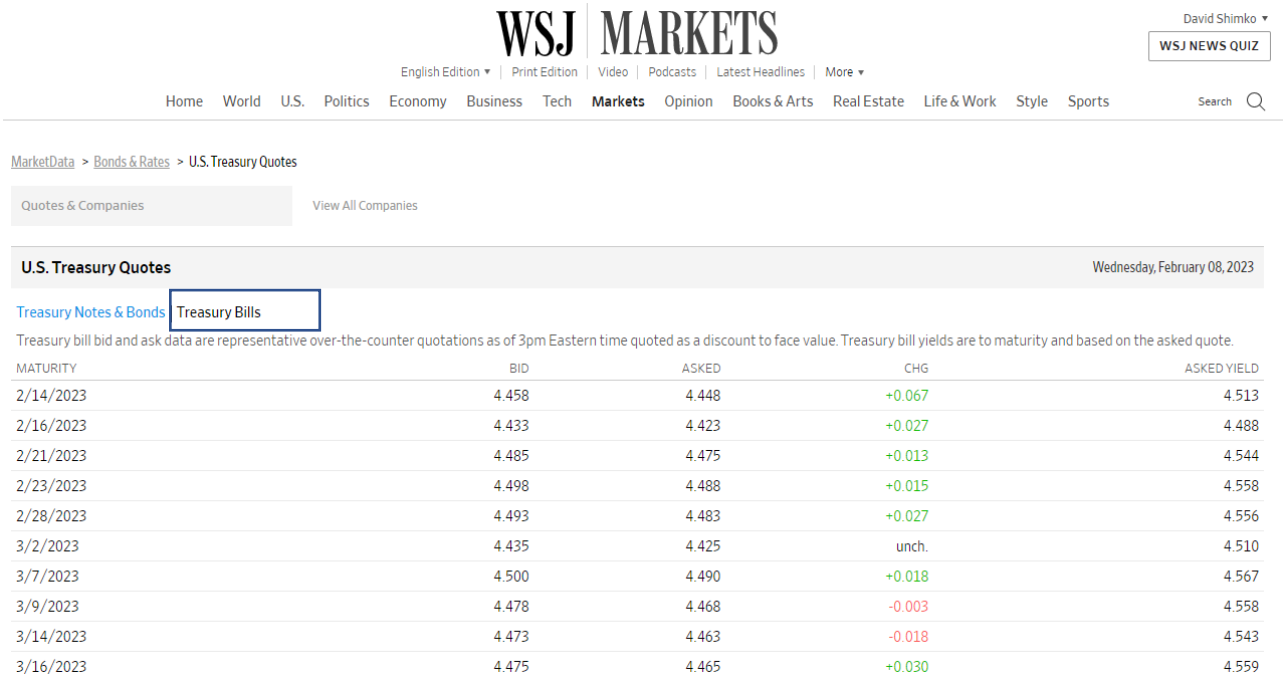
In terms of issuance, T-notes historically represented the largest single component with 2-10 years maturity at issuance. Note that STRIPs are stripped interest and principal payments from notes or bonds, and therefore are included in the notes or bond categories.

Marketable debt is unsustainably increasing at an increasing rate, the result of increased spending in both the Trump and Biden administrations, fueled in part by COVID-related expenditures, investment losses, reduced tax collections, increased entitlements, and the war in Ukraine. As interest rates rise, the cost of servicing the new debt is also increasing rapidly. Challenging economic conditions have important implications for bond markets and the risks associated with bond trading and bond portfolios generally. For these reasons, the financial engineer must be well-grounded in the valuation and risk assessment of fixed income investments.

4.3 TREASURY BILLS

With less than one year to maturity, and no coupons, T-Bills are an ideal ZCB benchmark for valuing short-term cash flows. They are issued by the U.S. government and therefore are considered risk free. The face value at maturity is $10,000 per T-Bill.

Here is a WSJ quote excerpt from 2/8/23:



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U.S. Treasury Quotes Wednesday, February 08, 2023

Treasury Notes & Bonds | Treasury Bills

Treasury bill bid and ask data are representative over-the-counter quotations as of 3pm Eastern time quoted as a discount to face value. Treasury bill yields are to maturity and based on the asked quote.

MATURITY	BID	ASKED	CHG	ASKED YIELD
2/14/2023	4.458	4.448	+0.067	4.513
2/16/2023	4.433	4.423	+0.027	4.488
2/21/2023	4.485	4.475	+0.013	4.544
2/23/2023	4.498	4.488	+0.015	4.558
2/28/2023	4.493	4.483	+0.027	4.556
3/2/2023	4.435	4.425	unch.	4.510
3/7/2023	4.500	4.490	+0.018	4.567
3/9/2023	4.478	4.468	-0.003	4.558
3/14/2023	4.473	4.463	-0.018	4.543
3/16/2023	4.475	4.465	+0.030	4.559

The legacy quotation system for trading and reporting T-Bill rates presents a minor issue. This can easily be resolved once we know the conversion from a rate quote into a price quote:

$$T-Bill\ Asking\ Price = 10{,}000\left(1 - Quoted\ asked\ \times \frac{Days}{360}\right)$$

From the price, we can compute the continuously compounded yield using:

$$r = \frac{-\ln\left(\frac{Price}{10{,}000}\right)}{\left(\frac{Days}{365}\right)}$$

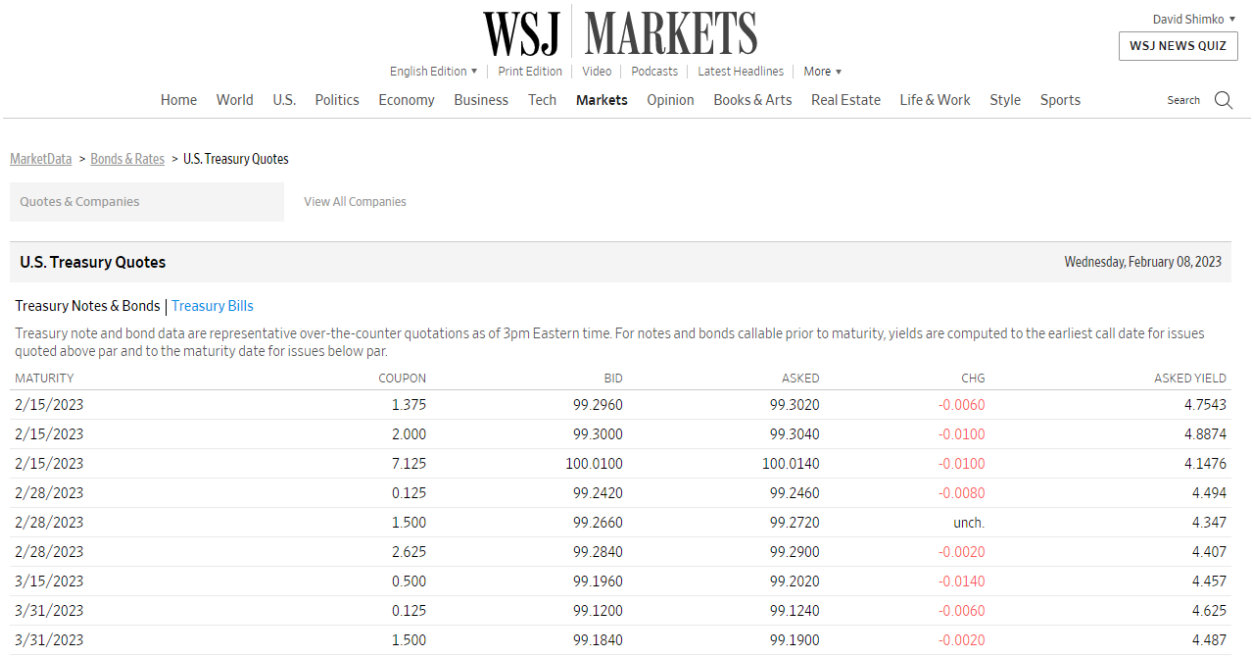
For example, the 3/16/23 T-Bill maturity (36 days) shows an asking quote of 4.465 percent. This means the price is $10{,}000(1 - 0.04465\left(\frac{36}{360}\right))$=\$9,955.35. The continuously compounded yield is 4.54% p.a., quite close to the reported asked yield of 4.559%.

Please note: The WSJ yield calculation differs slightly from ours. Please see https://www.treasurydirect.gov/instit/annceresult/press/preanre/2004/ofcalc6decbill.pdf for complete details.

4.4 U.S. GOVERNMENT COUPON BONDS – T-NOTES AND T-BONDS

U.S. government notes and bonds make coupon payments semiannually. Their quoted face value is \$100, and the semiannual coupon is given by 100c/2, where c is the annual coupon rate, so a 6% coupon bond pays \$3 every six months until the bond matures. On the maturity date, the bond pays the final coupon and the \$100 principal.

Here is a sampling from the Wall Street Journal's reported bond prices on 2/8/23. Altogether, there were 328 bond quotes posted that day for different maturities.



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U.S. Treasury Quotes Wednesday, February 08, 2023

Treasury Notes & Bonds | Treasury Bills

Treasury note and bond data are representative over-the-counter quotations as of 3pm Eastern time. For notes and bonds callable prior to maturity, yields are computed to the earliest call date for issues quoted above par and to the maturity date for issues below par.

MATURITY	COUPON	BID	ASKED	CHG	ASKED YIELD
2/15/2023	1.375	99.2960	99.3020	-0.0060	4.7543
2/15/2023	2.000	99.3000	99.3040	-0.0100	4.8874
2/15/2023	7.125	100.0100	100.0140	-0.0100	4.1476
2/28/2023	0.125	99.2420	99.2460	-0.0080	4.494
2/28/2023	1.500	99.2660	99.2720	unch.	4.347
2/28/2023	2.625	99.2840	99.2900	-0.0020	4.407
3/15/2023	0.500	99.1960	99.2020	-0.0140	4.457
3/31/2023	0.125	99.1200	99.1240	-0.0060	4.625
3/31/2023	1.500	99.1840	99.1900	-0.0020	4.487

The bid and ask prices are quoted here in decimal form, although prices used to be reported in 32^{nd}s. The *asked yield* equals 2 times the semiannual yield to maturity of the bond, and is also called a *bond equivalent yield*. The yield calculation is based on the asking price, since this is the cost of a bond to the bond buyer.

The bid and ask prices are known as *clean prices*, since they do not include *accrued interest*. For example, the bond maturing 2/15/23 paid its previous coupon six months before the expiration date, i.e. 8/15/22. Since that time, 177 days have *actually* passed out of 184 *actual* days in the coupon period. This is called the ACTUAL/ACTUAL convention for computing accrued interest accrued interest in the U.S. bond market. To compute the accrued interest, we take $177/184^{ths}$ of the semiannual coupon payment (1.375/2) to get the dollar amount, which is \$0.6613. Therefore, if you were to buy the bond, you would pay the *invoice price*, which would be the asking price of 99.3020 plus the accrued interest of 0.6613, for a total of \$99.9633. The total cost is also sometimes called the *dirty price* of the bond.

Bootstrapping the Yield Curve

We would like to use the pricing information from coupon-paying bonds to estimate the correct discount rate for ZCBs maturing any date in the future. This process is called *bootstrapping* the yield curve. There are many techniques to bootstrap, but we will only present one here.[10] Some methods rely on advanced interpolation and smoothness maximization techniques. Our method proposes a flexible functional form for the continuously compounded discount rate as a function of time. The function parameters are chosen so

[10] Many yield curve models are based on interpolation rather than nonlinear regression. Interpolation has the benefit of matching market reported prices exactly, but at a cost of functional complexity. The nonlinear regression approach is appropriate if the analyst wants smooth curves and is willing to assume that not all reported prices equal exact "market prices."

that the model's bond prices best approximate market prices. The best parameters minimize the sum of squared error (SSE), i.e. differences between market prices and model prices. This method can be referred to as a nonlinear regression method.

Functional Form Procedure for Bootstrapping

1. Select a function r(t) with undetermined parameters that is sufficiently flexible to provide a wide variety of ZCB term structures, for example

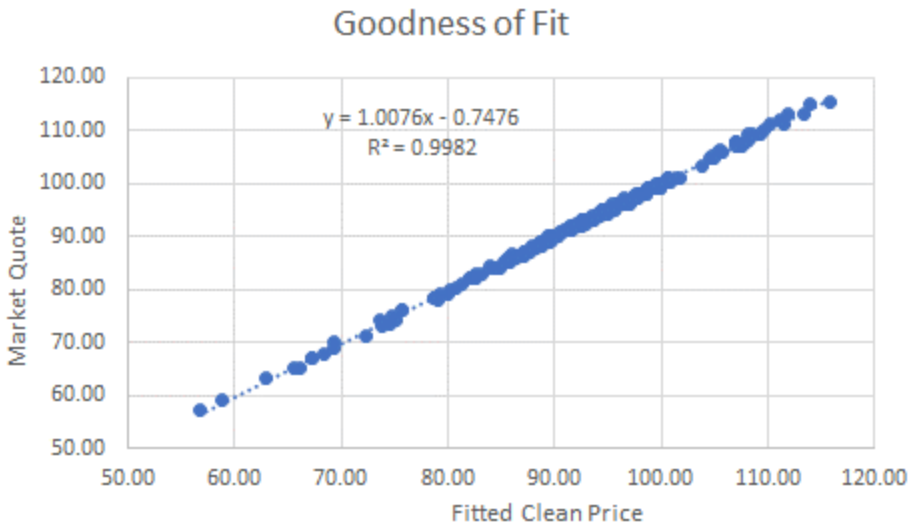
$$r(t) = a + bt + cc\ t^2 + d\ e^{ft} + g\ \ln(t - h)$$

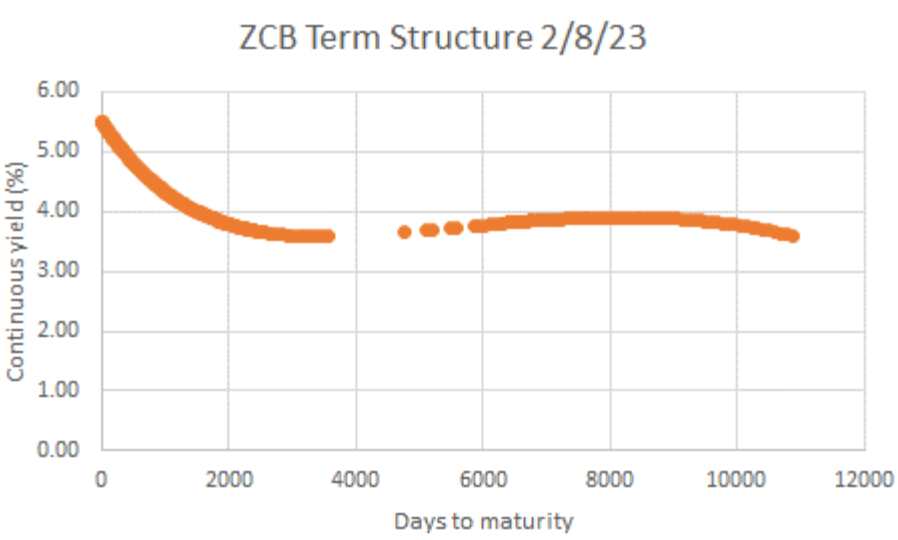
2. For each reported bond, determine the list of coupon dates
3. For each coupon date, determine the number of years (t) between the quotation date and the coupon date. Do the same for the maturity date (T).
4. For each coupon date, compute a discount factor $d_t = \exp[-r(t)t]$. For the principal payment, set $d_T = \exp[-r(T)T]$.
5. For each bond, compute the fitted bond dirty value and subtract accrued interest. The summation is taken over all coupon dates including the maturity date.

$$B = 100 d_T + \frac{c}{2}\sum_t d_t - accrued\ interest$$

6. Compute the sum of squared deviations between the fitted bond price and the market price (most analysts use the average of the bid and ask price as the market price)
7. Use a nonlinear solver (such as in Excel or Python) to minimize the sum of squared deviations between the fitted prices and the market prices by changing the parameters, i.e. {a, b, cc, d, f, g, h}

We leave this as an extremely important and highly recommended task for the reader to bootstrap their own ZCB yield curve. Here are the results we obtained for 2/8/23. The first graph shows the fit between predicted bond prices and the average of the bid and offer prices. The second shows the implied ZCB yield curve, i.e. r(t), using the implied parameters.





a	0.2720
b	0.0086
cc	-0.0001
d	0.0375
f	0.0150
g	-0.1225
h	-7.9728
SSQ	48.83

The parameters that provided the ZCB term structure are here for your reference, to compare with your results. You may like to experiment with different functional forms to see if you can reduce the SSE even further.

Bottom line, we now have a discount factor d_t for every possible date "t" in the future, based on our bootstrapping procedure. This method should not be extrapolated beyond 30 years. However, it can be very useful for valuing risk-free cash flows at time-dependent rates for up to 30 years.

The shape of the term structure

What determines the shape of the term structure? Forward rates are determined by two factors. First, the market has implicit expectations of future interest rates. Expectations of rising rates would be sufficient to produce an upward-sloping yield curve. However, even when the market expects rates to fall, we often see upward-sloping yield curves. Why is this the case?

If all interest rates were the same, bond investors would prefer to lend short-term rather than long-term. However, governments and corporations would prefer to borrow long-term than borrow short-term. Because of this natural imbalance in preferences, long-term rates generally exceed short-term rates to induce lenders (investors) to buy long term bonds. Of course, lower short-term rates give an incentive for corporations to borrow shorter-term. Therefore, there is a *term premium* in equilibrium, i.e. an additional component of the interest rate necessary to clear the risk-free capital markets. We usually write it in the form

$$Forward\ rate = Expected\ future\ spot\ rate + Term\ premium$$

This equation is implied by the *liquidity premium hypothesis*, or can be seen as the likely implication of the *preferred habitat theory*. In the *expectations hypothesis* of the term structure, the term premium is zero. However, few economists would support the expectations hypothesis any longer.

Explaining the ZCB term structure on 2/8/23

You may be wondering why there is a gap in the term structure, and why it takes the shape it does. The gap is caused by a temporary suspension of the issuance of 30-year bonds during the Clinton administration. Because of that suspension, and the fact that more short-term T-notes are issued than longer term T-bonds, there are some dates for which we do not have coupon payments.

But how do we explain the *inverted* yield curve we see in the chart? Suppose we use the equation above to help understand it. If the liquidity premium is positive, then the only way the yield curve can slope downwards is if people expect interest rates will fall in the future by more than the liquidity premium. Since the interest rate is the cost of money, this suggests that the cost of money will fall in the future, possibly due to lack of demand for money for investment. Lack of investment leads to lack of growth.

Of course, the argument is not foolproof. If the market expected high inflation simultaneously, the nominal rate would be higher, possibly leading to an upward-sloping nominal yield curve. It seems that as of 2/8/23, the market is expecting inflationary fears to subside and investment to fall. Time will tell!

4.5 TIPS AND INFLATION

"TIPS" stands for *Treasury Inflation-Protected Securities*. These securities were first issued by the U.S. government in 1997 partly in response to demand for long-term inflation-protected assets by pension funds and long-term investors.

From Treasurydirect.gov:

> "Treasury Inflation-Protected Securities, or TIPS, provide protection against inflation. The principal of a TIPS increases with inflation and decreases with deflation, as measured by the Consumer Price Index. When a TIPS matures, you are paid the adjusted principal or original principal, whichever is greater.

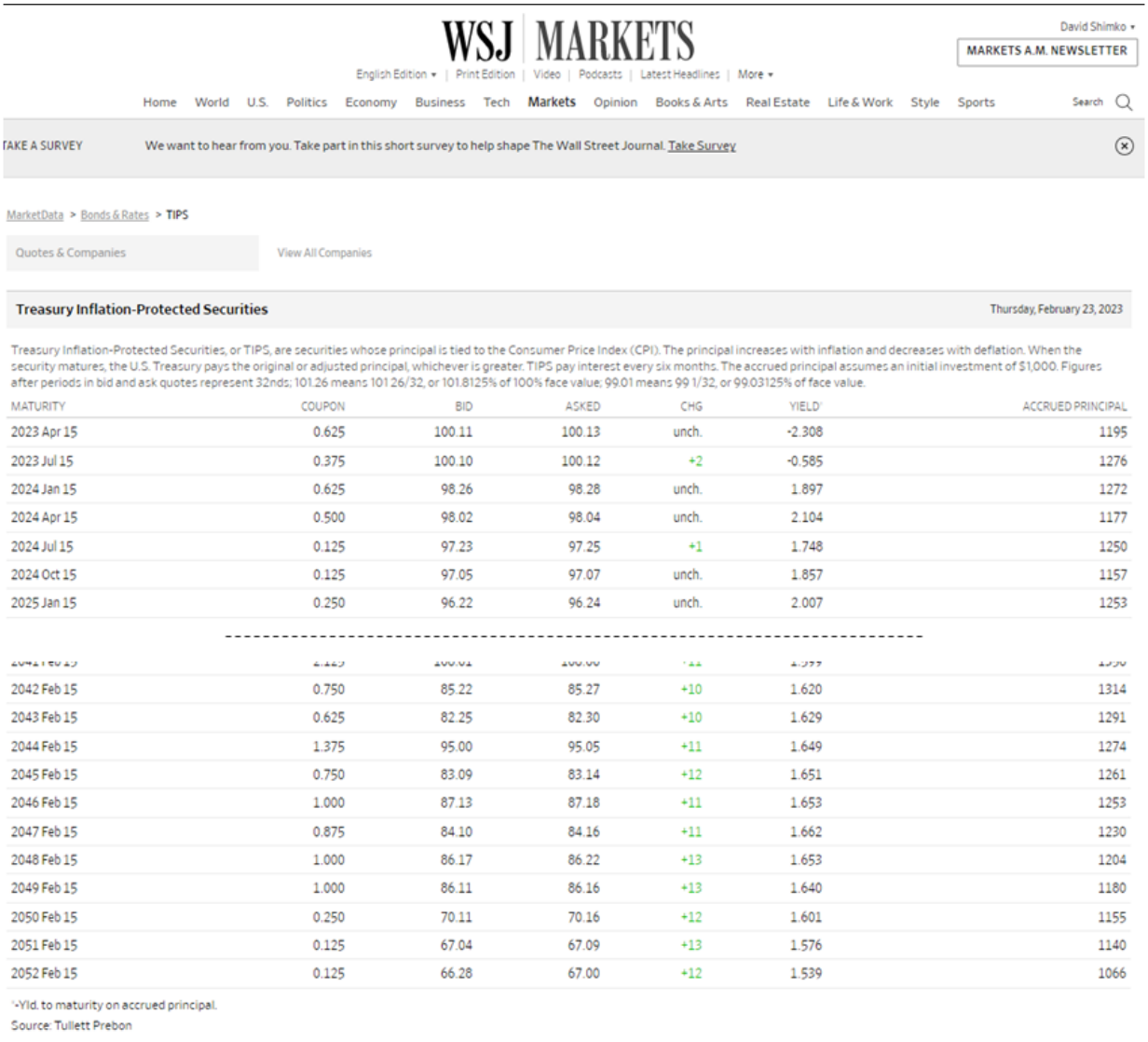
> "TIPS pay interest twice a year, at a fixed rate. The rate is applied to the adjusted principal; so, like the principal, interest payments rise with inflation and fall with deflation."

Investors who purchase TIPS can expect their coupons and principal to appreciate at the same rate as the CPI-U index. However, the tax code inserts a slight wedge in the inflation hedge:

> "Interest payments from Treasury Inflation-Protected Securities (TIPS), and increases in the principal of TIPS, are subject to federal tax, but exempt from state and local income taxes."

Whereas the return of principal in a regular bond is untaxed, the return of inflation-adjusted principal above the face value is taxed. Therefore, this bond is not a complete inflation hedge. However, all is not lost --- the principal also has an embedded option since it pays the higher of the original principal amount or the inflation adjusted principal. This protects against the seemingly remote risk of deflation.

The WSJ report for TIPS quotes on February 23, 2023, showed the following table.



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Treasury Inflation-Protected Securities Thursday, February 23, 2023

Treasury Inflation-Protected Securities, or TIPS, are securities whose principal is tied to the Consumer Price Index (CPI). The principal increases with inflation and decreases with deflation. When the security matures, the U.S. Treasury pays the original or adjusted principal, whichever is greater. TIPS pay interest every six months. The accrued principal assumes an initial investment of $1,000. Figures after periods in bid and ask quotes represent 32nds; 101.26 means 101 26/32, or 101.8125% of 100% face value; 99.01 means 99 1/32, or 99.03125% of face value.

MATURITY	COUPON	BID	ASKED	CHG	YIELD*	ACCRUED PRINCIPAL
2023 Apr 15	0.625	100.11	100.13	unch.	-2.308	1195
2023 Jul 15	0.375	100.10	100.12	+2	-0.585	1276
2024 Jan 15	0.625	98.26	98.28	unch.	1.897	1272
2024 Apr 15	0.500	98.02	98.04	unch.	2.104	1177
2024 Jul 15	0.125	97.23	97.25	+1	1.748	1250
2024 Oct 15	0.125	97.05	97.07	unch.	1.857	1157
2025 Jan 15	0.250	96.22	96.24	unch.	2.007	1253
2042 Feb 15	0.750	85.22	85.27	+10	1.620	1314
2043 Feb 15	0.625	82.25	82.30	+10	1.629	1291
2044 Feb 15	1.375	95.00	95.05	+11	1.649	1274
2045 Feb 15	0.750	83.09	83.14	+12	1.651	1261
2046 Feb 15	1.000	87.13	87.18	+11	1.653	1253
2047 Feb 15	0.875	84.10	84.16	+11	1.662	1230
2048 Feb 15	1.000	86.17	86.22	+13	1.653	1204
2049 Feb 15	1.000	86.11	86.16	+13	1.640	1180
2050 Feb 15	0.250	70.11	70.16	+12	1.601	1155
2051 Feb 15	0.125	67.04	67.09	+13	1.576	1140
2052 Feb 15	0.125	66.28	67.00	+12	1.539	1066

*-Yld. to maturity on accrued principal.

Source: Tullett Prebon

Details aside, the TIPS product is best for those investors looking to get protection from long-term CPI-U inflation increases. Also, we can use the WSJ quotes in the exhibit to estimate the market's expected inflation rate.

To first approximation, the TIPS yield resembles a real yield. From the treasury bond report the same day, we can take the nominal yield of the 30-year T-bond to be 3.875%. The real yield for a 30-year TIPS product is estimated at 1.539%, from the last row in the table above . Taking the expected inflation rate equal to the difference, we see that the 30-year expected CPI-U inflation rate is being priced at 2.236%.

It seems investors are optimistic about the U.S. Government's ability to control inflation. Like all things in finance, only the future will tell.

4.6 FLOATING RATE NOTES

Unlike fixed rate bonds, FRNs pay a quarterly variable coupon determined in arrears, implying that the first coupon payment is always known, but later payments are unknown until 3 months before each payment is made. The FRN coupon rate is the sum of the 13-week Treasury-bill auction rate (the index determination rate) and a spread determined by the auction process on the issue date. The spread is constant over the life of the FRN while future variable payments are determined by the fluctuations in the index.

Because the interest payment includes a floating component and a fixed component, it is not a pure floating instrument as described in Chapter 3. The instrument should be valued as a combination of a floating rate note and a fixed rate note. For example, the 2/23/23 FRN auction settled with a high discount margin of 0.200% and an index determination rate of 4.575%. This means that the total annual coupon is the sum, 4.775%. The actual floating index changes weekly. However, after every 3 months until maturity, a payment is made and the following payment is determined using the contemporaneous index determination rate plus the constant spread.

To value a floating rate note accurately, let's consider the auctioned note just mentioned. Suppose we have already completed a ZCB yield curve exercise and have values of r(t) for all values of t. Then, for a \$1000 face value bond, we have a quarterly fixed payment of $0.200\% \times 1000/4 = \$0.50$. The first payment date (t_1) is known, and so is the first payment amount, or $4.575\% \times 1000/4 = \$11.4375$. The present value of the first payment is therefore the sum, \$11.9375, discounted at the t_2 spot rate, i.e. multiplied by $\exp(-r(t_1)t_1)$.

The second payment date (t_2) is also known. We don't know the payment yet, but we can use the forward rate from t_1 to t_2 as a risk-neutral substitute for the expectation. Then the value of the second payment is given by $(0.50 + f(t_1,t_2)*1000/4)$ discounted by multiplying by $\exp(-r(t_2)t_2)$.

Let's perform the valuation using the 2/8/23 ZCB term structure we estimated earlier in the chapter. At the inception of the note, the timing of the payments in years is {0.25,0.50, … 2.00}. The spot discount factors are computed using $d_t = \exp(-r(t)t)$. The forward rates are easily computed from the spot rates:

$$f(t_1, t_2) = \ln\left(\frac{d_1}{d_2}\right)/(t_2 - t_1)$$

The fixed component of cash flow is risk-free, and the risk-neutral floating component can be computed from the forward rates, multiplied by 1000/4. The total risk-neutral cash flow (RNCF) is discounted at the spot rates to find the present value. The result is remarkably close to the face value of \$1000.

Date	2/23/2023					
Spread	0.2000%		**FRN Valuation on issue date**			
Index	4.5750%		**Using 2/8/23 ZCB term structure**			
Notional	1000		*No accrued interest*			

t	Spot d(t)	Forw	Fixed	RN(Float)	RNCF	PV(RNCF)
0.25	0.9867		0.5	11.4375	11.9375	11.7784
0.50	0.9742	0.0509	0.5	12.7183	13.2183	12.8773
0.75	0.9625	0.0484	0.5	12.0880	12.5880	12.1160
1.00	0.9515	0.0461	0.5	11.5210	12.0210	11.4376
1.25	0.9411	0.0440	0.5	11.0117	11.5117	10.8331
1.50	0.9312	0.0422	0.5	10.5553	11.0553	10.2944
1.75	0.9218	0.0406	0.5	10.1475	10.6475	9.8145
2.00	0.9128	0.0391	1000.5	9.7842	1010.2842	922.1825
					PV	1001.3338

In this case, there is no accrued interest, since the pricing is at issuance. However, it is a simple matter to get to the quoted FRN value by deducting the accrued interest.

Eurodollar bonds and LIBOR

FRNs also trade outside the U.S. in the Eurodollar market, using a structure similar to the U.S. Treasury FRNs.[11] In the past, issuers used LIBOR, or the London Interbank Offer Rate for US dollars as a reliable index and benchmark of the cost of short-term dollars. LIBOR also had the benefit of longer maturities being available, particularly 1-month, 3-month, 6-month and 12-month LIBOR. In fact, the 3-month LIBOR was so popular it was chosen as the benchmark for the Eurodollar futures contract by the Chicago Mercantile Exchange. Note that since LIBOR represented short-term unsecured dollar debt between London banks, part of the rate could be attributed to the credit risk of the borrowing bank. Hence, it was not a truly riskless benchmark dollar rate, since it includes a credit risk premium.

Regrettably, in 2012, "multiple criminal settlements by Barclays Bank revealed significant fraud and collusion by member banks connected to the [LIBOR] rate submissions, leading to the scandal."[12] . The LIBOR index is now managed by ICE, the Intercontinental Exchange, and was scheduled to be discontinued in 2021 or 2022. A new standard, the *Secured Overnight Funding Rate* (SOFR) has been recommended as an alternative benchmark. See https://www.financialresearch.gov/data/libor-alternatives/ for details.

According to Rosenblum and Ventura (Goldberg Kohn, 2023), the current situation for legacy LIBOR-based transactions can be summarized as follows:

- As of 12/31/21, no new LIBOR linked contracts could be signed with expiration dates later than 6/30/23
- SOFR is a widely accepted alternative to LIBOR.
- Since it is secured, credit risk is minimal; it is nearly a risk-free rate.
- The SOFR index is based on transactions exclusively, and does not rely on subjective input.
- It is estimated that after 6/30/23, $74 trillion of LIBOR-based contracts will still be in existence, mostly interest rate derivatives.

[11] The "Euro" component of "Eurodollar" has little to do with Europe. Any dollar-denominated instruments traded outside the U.S. are considered Eurodollar investments.
[12] See the Wikipedia entry "LIBOR scandal" for more details

- The LIBOR Law (3/15/22) regulates how disputes will be settled as LIBOR is legally replaced by SOFR plus a spread for the purpose of computing contract payments and receipts.

4.7 MUNICIPAL BONDS

Municipal bonds (munis) are issued by non-federal U.S. government entities, such as states, cities, and their agencies. For example, the Port Authority in New York has issued *authority bonds* to fund tunnels and bridges. These bonds are not guaranteed by the full faith and credit of the U.S. government, so default is a possibility. States and cities have tax authority, but do not have the power to create U.S. dollars. However, in many cases, municipal bonds are rated AAA due to the quality of the underlying collateral, the assumed ability of the municipality to collect taxes and usage fees, or insurance provided by third-party insurance companies.

We will study AAA munis so that we can focus more on implied tax rates as opposed to default risk. The following summary of Muni AAA rates on 2/24/23 was supplied by FMS Bonds Inc. Because municipal bonds trade unevenly in over-the-counter markets, reliable quotes are difficult to find. Nevertheless, this will give us some information.

AAA RATED MUNI BONDS

ISSUE	MATURITY RANGE	TODAY	LAST WEEK
National	10 Year	2.50	2.25
National	20 Year	3.30	3.10
National	30 Year	3.50	3.30

Source: FMS Bonds Inc. 2/24/23

The yields on AAA municipal bonds are lower than the yields on U.S. government bonds. For lower-rated bonds, we would expect that the yields will be higher to reflect some element of default risk and losses given default. However, favorable tax treatment allows muni yields to be lower. From Investopedia.com:

> "While interest income is usually tax-exempt for municipal bonds, capital gains are usually not tax-exempt. Capital gains realized from selling a bond are subject to federal and state taxes. ... These investors are subject to taxation on the interest income from some munis."

Assuming bonds are held to maturity, then, municipal coupons will be untaxed whereas U.S. government bond coupons are taxed at the federal level. The spread between the bonds would then be derived from the GVE:

$$\text{Municipal yield} =$$

$$= (1-\tau_e)\,(\text{Treasury yield} + \text{Expected loss rate} + \text{Additional risk premium})$$

Note that municipal bonds, other things equal, will attract investors in the highest tax brackets. At the time of the FMS chart above, U.S. government yields were 3.99%, 4.15% and 3.95% on 10, 20 and 30-year bonds. Assuming AAA bonds have negligible risk, the effective marginal tax rate could be computed for these terms using the equation. For the 10-year bond, $\tau_e = 37.34\%$, which in 2022 would be the 20% federal marginal tax

rate with 17.34% attributable state and local taxes. This might suggest that the marginal investor is in a highly-taxed U.S. state. For the 20-year bond, τ_e = 20.48%, perhaps suggesting that the marginal investor is in a tax-free state such as Florida or Texas! For the 30-year bond, the implied state/local tax rate is negative.

Clearly, our simple model is not capturing the complexity of this valuation problem, since we are not able to easily assess risks such as changes in the tax code and tax rates over the longer term. The generally declining values of τ_e do seem to be consistent with the idea that the expected tax benefits of municipal bonds decline with time to maturity.

4.8 CORPORATE BONDS

U.S. corporate bonds outstanding totaled about $10 trillion at the end of 2021, according to SIFMA, amounting to roughly 45% of the size of the marketable U.S. government bond market. The structure of corporate fixed rate bonds resembles that of the U.S. treasury bonds, except that some corporations make coupons payments quarterly and some make coupon payments semiannually. However, unlike U.S. government debt, we realize that corporations can default on principal and interest, and creditors are not guaranteed to recover their losses.

Corporate bond pricing models range from simple to complex. A simple reduced-form framework for valuing corporate bonds was introduced by the Bank for International Settlements (BIS) which provides regulatory guidance to financial institutions and governments globally. These standardized abbreviations are well-known by bankers, and we will use the Greek letters for formulas.

- PD = π = Probability of default per year
- LGD = λ = Loss given default, the percentage of a bond's market value that is lost, as of the time of default

To make the framework more realistic, we will allow a risk premium (θ) for a corporate bond in addition to compensation for expected losses. We will also continue to use "r" as the riskless interest rate, and "k" to represent the number of coupon payments per year. If the conditional probability of default is constant from one period to the next, and the LGD is assumed constant, we can easily price corporate bonds using the tools of Chapter 2.

To make this as simple as possible, let's adapt the Excel PV function for an annuity with a final payment, =PV(rate,nper,pmt,fv). Now divide the cash flows from the corporate bond into two parts. The first part consists of payments all equal to ($1 - \lambda$) times the promised values. Since λ is the loss rate, $1 - \lambda$ is the *recovery rate*. Since this amount is assumed to be recovered, it may be considered risk-free. The remaining fraction of the promised payments λ is subject to a constant default rate of π per year. Since nothing is recovered from these cash flows in default, we can value the promised payments by discounting at $r^* = (r + \theta + \pi)/(1 - \pi)$. The resulting valuation can be stated in Excel terms as:

$$-V_0 = (1 - \lambda)PV\left(\frac{r + \theta}{k}, Nk, \frac{cF}{k}, F\right) + \lambda\, PV\left(\frac{r^*}{k}, Nk, \frac{cF}{k}, F\right)$$

Example

VFE Corp's 6% coupon 5-year bond is estimated to have an annual default rate of 3.77%. The LGD is unknown, and therefore assumed to be the standard average value of 45%. The risk-free rate of interest is 5%, and the additional risk premium historically required for VFE Corp's debt has been 3%. The bond makes quarterly coupon payments. What is the value of the bond and its YTM?

Solution

Using the formula above, we have $V_0 = 0.55(918.24) + 0.45(780.63) = 505.03 + 351.28 = \$856.32$. The YTM is found with the IRR function and the promised coupon & principal payments. In this case, YTM = 9.66%.

This can be compared to the brute force calculation done in Excel:

VFE Corp 6% coupon 5-year note

Term	5	years
Payments per year	4	
Face value	1000	
Coupon rate	6.00%	
Actual coupon	15	
Risk-free rate	5.00%	annual
	1.25%	periodic
Pi	3.77%	annual
	0.94%	periodic
Lambda	45.00%	
Theta	3.00%	annual
	0.75%	periodic
r*	2.97%	

Quick Approximation of YTM	
Risk-free (r)	5.00%
Expected loss rate ($\pi\lambda$)	1.70%
Risk premium (θ)	3.00%
Sum	9.70%

Payment	Promised	Recovered	At risk
1	15	8.25	6.75
2	15	8.25	6.75
3	15	8.25	6.75
4	15	8.25	6.75
5	15	8.25	6.75
6	15	8.25	6.75
7	15	8.25	6.75
8	15	8.25	6.75
9	15	8.25	6.75
10	15	8.25	6.75
11	15	8.25	6.75
12	15	8.25	6.75
13	15	8.25	6.75
14	15	8.25	6.75
15	15	8.25	6.75
16	15	8.25	6.75
17	15	8.25	6.75
18	15	8.25	6.75
19	15	8.25	6.75
20	1015	558.25	456.75

Discount rate (qtrly)	2.00%	2.97%
PV	\$505.03	\$351.28
Value		\$856.32
YTM		9.66%

In practice of course, one would have to be concerned with the accrued interest adjustment and the variable risk-free discount rates that were explained in the section on U.S. government bonds.

The table also shows a famous YTM approximation that is popular among bond traders and analysts to decompose the YTM of a corporate bond. There are several ways to reach this approximation. We could compute the LGD-weighted YTM of the two components of the bond valuation formula for example:

$$Approx\ YTM = (1-\lambda)(r+\theta) + \lambda\left(\frac{r+\theta+\pi}{1-\pi}\right) \approx r + \theta + \pi\lambda$$

This approximation works best when π is small. The term $\pi\lambda$ is often called the *expected loss rate*. Therefore, the total YTM of a corporate bond can be represented (approximately) as the sum of the risk-free interest rate for a government bond of similar maturity, the expected loss rate of the bond, and a bond-specific additional risk premium.

Bond traders also split the total YTM into the risk-free rate and the *credit spread*. Using this nomenclature, the credit spread is also the expected loss rate plus the bond-specific risk premium. For investment grade bonds, the expected loss rate is quite small, so the credit spread is dominated by the market risk premium term.

In practice, it is of course useful to be able to value bonds for different sets of parameters. However, we can also move backward. Suppose would like to learn about a company's implied credit quality from the market value of the bonds it has issued. In this case, we know the YTM and simply need to back out the components.

Here are quotes for the most actively traded corporate bonds as of 3/1/23 from FINRA:

Most Active Investment Grade Bonds

Issuer Name	Symbol	Coupon	Maturity	Moody's®/S&P	High	Low	Last	Change	Yield%
AMGEN INC	AMGN5542113	5.250%	03/02/2030	Baa1/BBB+	100.52300	98.82500	99.12300	-0.299000	
AMGEN INC	AMGN5542390	5.250%	03/02/2033	Baa1/BBB+	99.94768	98.45600	98.70500	-0.460000	
CIGNA GROUP	CI4866403	3.750%	07/15/2023	Baa1/A-	99.79433	99.27100	99.43200	0.110800	5.302697
AETNA INC NEW	CVS4370492	2.800%	06/15/2023	Baa2/BBB	99.40300	99.15300	99.24600	-0.093000	5.469234
PEPSICO INC	PEP4551335	3.000%	10/15/2027	A1/A+	94.05900	92.57800	92.67200	-0.387000	4.786204
AT&T INC	T4763018	4.100%	02/15/2028	Baa2/BBB	96.26000	94.31430	94.64600	-0.719700	5.344739
U S BANCORP MEDIUM TERM NTS-BOOK ENTRY	USB4792226	3.375%	02/05/2024	A2/A+	99.21100	97.99100	98.18300	0.028000	5.418111
AMGEN INC	AMGN5542392	5.650%	03/02/2053	Baa1/BBB+	98.95900	97.83100	98.40500	-0.682000	
META PLATFORMS INC	FB5522214	4.450%	08/15/2052	A1/AA-	83.19800	81.87300	82.10700	-0.544000	5.712253
COMCAST CORP NEW	CMCS5406980	2.887%	11/01/2051	/A-	64.47200	63.82800	63.82800	-0.673000	5.374975

Investment grade bonds are those higher in credit quality than other bonds, with a minimum rating of Baa/BBB on Moody's/S&P bond rating services. Bonds with lower ratings are called *high yield* bonds by marketers and *junk bonds* by investors.

Ratings are chosen by the ratings agencies to give their best possible estimate of the likelihood of default of the bond. They do not attempt to measure loss given default, but estimates of LGD can be built from models or purchased from analytic firms. This S&P default study was published in April 2022:

Table 26

Global Corporate Average Cumulative Default Rates By Rating Modifier (1981-2021)

(%)	--Time horizon (years)--														
Rating	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
AAA	0.00	0.03	0.13	0.24	0.34	0.45	0.50	0.58	0.64	0.69	0.72	0.75	0.78	0.83	0.89
AA+	0.00	0.05	0.05	0.09	0.14	0.19	0.25	0.30	0.35	0.41	0.46	0.52	0.58	0.64	0.70
AA	0.02	0.03	0.08	0.21	0.34	0.46	0.58	0.69	0.77	0.86	0.93	0.99	1.08	1.14	1.20
AA-	0.03	0.08	0.16	0.23	0.30	0.40	0.46	0.51	0.55	0.60	0.66	0.71	0.73	0.77	0.81
A+	0.05	0.08	0.18	0.30	0.39	0.48	0.58	0.68	0.80	0.93	1.04	1.17	1.31	1.47	1.62
A	0.05	0.13	0.21	0.31	0.43	0.59	0.75	0.90	1.07	1.27	1.42	1.54	1.65	1.72	1.87
A-	0.05	0.15	0.24	0.34	0.48	0.63	0.83	0.98	1.10	1.20	1.30	1.42	1.54	1.65	1.74
BBB+	0.09	0.25	0.45	0.64	0.85	1.09	1.28	1.49	1.75	1.99	2.21	2.36	2.53	2.75	2.98
BBB	0.14	0.36	0.56	0.89	1.21	1.54	1.86	2.15	2.45	2.75	3.07	3.32	3.54	3.64	3.83
BBB-	0.23	0.66	1.23	1.87	2.53	3.13	3.63	4.10	4.48	4.83	5.20	5.52	5.81	6.26	6.60
BB+	0.31	0.94	1.69	2.44	3.19	3.95	4.59	5.06	5.62	6.18	6.57	7.03	7.50	7.85	8.39
BB	0.46	1.45	2.82	4.12	5.47	6.57	7.58	8.46	9.32	10.06	10.85	11.43	11.90	12.22	12.56
BB-	0.92	2.86	4.89	6.97	8.79	10.54	12.02	13.41	14.53	15.48	16.22	16.96	17.67	18.34	18.97
B+	1.94	5.31	8.64	11.48	13.73	15.46	17.02	18.37	19.60	20.73	21.70	22.39	23.13	23.80	24.45
B	2.99	6.99	10.59	13.54	15.91	18.05	19.56	20.69	21.80	22.91	23.64	24.37	24.92	25.37	25.87
B-	5.89	12.59	17.88	21.56	24.26	26.23	27.69	28.84	29.65	30.34	31.41	32.10	32.71	33.34	33.77
CCC/C	26.55	36.74	41.80	44.74	46.91	47.95	49.08	49.82	50.48	51.05	51.49	51.92	52.45	52.91	52.97
Investment-grade	0.08	0.23	0.40	0.61	0.83	1.05	1.26	1.45	1.63	1.81	1.98	2.13	2.27	2.40	2.55
Speculative-grade	3.60	6.97	9.86	12.23	14.16	15.75	17.06	18.16	19.14	20.04	20.80	21.44	22.05	22.58	23.09
All rated	1.50	2.93	4.17	5.22	6.10	6.83	7.45	7.97	8.43	8.86	9.23	9.54	9.84	10.10	10.36

Sources: S&P Global Ratings Research and S&P Global Market Intelligence's CreditPro®.

This chart shows the cumulative default rates of global issuers measured from the first ratings date. For example, in the BBB- category (lowest investment grade), the historical one-year default rate was 0.23% or 23 bppa. For a 15-year cumulative period, the default rate was 6.60%, or 44 bppa. For investment grade bonds overall, the one-year default rate was 8 bppa, or (255/15 =) 17 bppa over a 15-year period.

Clearly, there are many assumptions one could make about default probabilities, losses given default and the risk premium, and it is impossible to get estimates of all three from a single spread observation. Nevertheless, skilled corporate bond analysts use all the means at their disposal to estimate these parameters as carefully as possible.

The bottom line: Corporate bond prices hold information about default risk and risk premia that can be useful not only for bond buyers, but for any analyst needing to understand more about the companies they are studying.

Example

Suppose we have a corporate bond that makes 10 annual payments of $50, returning $1000 in principal at the end of 10 years. The risk-free rate is 5%, but the price of this bond is $828.40, not $1000. What can we say about the implied default probability, assuming that the loss given default λ=60%, and there is no additional risk premium?

Solution

The rate that makes the present value of these payments equal to $828.40 is 7.5%. Therefore the "credit spread" is 2.5%. Using the rule of thumb, and assuming no additional risk premium, this is the same as the expected loss premium ($\pi\lambda$), and λ=60%. Therefore π=4.17%, implying an average of about 4.17% expected default rate per year. The exact valuation and yield can be computed using the exact formulae provided above.

4.9 REFERENCES AND ADDITIONAL READINGS

Brumma, Nina, Nunzia Rainone and Richard Crecel, "Global Credit Data LGD Report 2020", May 2020

Garbade, Kenneth, "Why the U.S. Treasury Began Auctioning Treasury Bills in 1929", FRBNY Economic Policy Review, July 2008.

Kraemer, Nick and Jon Palmer, 2021 Annual Global Corporate Default and Rating Transition Study, S&P Global Ratings, April 13, 2022

Rosenblum, James and Kevan Ventura, "The Demise of and Transition from LIBOR", Goldberg Kohn Ltd., January 21, 2023.

Tuckman, Bruce and Angel Serrat, **Fixed Income Securities: Tools for Today's Markets**, Wiley Finance 2022.

4.10 EXERCISES

1. A 5 year idealized U.S. government bond with a 1.75% coupon is valued at 99.81.
    a. What is its YTM (i.e. twice the semiannual yield)?
    b. If the YTM increases by 1%, what is the new price?
2. A Treasury bill with 135 days to maturity has an asking yield of 0.95%.
    a. What is the price of the T-Bill?
    b. What is the effective annual return?
3. Suppose the workers at Gideon Industries negotiated a contact that paid out $10,000,000 per year for the next five years.
    a. If the current risk-free rate is 5%, what would be the present value of a COLA (cost of living adjustment) clause in their contract for the five year period? Use the TIPS data provided in the chapter.

b. What is the fixed wage increase that would make the union indifferent between the COLA and a fixed wage structure?

2. A five-year AAA rated municipal bond yields 1.5% when a five-year government bond yield 2.0%. You have two investor clients, one in the 20% tax bracket and one in a 45% tax bracket. What would you advise them about buying this particular municipal bond compared to the U.S. government bond?

3. Download current TIPS yields and yields for U.S. government bonds. What can you infer about the
   a. Expected inflation rate
   b. Real rate of interest

4. A corporate junk bond is rated CCC by S&P and yields 14% when the yield of a comparable government bond is 2%.
   a. What can you infer about loss given default?
   b. If the additional risk premium is 5%, what can you infer about LGD?

MINIPROJECT 4: THE INFLATION STORY

Following Miniproject 3, we would like to build a ZCB term structure for inflation-indexed bonds. Download TIPS prices and Treasury bond/notes quotes to build a simultaneous nominal ZCB term structure, and real ZCB term structure.

This project should be completed in Python, and you should use APIs to download pricing data directly from the Wall Street Journal or other source. Once completed, you may use your model to build nominal and real term structures for any date, at any time in the future.

The outputs are the same as Miniproject 3. Your deliverable is a Jupyter Notebook file.

Chapter 5: Simulation

INTRODUCTION

As we transition from known cash flows to unknown cash flows in Part II, simulation increases in importance. As long as *laws of motion* can be prescribed for financial variables of interest, we can simulate possible movements in those variables for the purpose of determining expected cash flows, risk and ultimately, valuation. For example, we may want to assume that a security's future returns are normally distributed, or that corn prices follow a mean-reverting process on average, but with volatility.

Simulation provides one way to value assets, and in many cases, other methods can be used. These methods may include any of the following

- Closed-form solutions
- Binomial and trinomial trees
- Numerical integration
- Numerical solution of partial differential equations

All of these methods can be faster and more accurate than simulation, but unfortunately, each of these methods requires restrictions on the model complexity. Simulation is often the slowest and the least accurate method, but it is by far the most flexible, particularly when there are many stochastic variables involved – e.g. interest rates, exchange rates, commodity prices, event risks, or anything else that one might use to simulate corporate income. However, with advances in computing software and hardware, especially cloud computing, simulation can become much more accurate at a fraction of its previous cost, making it a viable contender as a valuation method.

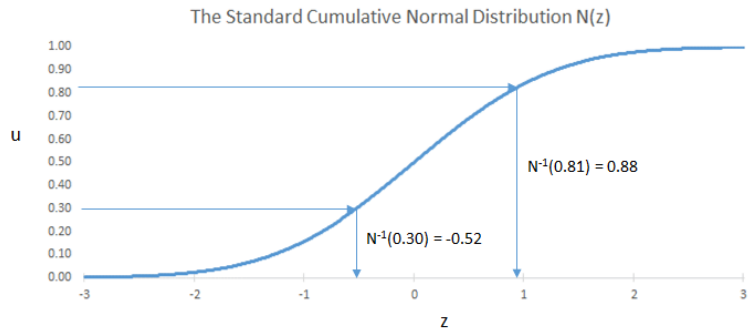
Simulation requires that we generate a large number of random outcomes of our model, and estimate the result of the model using the average result of our outcomes. We first generate random numbers, then that we compute the average outcome, and find some measure of the reliability of the result. This is the core of simulation, the rest is detail.

5.1 RANDOM NUMBER GENERATION

Of course, we cannot generate numbers that are truly random. When we say we are "generating random numbers", we really mean that we want to generate a series of numbers which have the key *properties* of random numbers. For example, the generated numbers should follow a prescribed probability distribution. Also, the numbers should be statistically independent of each other, meaning that I shouldn't be able to determine the elements of the sequence using some predictive model.

Numbers that meet these properties are called *pseudorandom numbers*, but we still call them random numbers for short. Entire books have been written about the creation of pseudorandom numbers, see for example von Neumann (1951) for an early example. The importance of these numbers extends deeply into the fields of computer science, cryptography and computer security. In finance, we use pseudorandom numbers to generate price processes, default states, and the occurrence of all sorts of unpredictable events. For many applications, we require uniform and/or normally distributed random numbers, all independent of each other.

Suppose for example that we wish to generate a large number of normally distributed variables (z) with the standardized mean of 0 and standard deviation of 1. We know the probability density function or PDF, n(z), and also the cumulative density function or CDF, N(z) for values of z. Although z can take any value on the real line, the most common values range from -3 to 3, i.e. within 3 standard deviations of the mean. The graph below shows the CDF of N(z):



Notice that the range of N(z) is (0,1). To generate random values of z, therefore, it is easiest to generate pseudorandom numbers (u) that are uniformly distributed on the interval (0,1), and map them to the corresponding values of z. For example, if a uniform random number generator provided 0.81 as a value, we could map it to z by computing the inverse of the CDF at 0.81, obtaining the z value of 0.88. Similarly, 0.30 would map to -0.52.

There are many techniques to generate values of u. The only real challenge with this method is that the normal CDF is not analytic. However, this can easily be addressed in Excel or Python. In Excel, uniformly distributed random numbers can be generated with the function *u=rand()*. The inverse standard normal CDF in Excel is given by $N^{-1}(u)$ = *normsinv(u)*. Therefore, the single line *=normsinv(rand())* will generate a standard normally distributed random number. When the formula is copied, each cell provides a new random number, allowing us to generate as many random numbers as we like. A similar function is available in Python: numpy.random.normal(mean, stdev, size).

Important: This technique works for all CDFs. If you would like to generate random number of any standard distribution, you can do so as long as you have some way to compute the inverse CDF and some way to generate uniform (0,1) random numbers. In some cases, this leads to a simple numerical result, such as the exponential distribution, which we will see later in this chapter.

Exercise

Using the uniform random numbers 0.36 and 0.98, generate two random numbers from a Chi-squared distribution with 1 degree of freedom using Excel functions.

Solution

Using the inverse Chi-squared CDF function, this is computed as =chisq.inv(u,1), resulting in 0.22 and 5.41.

The underlying core logic of simulation is simple. These steps are common to all simulations. Let's say we are going to simulate a large number of values of z, and we want to know the expected value of f(z), a function of z. We also want to define our maximum likely error based on a 2 standard deviation range (though we might like to use 3 or more standard deviations for critical applications)

Core logic of simulation

1. Generate a large number N of random numbers z
2. Compute outcome values f(z)
3. Compute the average value $\bar{X}$ of the f(z) results --- this is your estimate of E[f(z)]
4. Compute the sample standard deviation of the values of f(z). Let's call this "s"
5. Compute the *standard error*, which is the standard deviation divided by $\sqrt{N}$, i.e. $s/\sqrt{N}$
6. Compute the maximum likely error, which is 2 times the standard error = $2s/\sqrt{N}$
7. If the maximum likely error is too large, increase the size of N in proportion to the square of the desired increase in precision.
8. The solution is then given by $E[f(z)] = \bar{X} \pm 2s/\sqrt{N}$

The only tricky step in the process is to remember to compute the standard error by dividing by the square root of N. We use the words standard *error* to remind ourselves that it is the standard *deviation* of the measure of the average.

Note that the variance of a sum of N independent identically distributed random variables is N times the individual variance. Also, the variance of (sum/N) is the variance of the sum divided by N^2. Accordingly, the variance of the average is the variance of the underlying data divided by N, and the standard deviation is the square root of that. Furthermore, as N increases, the distribution of $\bar{X}$ asymptotically approaches a normal distribution has a mean equal to the true value of the population mean, and a *standard error* approaching $\sigma/\sqrt{N}$. This is known as the *Central Limit Theorem*.

A detailed example

Let's suppose we want to compute the expected value of max(z,c), where z is the standard normally distributed random variable and c is a constant. In this case, we actually know the formula for this, so we can check the result when we are finished:

$$E[\max(z, c)] = n(c) + c\, N(c)$$

For example, when c = 0.5, the result is 0.6978.

To begin, let's generate 1000 values of z in Excel by copying the formula "*=normsinv(rand())*" into 1000 cells in a column. Then for each cell, we compute the value of max(z,c) with a given value of c of 0.50. Finally, we compute the mean, standard deviation, standard error and maximum likely error. Here is an illustration of how your spreadsheet might look (only showing the first 10 simulations):

First 10 simulations shown of 1000

c		0.5
	z	max(z,c)
1	-0.45	0.50
2	-1.84	0.50
3	-0.31	0.50
4	1.54	1.54
5	0.02	0.50
6	-0.66	0.50
7	0.30	0.50
8	-0.26	0.50
9	-0.58	0.50
10	0.50	0.50

Statistics for 1000 simulations

	z	max(z,c)
Average	-0.0373	0.6874
StDev	1.0006	0.3911
n	1000	1000
StErr	0.0316	0.0124
#StErr	2	2
MaxErr	0.0633	0.0247
Min		0.5000
Max		2.9665
True value		*0.6978*

Our estimate of the value of E[max(z,0.5)] is 0.6874 ± 0.0247. The true computed value of 0.6978 falls in the lower end of that range. If we need to reduce the size of the maximum likely error by a factor of 30, we can multiply the number of simulations by 900 or 30^2 for a total of 900,000 simulations.

You may have noticed that the randomized z values have a mean of -0.0373 and a standard deviation of 1.0006, not the prescribed values of 0 and 1. Of course, if we are simulating any set of random numbers, we will get values of the mean and standard deviations changing with every simulation, i.e. every recalculation of the spreadsheet. We will see in the next section how we can improve on the simulated values of z to get more accurate simulation results.

Exercise

Before proceeding to the next section, replicate this example in Excel, but adjust the random numbers so that the mean is zero and the standard deviation is 1. Does the adjusted set of random numbers seem to provide a more accurate result?

5.3 VARIANCE REDUCTION TECHNIQUES

Much has been written about methods to increase simulation accuracy or computation speed, i.e. the reduction in the sample variance of f(z) in the previous example. In this section, we will discuss two of them, known as *moment matching* and *antithetic sampling*. Please refer to Botev and Ritter (2017) for a comprehensive treatment of these and other techniques. *Control variate* techniques can also be used to reduce variance if the variables correlate with the simulation results.

One simple and effective way to reduce sampling variance is to force the first two moments, i.e. the mean and standard deviation of the random sample of z values to be equal to their theoretical values, i.e. 0 and 1. This can be accomplished by replacing each simulated value of z with its standardized equivalent, that is,

$$z \leftarrow \frac{z - \bar{z}}{s_z}$$

where we subtract the sample mean from each value of z, and divide by the sample standard deviation. This technique, *moment matching*, eliminates two sources of uncertainty from the random sample and can have a positive impact on simulation performance.

Antithetic sampling means that each time we choose a uniformly distributed random number, we include the value of (1-u) as a separate random number. Notice that for the normal distribution, this has the effect of forcing the mean of the sample random numbers to be zero, but does not affect the variance, since z and –z are included in every simulation. However, the antithetic sample creates negative correlations between observations in the dataset, allowing for some variance reduction. Antithetic sampling used to be considered beneficial because the computation time for generating an antithetic number was less than the time of creating a new pseudorandom number. Nowadays, this benefit is not as highly valued due to rapid advances in computing speed and parallel processing.

To test the impact of moment matching and antithetic sampling, we will now perform 1000 iterations of our simulation of 1000 values, for a total of 1,000,000 values. To judge the efficiency of the variance reduction, we will compute the sample standard deviation of the results of every 1000 sims and then compile the results in a two stage process. In Excel, therefore, we will use our original setup to simulate 1000 outcomes, and then use a *what-if table* to generate 1000 results of the 1000 individual simulations. The results are as follows:

							Basic	Moment Matching	Antithetic
Mean	-0.0233	0.0000	0.0000	0.6911	0.6986	0.6926	0.6974	0.6977	0.6970
StDev	0.9992	1.0000	0.9806	0.3956	0.4039	0.3955	0.0127	0.0035	0.0116
StErr	0.0316	0.0316	0.0310	0.0125	0.0128	0.0125	0.0004	0.0001	0.0004

	Single simulaton of 1000 outcomes						*What-if table of 1000 simulations*			
	z	moment	antith	f(z)	moment	antith	means			
1	0.93	0.96	0.93	0.93	0.96	0.93	1	0.69	0.70	0.68
2	-2.45	-2.43	-2.45	0.50	0.50	0.50	2	0.70	0.70	0.68
3	-1.16	-1.14	-1.16	0.50	0.50	0.50	3	0.68	0.70	0.67
4	0.53	0.55	0.53	0.53	0.55	0.53	4	0.69	0.70	0.68
5	1.35	1.38	1.35	1.35	1.38	1.35	5	0.68	0.69	0.69
6	-0.46	-0.44	-0.46	0.50	0.50	0.50	6	0.72	0.70	0.69
7	-0.98	-0.96	-0.98	0.50	0.50	0.50	7	0.69	0.70	0.70
8	-1.04	-1.01	-1.04	0.50	0.50	0.50	8	0.69	0.70	0.67
9	0.40	0.42	0.40	0.50	0.50	0.50	9	0.69	0.70	0.68
10	1.07	1.09	1.07	1.07	1.09	1.07	10	0.71	0.70	0.69

The first three columns contain the raw z-values, the z values with corrected mean (0) and standard deviation (1), and the antithetic values; entries 501-1000 are the negative values of 1-500. The next 3 columns show the values of f(z) for the three methods. The final table on the right tabulates 1000 simulations of the results of the single simulation done in the spreadsheet. For each column, the statistics are computed on the top.

Using the basic technique, or the z column, the standard deviations of the 1000-sim values was 0.0127, which as expected, were lower by a factor of about sqrt(1000) from the original f(z) simulation result of 0.3956. Remarkably, the moment matching technique reduced the standard error by a factor of more than 3.5 times. The antithetic technique did not have much success with this particular problem, though in other problems it may be much more successful. In all cases, the large number of final simulations leads to good results for all three techniques.

Perhaps needless to say, different variance reduction techniques work better with different simulations, but the results can be significant.

5.4 CONTROL VARIATE TECHNIQUE

In some situations, we may be able to condition the simulated variable (let's say "Y") on values of an explanatory vector of other observed variables "X", one of which may be a constant. That is, with a constant, all the first column elements of the X matrix equal "1". Furthermore, let's assume the relationship between Y and X is linear for the time being, so Y = bX, and b is a coefficient vector derived from a linear regression.

In this situation, if "b" were known with certainty, we would expect that the variance of Y would drop to the variance of the residuals. But in most practical cases, b is estimated, so we must also consider the variance in those estimates. For any given vector X, let's call it X_h, then the *variance of the mean response* conditional on X_h is given by:

$$Var(X_h'b) = s^2 X_h'(X'X)^{-1}X_h$$

where "s" is the residual standard deviation from the regression. The further X_h deviates from its mean, the greater the mean response variance. In some cases, we know the theoretical value of the expected value of X, in which case, we can set this vector equal to X_h to minimize the mean response variance.

Example

For the sake of demonstrating the calculation, let's assume that the X variable is a vector of the numbers 1-10. The Y values are secretly equal to X^2, but we are pretending we don't know this and are using X as a control variate to predict the theoretical mean E[Y|X]. The calculations in Excel are shown below:

Y	1	X	E[Y|X]	MR std
1	1	1	-11	4.77
4	1	2	0	4.05
9	1	3	11	3.41
16	1	4	22	2.90
25	1	5	33	2.61
36	1	6	44	2.61
49	1	7	55	2.90
64	1	8	66	3.41
81	1	9	77	4.05
100	1	10	88	4.77

Regression statistics	
a	-22
b	11
s	8.12
sy	10.81
X'X	
10	55
55	385
Inv(X'X)	
0.4667	-0.0667
-0.0667	0.0121

The calculation demonstrates that the mean response (MR) standard deviations are indeed lowest near the mean value of X (5.5), and increase as one moves away from the mean. More importantly for our purposes, for values near the mean, the mean response standard deviation is much smaller than the standard error of the Y values without regressing on X (10.81). In this case, the control variate technique is likely to reduce error standard deviations by more than 50%.

We will use regression in Part II in our effort to value assets by benchmarking them to others. In these cases, we will compute the error variance from the mean response variance rather than the unconditional variation of the simulated asset values.

5.5 SIMULATION AND THE DEFAULTABLE BOND VALUE

This section provides a second simulation example, using the defaultable bond valuation from Chapter 4. In this case, we will actually use simulation to do a valuation, and we can check the results with the pricing from Chapter 4. Also, we will work the geometric distribution to simulate the bond's default date and therefore its value conditional on the default date.

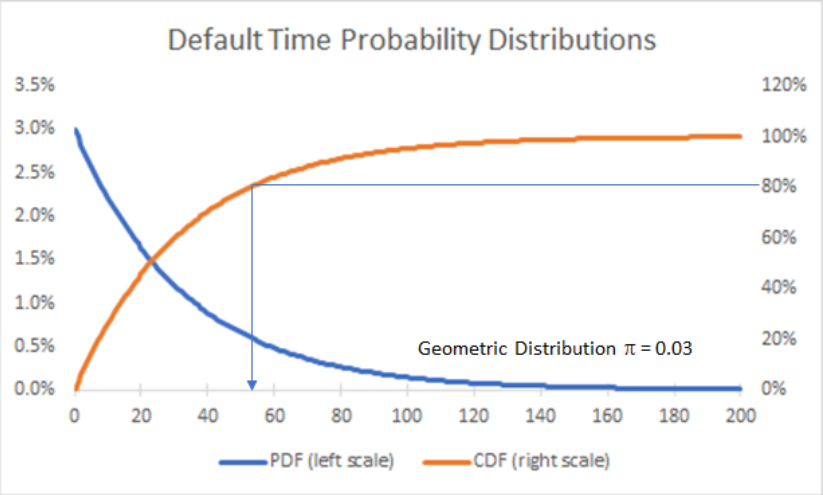
To simulate the bond's value, we take the following steps, adapted for this problem:

1. Determine the bond's value conditional on defaulting at time t for each t.
2. Simulate default dates, and assign a bond value to each simulation using step 1.
3. Calculate the average of the values as an estimate of the bond value
4. Calculate the maximum error of the simulation

Step 1. We already know the conditional bond value based on the default dates, from our earlier spreadsheet analysis:

Date	PV
1	426.71
2	458.44
3	488.95
4	518.29
5	546.50
over	1,066.78

Step 2. To simulate dates, examine the following chart. The geometrically distributed default dates can be converted to a uniformly distributed random number on [0,1] when the cumulative density is computed.



For example, the value of 0.8 maps to a default time of 53 periods. We can generate geometrically distributed random numbers from uniformly distributed random numbers. Letting $u=1-(1-\pi)^t$, we can solve for t as a function of u:

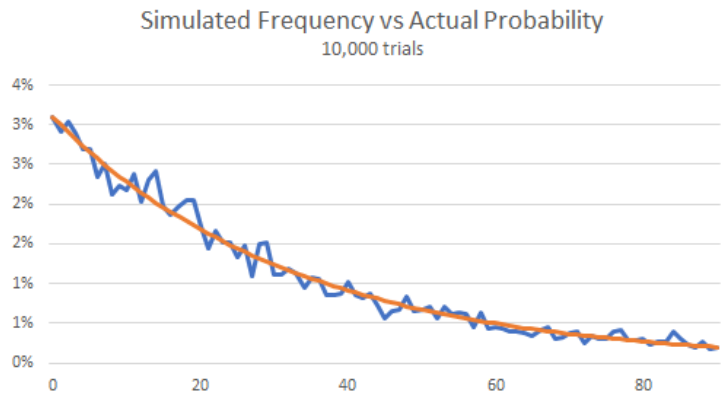
$$t = \frac{\ln(1-u)}{\ln(1-\pi)}$$

For our purposes, we want to generate discrete random times, so we have to round up. In Excel, we can use the INT function or the ROUNDUP function:

$$t = INT\left(\frac{\ln(1-u)}{\ln(1-\pi)}\right) + 1$$

$$t = ROUNDUP\left(\frac{\ln(1-u)}{\ln(1-\pi)}, 0\right)$$

Step 3. We generated 10,000 equally likely possible default dates using this formula. The empirical frequency distribution is shown alongside the theoretical distribution in the following chart:



Here's a small sample of 25 of the 10,000 sims, with the matching conditional bond values on the right:

Simulations				
95	88	10	29	15
23	4	28	14	29
6	47	19	2	145
16	109	69	39	1
5	37	12	128	45

Valuations				
1,067	1,067	1,067	1,067	1,067
1,067	518	1,067	1,067	1,067
1,067	1,067	1,067	458	1,067
1,067	1,067	1,067	1,067	427
546	1,067	1,067	1,067	1,067

The values should be 1,067 about 85% of the time. The valuations were matched using the VLOOKUP function, and the cells with defaults before 5 years were shaded using conditional formatting. Excel note: Be sure to replace the word "over" with the number 6 in the lookup table to guarantee the right VLOOKUP results.

Step 4. We already know the computed bond value is $984.73, and the standard deviation is $202.92. Make sure you know how to compute the standard deviation on your own. When we consider the results of 10,000 simulations, we will all get different but similar results. In one case, the results were:

Average	987.77
StDev	200.04
Count	10,000
StErr	2.00

The standard deviation is calculated using the STDEV function. AVERAGE and COUNT are also Excel functions.

Since this is a simulation, our estimate of $987.77 has a natural error range. To compute the maximum likely error range, recall that the distribution of a sample average of values from any distribution approaches a normal distribution. The simulated standard deviation was quite close to the theoretical value of 202.92. The sample standard error was 2.03, i.e. $202.92/\sqrt{10{,}000}$. Conservatively then, we could go 3 standard deviations around our estimate to get a range where the true value should lie:

$$\text{Estimate range} = 987.77 \pm 3(2.03) = \{981.68, 993.86\}$$

Of course, the value 984.73 falls comfortably within that range. To increase our accuracy by any factor A, the sample size must be multiplied by A^2.

Example: Suppose a simulation of a defaultable bond value yielded an estimate of $1040 for the bond. The standard deviation of the simulations was $35, and 1,000 simulations performed. What is the 3 standard deviation confidence estimate of the true bond value? {1036.68, 1043.32}

5.6 MULTIVARIATE SIMULATIONS

In most problems of practical interest, we will want to simulate many different random variables simultaneously. If the variables were statistically independent, we could simply simulate them separately, but of course, there is bound to be some kind of interaction required between the variables. Most frequently, we expect random variables to have specified correlations. Since the (Pearson) correlation is the measure of the strength of a linear relationship applied to jointly normally distributed random variables, it makes sense to begin there.

Suppose we wish to simulate two standardized normal random numbers z_1 and z_2 with a given correlation ρ. One method is to begin with two independent random numbers z_1 and z_3. We can keep z_1 and then compute z_2 as a linear combination of z_1 and z_3 that meets these requirements:

- The mean of z_2 must be 0 (this is guaranteed)
- The variance of z_2 must be 1
- The correlation between z_1 and z_2 must be ρ

These requirements are uniquely met by the equation

$$z_2 = \rho z_1 + z_3\sqrt{1-\rho^2}$$

To implement this, we could do so one line at a time in the data. However, it may be more convenient to express these equations in matrix form, especially when we have more than two variables:

$$(z_1 \quad z_2) = (z_1 \quad z_3)\begin{pmatrix} 1 & \rho \\ 0 & \sqrt{1-\rho^2} \end{pmatrix}$$

We can think of $(z_1 \quad z_3)$ as the uncorrelated data, and $(z_1 \quad z_2)$ as the correlated result. The 2x2 matrix is known as the transpose of the (lower diagonal) Cholesky decomposition of the correlation matrix $\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$. In fact, if you matrix multiply the Cholesky decomposition by its transpose, you will obtain the correlation matrix as the result.

Note that for standardized z values, the correlation matrix is the same as the covariance matrix. When nonstandard mean and variance are required, the covariance matrix may be used instead of the correlation matrix

to account for different variance levels for the k variables, and the means are simply added to the result. The transposed Cholesky decomposition of the covariance matrix above is $\begin{pmatrix} \sigma_1 & \rho\sigma_2 \\ 0 & \sigma_2\sqrt{1-\rho^2} \end{pmatrix}$.

When simulating more than two correlated normal random numbers, the Cholesky decomposition is invaluable. In this case, let's suppose we have k random variables with correlation matrix R, and we want to simulate N sets of random variables. We first produce an Nxk matrix of uncorrelated random z values. We then post-multiply the data by the transpose of the Cholesky decomposition of R. The resulting matrix of random numbers has the desired correlation matrix.

The appendix covers the construction of a Cholesky decomposition, moment matching to "purify" the sample data, and the Cholesky transformation to create data with the specified covariance matrix.

5.7 COPULAE AND THE CORRELATION OF NON-NORMAL RANDOM VARIABLES

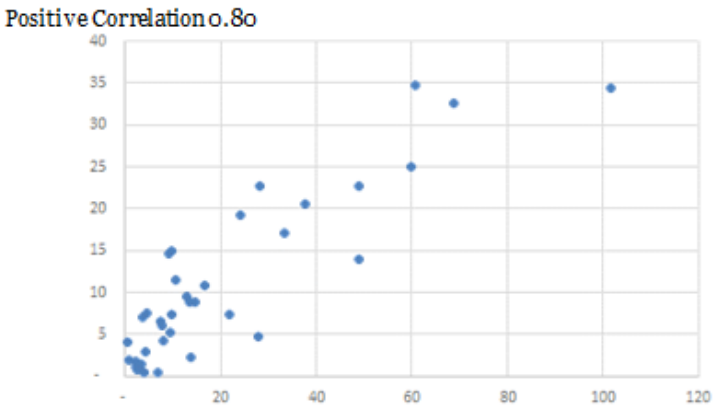
In the last section, we showed how to convert simulated normal independent random numbers into correlated random numbers. Technically, this approach works only normally distributed variables, so the question naturally arises, "how do we deal with non-normal variables?"

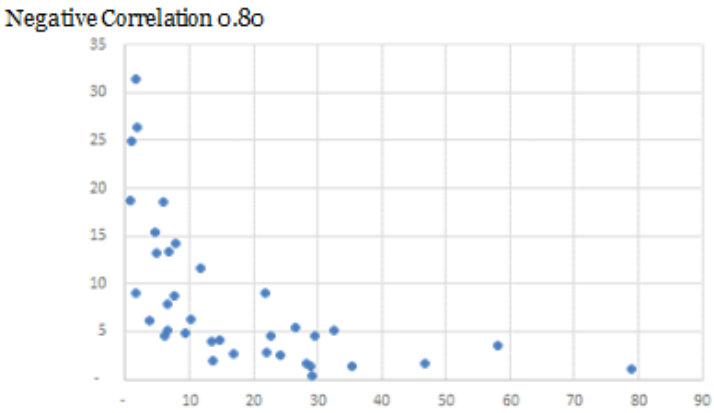
To illustrate our approach, consider the geometrically distributed default dates for the bonds discussed earlier in this chapter. Suppose we have two bonds. Bond A has a default rate of 5% per year, and Bond B has a default rate of 10% per year. We would like to introduce systemic risk to the analysis of our bond portfolio, so we will assume an 80% correlation in the default dates. What does that mean from a computational perspective?

We can generate normally distributed random numbers with a correlation of 80% easily using the Cholesky method described in the last section. To convert these to geometrically distributed default dates, we may first convert them to uniform numbers by computing the normal CDF, and then convert them to geometric distributions using the inverse CDF of the geometric distribution. This is an example of a Gaussian copula, the calculations for which are shown below.

The first two columns are random z values with a correlation of 0.80. The second two columns are the CDFs or uniform equivalents of those values. The third set of two columns are the transformed default date simulations using the inverse function of the geometric CDF.

Correlated Normal		Uniform		Simulated Defaults	
	80%			5%	10%
0.19	(0.23)	0.58	0.41	17	5
(1.29)	(2.00)	0.10	0.02	2	0
0.70	0.94	0.76	0.83	28	17
0.53	0.32	0.70	0.63	24	9
1.18	0.75	0.88	0.77	42	14
0.29	0.58	0.62	0.72	19	12
2.59	2.78	1.00	1.00	104	56
1.03	0.64	0.85	0.74	37	13
1.12	1.59	0.87	0.94	40	27
0.89	(0.19)	0.81	0.42	33	5
(1.37)	(0.73)	0.09	0.23	2	3
(1.55)	(1.10)	0.06	0.14	1	1
0.11	0.71	0.55	0.76	15	14
0.15	(0.51)	0.56	0.30	16	3
(0.65)	(0.85)	0.26	0.20	6	2
(2.37)	(2.50)	0.01	0.01	0	0
(1.02)	(0.60)	0.15	0.27	3	3
(1.74)	(0.60)	0.04	0.28	1	3
(1.01)	(0.06)	0.16	0.48	3	6
(0.86)	(0.37)	0.20	0.36	4	4
0.77	0.39	0.78	0.65	29	10
0.71	0.34	0.76	0.63	28	10
(0.71)	(1.13)	0.24	0.13	5	1
(0.44)	(0.47)	0.33	0.32	8	4
2.34	1.62	0.99	0.95	90	28
0.39	0.22	0.65	0.59	21	8
0.41	0.12	0.66	0.55	21	8
0.67	0.37	0.75	0.64	27	10
(0.56)	1.24	0.29	0.89	7	21
0.20	0.14	0.58	0.55	17	8
1.23	0.87	0.89	0.81	43	16
(0.62)	(1.23)	0.27	0.11	6	1
0.06	(0.37)	0.52	0.36	14	4
0.49	(0.10)	0.69	0.46	23	6
2.29	0.75	0.99	0.77	88	14
1.47	1.24	0.93	0.89	52	21





The graphs illustrate the impact of correlation in defaults on the timing of joint default. As expected, when defaults are positively correlated, we see many more outcomes where both defaults occur within 10 years of the starting date, as compared to the simulations with the negative correlation. The copula can be used to capture the essence of correlation while remaining true to the geometric distribution model of bond default dates.

5.8 SIMULATION AND OPTIMIZATION

In some cases, we may be required to optimize the expected value of some function of one or more random variables. If the function is complex or the number of random variables is large, it may be necessary to find the optimal value(s) of the choice variable(s) using simulation.

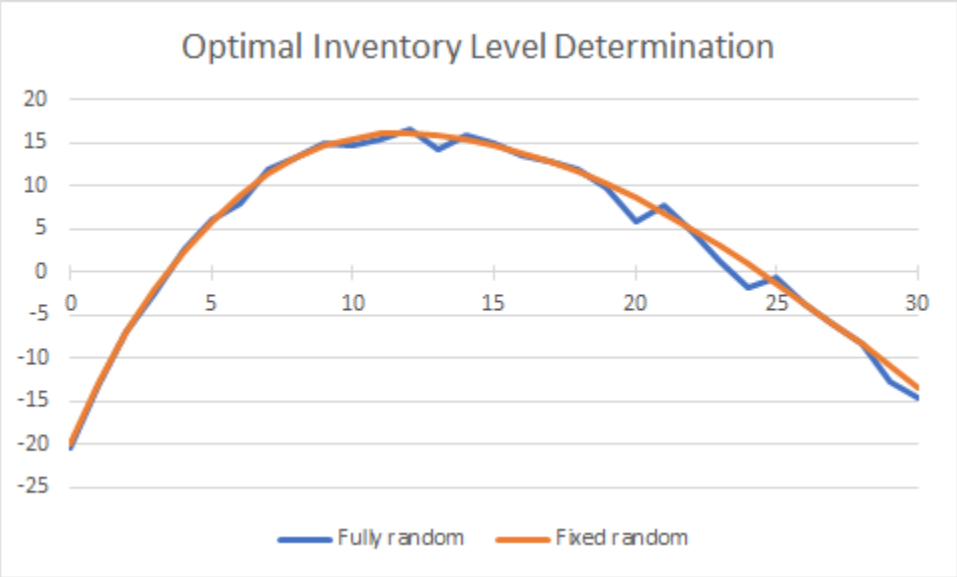
For example, consider a simplified inventory management problem. Assuming a one-period problem, we purchase or stock "s" units of a product, but we do not know the demand "d". The demand level is given by a geometric distribution with parameter 0.10, so expected demand is 10 units. The actual number of units sold is the minimum of s and d. For each unit we sell, we obtain a gross profit of $5.00. Let's suppose there is a cost of overstocking of $3.00 per unit, and a cost of understocking of $7.00 per unit. Overstocking leads to return or storage costs, while understocking leads to disappointed customers. Overstocked quantity is given by max(s-d,0) and understocked quantity is given by max(d-s,0).

Therefore net profit is given by the equation

$$Net\ profit = 5\min(s,d) - 3\max(s-d,0) - 7\max(d-s,0)$$

Our objective is to choose the stock level to maximize expected net profit. We could go with s = E(d) = 10, but since the cost of overstocking is lower than the cost of understocking, it would be better to err on the side of having s > 10. But how much greater? With the geometric distribution, the probability of higher demand levels is decreasing. Also we have the problem of optimizing a discrete choice, so calculus is unlikely to provide an easy answer. Therefore we turn to simulation.

The obvious approach seems to estimate expected profits for several different values of s, and then choose the one that gives the highest value. However, because of the natural errors in the simulation process, the result might appear like the choppy curve in the diagram below. While it does not cause much harm in this case, this simulation error could cause serious problems in other applications. Therefore, we need a better method.



Fortunately, another variance reduction method can be employed. If we fix the random numbers for our experiment, and then calculate expected profit for each given level of s, we obtain the smooth line is the chart, which is easily optimized at s = 12.

This is known as the method of *common random numbers*, an example of which appears in Hamrick (2016). In general, the technique of common random numbers requires that the same set of random numbers be used in comparisons when an inference will be drawn about the *change* in the simulation result. One weakness of this approach is that it uses fewer random numbers than the fully random method, so any analysis you do of this type should be checked with multiple sets of fixed random numbers, each of which will provide a specific optimal solution. For example, if we had a "bad" set of random numbers in our optimization above, we may have arrived at the incorrect optimal value, having biased all the profitability estimates in the same way.

5.9 SUMMARY

From this chapter, you should have learned the following.

1. Many pseudorandom number generation techniques exist to produce statistically independent uniformly distributed random numbers.
2. Uniformly distributed random numbers (u) can be converted to any other distribution by computing the inverse CDF of the target distribution for each value of u.

3. Simulation is used to estimate the expected value of an arbitrary function of an arbitrary number of random variables.
4. Simulation's core logic requires the computation of an average outcome, the standard error of the average outcome, and the maximum likely error based on a number of standard deviations needed to produce the desired confidence interval. $E[f(z)] = \bar{X} \pm 2s/\sqrt{N}$
5. Simulation results can be improved with variance reduction techniques. The most notable and easiest to implement are the moment matching technique and the antithetic variable technique.
6. When available, control variate techniques can be used when predictor variables X explain some of the variance of the prediction Y. In this case, the confidence intervals conditioned on X are given by a Mean Response variance.
7. Multivariate normal simulations are achieved by simulating independent normal variables and post-multiplying the dataset by the transpose of the Cholesky decomposition of the desired covariance matrix to obtain simulated data with the correct covariance matrix.
8. For non-normal variables, the copula can be used as a linkage between two random variables. Correlated normal variables lead to correlated uniform CDF variables, which can be converted to any distribution using #2 above.
9. In optimization problems involving simulation, it is normally best to use fixed (or common) random numbers as a variance reduction technique.
10. (Appendix) The Cholesky decomposition can be computed in Excel or called as a function in Python.

5.10 REFERENCES AND FURTHER READING

Botev, Z.; Ridder, A. (2017). "Variance Reduction". Wiley StatsRef: Statistics Reference Online: 1–6. doi:10.1002/9781118445112.stat07975.

Haugh, Martin, (2017) "Simulation Efficiency and an Introduction to Variance Reduction Methods", IEOR #4703 Columbia university

Higham, Nicholas (2008), Cholesky Factorization, MIMS EPrint 2008.116, http://www.math.drexel.edu/~tolya/Higham_Cholesky.pdf

Hamrick, Jeff. "The Method of Common Random Numbers: An Example". Wolfram Demonstrations Project. Retrieved 29 March 2016.

von Neumann J., "Various techniques used in connection with random digits," in A.S. Householder, G.E. Forsythe, and H.H. Germond, eds., Monte Carlo Method, National Bureau of Standards Applied Mathematics Series, 12 (Washington, D.C.: U.S. Government Printing Office, 1951): 36-38.

5.11 EXERCISES TO BE COMPLETED IN EXCEL OR PYTHON

A. Calculate 1000 values of the random variable z, a standard normally distributed random number. Find the expected value range of z^4 within two standard deviations. Does your range include the correct value? (Hint: What is the kurtosis of the normal distribution?)

B. Repeat A, but calibrating the normally distributed numbers so that the mean is zero and the standard deviation is 1. How does your result compare to A?

C. For a given correlation matrix of three standard normally distributed random variables, estimate the expected maximum value of the set of three variables. (Hint: Requires computation of a Cholesky decomposition)

D. Design a spreadsheet to produce 100 outcomes of two dice. The dice are independent, and the probabilities of the outcomes {1, 2, … 6} are equally likely.

E. Repeat D for a pair of dice with a "correlation" of 0.50 between the two outcomes. Show the simulated frequencies of the uncorrelated dice pairs from part D with the correlated dice pairs in part E.

F. For the inventory management problem described in the optimization section, show how the optimal stock level depends on gross profit, overstock costs and understock costs.

Appendix: Computing the Cholesky decomposition of a covariance matrix

This appendix shows a worked example of how to compute a Cholesky decomposition for a 5x5 matrix, and how to create a random dataset with the properties of a given covariance matrix. In Python, use numpy.linalg.cholesky(A) or scipy.linalg.cholesky(A). The scipy function returns the upper triangular decomposition by default, and the numpy function returns the lower triangular decomposition.

Here is an algorithm you can implement in Excel, or another programming language from Higham (2008):

The Cholesky factorization can be computed by a form of Gaussian elimination that takes advantage of the symmetry and definiteness. Equating (i, j) elements in the equation $A = R^T R$ gives

$$j = i: \quad a_{ii} = \sum_{k=1}^{i} r_{ki}^2,$$

$$j > i: \quad a_{ij} = \sum_{k=1}^{i} r_{ki} r_{kj}.$$

These equations can be solved to yield R a column at a time, according to the following algorithm:

for $j = 1{:}\,n$
  for $i = 1{:}\,j-1$
    $r_{ij} = (a_{ij} - \sum_{k=1}^{i-1} r_{ki} r_{kj})/r_{ii}$
  end
  $r_{jj} = (a_{jj} - \sum_{k=1}^{j-1} r_{kj}^2)^{1/2}$
end

Here is a numerical example:

Covariance matrix

100	80	70	60	50
80	144	80	70	60
70	80	196	80	70
60	70	80	256	80
50	60	70	80	324

Step 1: First column

10	- Square root of the first element
8	- Each element of this column is divided by 10
7	
6	
5	

Step 2: Second column

10	0	0	0	0
8	8.94	- Sum of squares of this row must be 144		
7	2.68	= (80-8x7)/8.94		
6	2.46	= (70-8x6)/8.94		
5	2.24	= (60-8x5)/8.94		

Step 3: Third column

10	0	0	0	0
8	8.94	0	0	0
7	2.68	11.82	- Sum of squares of this row must be 196	
6	2.46	2.66	= (80-7x6-2.68x2.46)/11.82	
5	2.24	2.45	=(70-7x5-2.68x2.24)/11.82	

Step 4: Fourth column

10	0	0	0	
8	8.94	0	0	
7	2.68	11.82	0	
6	2.46	2.66	14.38	- Sumsq = 256
5	2.24	2.45	2.64	=(80-6x5-2.46x2.24-2.66x2.45)/14.38

Step 5: Last row sum of squares must be 324

10	0	0	0	0
8	8.94	0	0	0
7	2.68	11.82	0	0
6	2.46	2.66	14.38	0
5	2.24	2.45	2.64	16.76

Step 6: Multiply by transpose to recover original matrix

100	80	70	60	50
80	144	80	70	60
70	80	196	80	70
60	70	80	256	80
50	60	70	80	324

Now we wish to simulate data with the desired covariance matrix. Begin by generating a 20x5 dataset of "independent" z values. Of course, the correlations will not be zero because this is a sample. Nevertheless, for illustration purposes, we'd like to know how to "purify" the data so that the correlation matrix is the identity matrix.

This is accomplished by post-multiplying the data matrix by the inverse of the Cholesky decomposition of the sample covariance matrix, essentially undoing what a normal moment matching exercise would do. The results are shown below.

Random "independent" data

	A	B	C	D	E
1	1.21	-1.72	0.85	-1.52	0.89
2	0.09	1.89	-0.22	-0.71	0.39
3	0.43	0.97	-0.42	0.41	0.55
4	-0.18	-0.51	1.84	-1.02	-0.39
5	-0.25	0.49	-0.41	-0.49	-1.23
6	-0.05	0.03	0.56	-0.08	-0.19
7	0.11	0.24	0.61	0.62	-0.35
8	0.32	-0.01	0.19	-1.93	-1.99
9	1.45	0.34	-0.55	0.44	-0.15
10	-0.20	0.13	1.52	0.34	2.13
11	1.05	1.08	-0.35	-0.32	2.17
12	-0.17	-0.06	-1.35	1.83	-0.36
13	0.51	0.37	-0.73	1.98	0.23
14	-0.51	0.82	1.34	2.98	1.49
15	1.17	1.92	-0.24	-0.22	-0.51
16	0.76	0.38	0.97	1.51	0.22
17	-0.64	0.92	0.35	-0.28	1.46
18	-0.57	0.00	-0.44	-1.35	-0.10
19	-1.69	0.39	-0.45	-1.17	1.53
20	-1.60	0.37	1.27	-1.37	0.09
Mean	0.06	0.40	0.22	-0.02	0.29

Covariance					
	0.68	0.00	-0.13	0.17	-0.12
	0.00	0.59	-0.20	0.21	0.09
	-0.13	-0.20	0.71	-0.08	0.17
	0.17	0.21	-0.08	1.62	0.29
	-0.12	0.09	0.17	0.29	1.09

Cholesky					
	0.83	0.00	0.00	0.00	0.00
	0.00	0.77	0.00	0.00	0.00
	-0.16	-0.26	0.79	0.00	0.00
	0.21	0.27	0.02	1.23	0.00
	-0.14	0.12	0.23	0.23	0.97

Moment matching
- De-mean & post-multiply by inv of Cholesky transpose

	A	B	C	D	E
	1.39	-2.76	0.19	-0.87	1.31
	0.04	1.93	0.07	-0.99	0.08
	0.45	0.73	-0.48	0.12	0.32
	-0.29	-1.18	1.61	-0.54	-0.85
	-0.37	0.11	-0.83	-0.33	-1.36
	-0.14	-0.49	0.25	0.08	-0.53
	0.06	-0.21	0.45	0.54	-0.86
	0.31	-0.53	-0.14	-1.49	-1.85
	1.68	-0.09	-0.66	0.12	-0.08
	-0.32	-0.35	1.47	0.40	1.45
	1.19	0.87	-0.19	-0.64	2.18
	-0.28	-0.60	-2.23	1.72	-0.52
	0.55	-0.05	-1.10	1.57	-0.09
	-0.69	0.55	1.46	2.42	0.15
	1.34	1.97	0.33	-0.82	-0.76
	0.85	-0.04	1.11	1.09	-0.47
	-0.85	0.67	0.22	-0.21	0.99
	-0.76	-0.51	-1.16	-0.82	0.02
	-2.12	-0.01	-1.28	-0.55	1.40
	-2.01	-0.03	0.92	-0.77	-0.53
	0.00	0.00	0.00	0.00	0.00

	1.00	0.00	0.00	0.00	0.00
	0.00	1.00	0.00	0.00	0.00
	0.00	0.00	1.00	0.00	0.00
	0.00	0.00	0.00	1.00	0.00
	0.00	0.00	0.00	0.00	1.00

In this exercise, we computed the sample covariance matrix for the simulated “independent” data. After de-meaning the data, we multiplied by the inverse of the transpose of the Cholesky decomposition to obtain the dataset on the right. These data have zero mean and a correlation matrix equal to the identity matrix.

Now, to create a dataset with the prescribed covariance matrix, we take the purified data and post-multiply by the transpose of the Cholesky decomposition, as follows:

Moment matching to target correlation

- De-mean & post-multiply by inv of Cholesky transpose

A	B	C	D	E
1.39	-2.76	0.19	-0.87	1.31
0.04	1.93	0.07	-0.99	0.08
0.45	0.73	-0.48	0.12	0.32
-0.29	-1.18	1.61	-0.54	-0.85
-0.37	0.11	-0.83	-0.33	-1.36
-0.14	-0.49	0.25	0.08	-0.53
0.06	-0.21	0.45	0.54	-0.86
0.31	-0.53	-0.14	-1.49	-1.85
1.68	-0.09	-0.66	0.12	-0.08
-0.32	-0.35	1.47	0.40	1.45
1.19	0.87	-0.19	-0.64	2.18
-0.28	-0.60	-2.23	1.72	-0.52
0.55	-0.05	-1.10	1.57	-0.09
-0.69	0.55	1.46	2.42	0.15
1.34	1.97	0.33	-0.82	-0.76
0.85	-0.04	1.11	1.09	-0.47
-0.85	0.67	0.22	-0.21	0.99
-0.76	-0.51	-1.16	-0.82	0.02
-2.12	-0.01	-1.28	-0.55	1.40
-2.01	-0.03	0.92	-0.77	-0.53
0.00	0.00	0.00	0.00	0.00
1.00	0.00	0.00	0.00	0.00
0.00	1.00	0.00	0.00	0.00
0.00	0.00	1.00	0.00	0.00
0.00	0.00	0.00	1.00	0.00
0.00	0.00	0.00	0.00	1.00

Moment matching to original target covariance

A	B	C	D	E
13.92	-13.58	4.60	-10.43	20.98
0.37	17.55	6.31	-9.06	3.36
4.45	10.13	-0.59	4.92	8.33
-2.92	-12.88	13.84	-8.17	-15.81
-3.75	-1.99	-12.14	-8.93	-27.35
-1.43	-5.51	0.62	-0.29	-9.95
0.62	-1.36	5.17	8.84	-12.01
3.10	-2.28	-0.94	-21.31	-34.88
16.76	12.64	3.75	9.78	5.52
-3.24	-5.76	14.16	6.77	26.50
11.90	17.31	8.40	-0.38	42.36
-2.79	-7.55	-29.93	15.72	-12.34
5.46	3.95	-9.35	22.76	2.56
-6.94	-0.62	13.87	35.81	10.20
13.41	28.31	18.53	1.92	-2.96
8.45	6.43	18.95	23.59	1.87
-8.52	-0.80	-1.61	-5.96	13.76
-7.63	-10.68	-20.41	-20.78	-9.65
-21.18	-16.99	-29.98	-24.03	8.20
-20.06	-16.30	-3.27	-20.77	-18.69
0.00	0.00	0.00	0.00	0.00
100.00	80.00	70.00	60.00	50.00
80.00	144.00	80.00	70.00	60.00
70.00	80.00	196.00	80.00	70.00
60.00	70.00	80.00	256.00	80.00
50.00	60.00	70.00	80.00	324.00

Cholesky transpose of original target cov

10.00	8.00	7.00	6.00	5.00
0.00	8.94	2.68	2.46	2.24
0.00	0.00	11.82	2.66	2.45
0.00	0.00	0.00	14.38	2.64
0.00	0.00	0.00	0.00	16.76

Chapter 6: Stochastic Processes in Finance

INTRODUCTION

In Chapter 5, we discussed the simulation of one or more random variables at a point in time. In this chapter, we will discover how simulation can be applied to generate future financial asset prices, interest rates, commodity prices and cash flows. We will also learn how to estimate the parameters of these processes using historical data, and how to calibrate these processes to assumed future probability distributions. By the way, "stochastic" is a synonym of "random."

Many financial valuations derive from assumed stochastic processes for cash flows, so the ultimate objective is to have all the needed tools at our disposal to generate cash flows that can be valued using one or more of the techniques we will learn in Part II of this text.

6.1 THE RANDOM WALK

The random walk is one of the simplest stochastic processes, yet it forms the basis for many of the more complex ones. In fact, the French mathematician Louis Bachelier used this process in 1900 to describe stock price movements on the Paris Bourse, or stock exchange. Now clearly, there are many factors causing stock prices to change, but the most important factor comprises the unknown desires of investors looking to buy or sell in the future. Traders attempt to forecast these demands and take advantage of their information to buy or sell now and thereby profit. If they collectively do a good job, it should be impossible for us to predict the short-term future price movement, and we should not expect to earn more or less than the normal compensation for owning stocks in the long term.

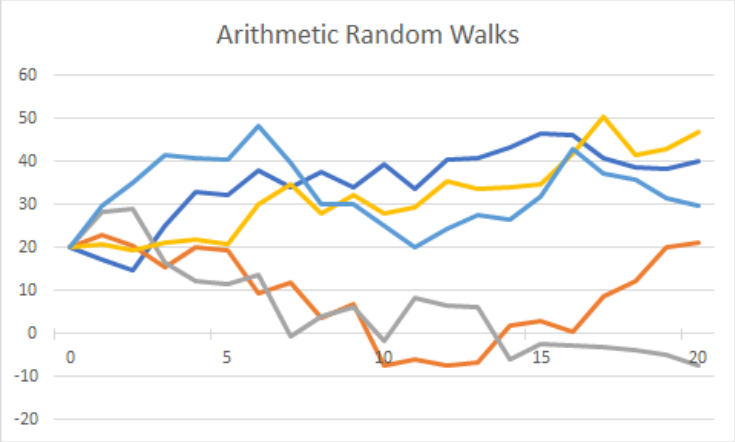
In other words, even though we don't believe the price changes are random, we may believe that based on the information available to us, there is no better model to explain their behavior. This observation has come to be known as the Efficient Market Hypothesis --- the idea that we cannot forecast prices better than the market if our available information is the same or less. However, price forecasting is not our only goal. We also want to capture potential future volatility and correlation between prices.

Mathematically, we could write down this relationship as a word-equation and then formalize it. For simplicity and convenience, as before, we will use the standardized normal distribution to generate the random variables "z". Also, time t represents any specified time, but can be thought of as the present, implying that the price at time t+1 is unknown to us.

Future change in stock price = Expected compensation for stock price risk + Random shock

$$S_{t+1} - S_t = \mu + \sigma z_{t+1}$$

This is a stochastic process known as an Arithmétic Random Walk, or ARW. (The accent mark indicates the stressed spoken syllable.) If the time step is continuous dt instead of 1, this is also called Arithmétic Brownian Motion or ABM. The non-stochastic version of this model appeared in Chapter 2: a linearly growing perpetuity. The term μ is called the drift of the process, and σ is called the volatility. The stochastic (random) term is also sometimes referred to as the *innovation* in the process. The random variable z is indexed based on the time it is known, so z_{t+1} is unknown until time t+1. The drift of the process reflects the expected growth in S over time, and the volatility relates to how much we may deviate over time from that straight-line growth in a single period. Here is a sample of five Arithmetic Random Walk outcomes, with a starting value of 20, a drift of 0, and volatility of 5.



The graph shows five simulations of 20 periods, all starting from the same point. The upper lines may appear to have a positive drift, but they do not. As you can see, the average growth rate appears to be about zero, and the standard deviation of the outcomes seems to increase over time. Can we quantify these?

Using repeated applications of the one-step stochastic process, $S_{t+1} - S_t = \mu + \sigma z_{t+1}$, we can extrapolate to a j period forecast made at time t. In particular,

$$S_{t+j} = S_t + j\mu + \sigma \sum_{i=1}^{j} z_{t+i}$$

From this equation we can determine the properties of S_{t+j}. Except for the constants, S_{t+j} is the sum of k normally distributed independent values of z. The sum of normally distributed random variables is also normal. We can compute the mean and variance conditional on S_t easily due to the independence of the z values:

Moments of arithmetic random walk

$$E[S_{t+j}|S_t] = S_t + j\mu$$

$$Var[S_{t+j}|S_t] = j\sigma^2$$

The mean of the process is expected to grow linearly with time. However, the standard deviation of the process outcomes increases indefinitely with the square root of time --- the stochastic process values become more and more uncertain as we forecast longer into the future.

The covariance terms are also straightforward in this context, since each value of z correlates perfectly with itself, and does not correlate to any other value of z. Therefore,

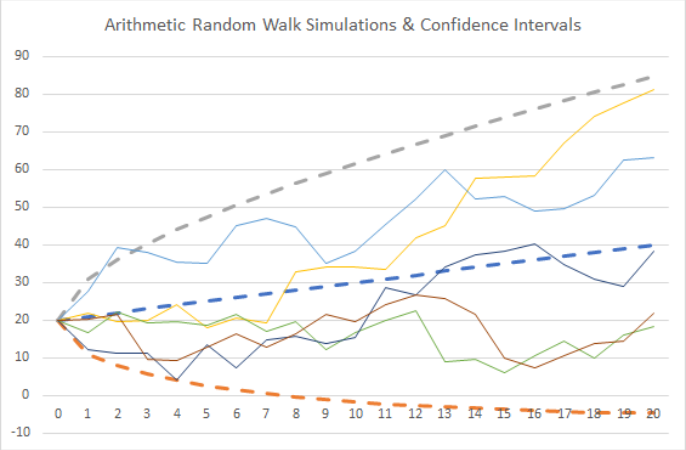
$$Cov[S_{t+j}, S_{t+m}|S_t] = \sigma^2 \min(j, m)$$

In other words, we can think of S_t as a path, or if it is more convenient, we may also think of S_t as a vector of jointly normally distributed variables. That is, we could generate random outcomes for the vector by taking a dataset of independent standard normally distributed numbers and multiplying by the transpose of the Cholesky decomposition of this covariance matrix. This is particularly convenient in this case because the Cholesky decomposition of the matrix defined by $C_{jm} = \min(j,m)$ is simply a lower diagonal matrix of ones.

Exercise

Using S_0=20, μ=1, σ=5 and N=20, simulate and graph five Arithmetic Random Walk outcomes, and provide the mean and two standard deviation confidence intervals around the simulated processes in Excel.

Solution



6.2 FINANCIAL APPLICATIONS OF THE ARITHMETIC RANDOM WALK

Although Bachelier used this model for stock price fluctuations in the short term, a quick inspection of the simulations shown in the figure above reveals that the process may become negative over longer time periods. Clearly, for stocks with limited liability, prices can never become negative, so the choice of this process to model asset prices may be questionable.

However, there are financial variables that can become negative, for which the arithmetic random walk can be used as a stochastic process model. For example, the cash flows of a company may be modeled as an Arithmetic Random Walk, since cash flow can become negative. If cash flows become negative enough, then the company may decide to go bankrupt or go out of business, but nevertheless, the cash flow model could be reasonably modeled as an Arithmetic Random Walk.

Another commonly cited possible application for the Arithmetic Random Walk is a basis, or the difference between two prices. For example, suppose S represents the difference in oil prices between West Texas (WTI) and in the North Sea (Brent). This difference can become negative, but in the long term, we would not expect the standard deviation of a forecast to increase indefinitely --- there should be some fundamental relationship between the two that draws prices back together. Hence we would like to be able to model some variables in a way that recognizes their long-term joint relationship. For this purpose, we turn to the mean-reverting process, also known as the Ornstein-Uhlenbeck process.

6.3 MEAN REVERSION AND THE ORNSTEIN-UHLENBECK PROCESS

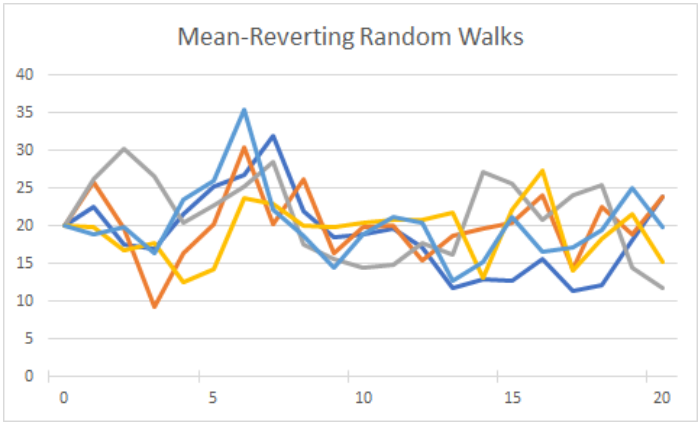
We have already seen the non-stochastic version of the OU process in Chapter 2: the mean-reverting perpetuity. Restating the deterministic process, $S_{t+1} = S_t + \kappa(\mu - S_t)$. The term μ represents the long-run mean of the process, and $0 \leq \kappa \leq 1$ represents the speed of mean reversion. When κ=0, there is no mean-reversion, and when κ=1, there is 100% mean reversion in one period. When μ > St, the process drifts positively, and when μ < St the process drifts negatively --- in both cases, in the direction of the long-run mean.

The stochastic version of mean-reversion is credited to Ornstein-Uhlenbeck, adding an innovation identical to the Arithmetic Random Walk:

$$S_{t+1} = S_t + \kappa(\mu - S_t) + \sigma z_{t+1}$$

Because the innovation is normally distributed, there is a possibility of a negative value of S when μ>0. However, mean-reversion will tend to pull the process back from negative territory so it won't stay negative indefinitely. Some modelers will avoid this "problem" by replacing σ with σS_t^{γ} with γ>0 which reduces the likelihood of a negative outcome by forcing the volatility lower as the value of S falls.[13] Of course, this may not be a problem at all for some variables, such as interest rates, that can become slightly negative for short periods of time.

The figure below shows some sample mean-reverting processes, with μ=20, κ=0.50 and σ=5.



Unlike the Arithmetic Random Walk, it appears that the variance of the outcomes does not increase in a linear fashion, but rather, seems to reach a maximum level. This can be verified by constructing the j-step forecast of S in an OU mean-reverting process:

$$S_{t+j} = \mu + (S_t - \mu)(1-\kappa)^j + \sigma \sum_{i=0}^{j-1} (1-\kappa)^i z_{t+j-i}$$

Note that the sum runs backwards, giving greater weight to later innovations. Earlier innovations are dampened by the mean reversion; the higher the mean reversion speed, the more dampened the innovations. As before, we can use this expression, involving a sum of normally distributed random variables, to determine the parameters of the normally distributed S_{t+j}.

Moments of the mean-reverting OU process

$$E[S_{t+j}|S_t] = \mu + (S_t - \mu)(1-\kappa)^j$$

$$Var[S_{t+j}|S_t] = \sigma^2 \left(\frac{1-(1-\kappa)^{2j}}{\kappa(2-\kappa)} \right)$$

[13] Note that in continuous time models, this prevents S from becoming negative, but in discrete models, there is still a possibility.

$$Cov[S_{t+j}, S_{t+m}|S_t] = \sigma^2 \left(\frac{1-(1-\kappa)^{2j}}{\kappa(2-\kappa)} \right) (1-\kappa)^{m-j} \; for \; m \geq j$$

The expected value follows the expected trajectory of exponential decay to the long-run mean. The variance increases at a decreasing rate, and reaches a maximum value at $\sigma^2/(\kappa(2-\kappa))$. As $\kappa \to 0$, the covariance approaches $\sigma^2 \; min(j,m)$, the value for the Arithmetic Random Walk.

The Cholesky decomposition of the covariance matrix of the OU process (with σ=1) has ones on the diagonal, and entries below the diagonal shrinking at a constant rate of κ . For example, if we have a 5-period forecast of a mean-reverting process with σ=1 and κ=0.20, we obtain the following covariance matrix and Cholesky decomposition:

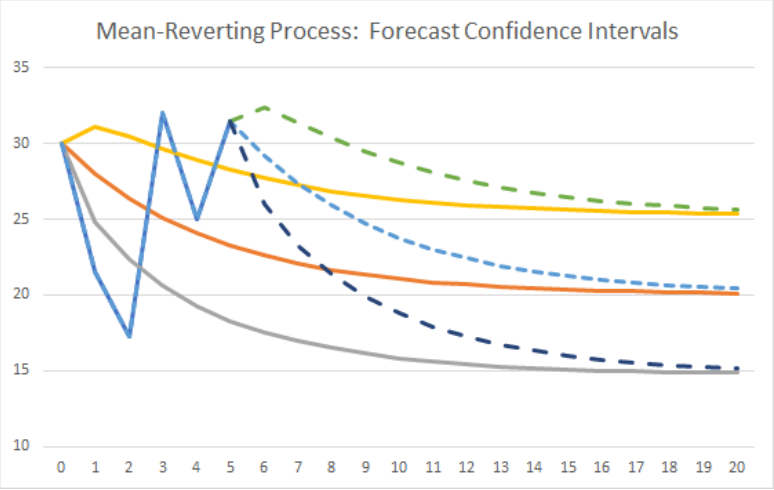
Covariance matrix (σ=1, κ=20%)

1	0.8	0.64	0.512	0.4096
0.8	1.64	1.312	1.0496	0.83968
0.64	1.312	2.0496	1.63968	1.311744
0.512	1.0496	1.63968	2.311744	1.849395
0.4096	0.83968	1.311744	1.849395	2.479516
			sum	30.20826

Cholesky decomposition (σ=1, κ=20%)

1	0	0	0	0
0.8	1	0	0	0
0.64	0.8	1	0	0
0.512	0.64	0.8	1	0
0.4096	0.512	0.64	0.8	1

Recall the distinguishing characteristic of the mean-reverting process is that the long-run variance approaches a maximum level of $\sigma^2/(\kappa(2-\kappa))$, regardless of the trajectory of the value of S. Imagine graphically that we can put a 2-standard deviation bound around future values of the mean-reverting process, and examine our future forecasts from the standpoint of the present. In the following figure, we have S_0=30, σ=5, κ=0.2 and μ=20. The jagged line represents one simulation of the process for the next 5 periods, exaggerated for graphical effect. Notice that wherever the jagged line ends up, the long-term mean and confidence interval for future values of S are unaffected.



This makes the mean-reverting process a natural choice for modelling the WTI-Brent oil basis, or interest rates. However, mean reversion would not be suitable for modeling the price of a share of stock or a stock index. Why? Mean-reversion implies predictability in the mean. If everyone believed share prices would fall in the future, they would end up falling now, as people sell their shares. The same could be said if prices were expected to rise by more than their equilibrium expected rates of return. For this reason, we need a different process, which is discussed in the next section.

Exercise

Simulate the annual price of corn for the next five years, using a current price of $5, a long-run mean of $4, annual mean reversion of 20% and volatility of $0.75 per year. Find the average of the next five years' prices within a 2 standard deviation range of $0.10. Can you find the exact values without simulation using the information only in this chapter and a hand calculator?

Solution

The simulation itself is left to the reader, using the methods presented in Chapter 5. The analytic mean of the five years is $4.54. The variance of the sum of five years' prices is the sum of all the covariance elements above (conveniently having the same κ=20% ☺), or 30.21, multiplied by the variance of 0.75^2, or 16.99. The variance of the average is therefore $16.99/5^2$, and its square root, the standard deviation of the 5-year average, is 0.82.

6.4 THE GEOMETRIC RANDOM WALK

Harry Markowitz of Modern Portfolio Theory fame assumed the expected percentage return and standard deviation of percentage returns were constant, and used this as a basis for determining efficient and optimal portfolios. We will modify this specification to assume that *continuously compounded* returns are normally distributed with mean α and standard deviation σ per period. This assumption allows us to work more easily with the natural logarithms of the prices, and also to ensure that security prices will not become negative.

The process we are developing is called the Geometric Random Walk, or GRW. This is also known as *Geometric Brownian Motion (GBM)* when the time step is infinitesimal. The specific equation used to generate the stochastic process in a single step from t to t+j is the following:

$$S_{t+j} = S_t \, exp\left[\left(\alpha - \frac{\sigma^2}{2}\right)j + z\sigma\sqrt{j}\right] \qquad "the\ GBM\ generator"$$

The equation exhibits exponential growth, but there is a puzzling adjustment to α. Why are we using a lower drift? The reason is a convexity adjustment due to the nonlinearity of the exponential function. In particular, the expected value of $\exp(z\sigma\sqrt{j})$ is $\exp(\frac{1}{2}\sigma^2 j)$ when z is a standard normally distributed random variable. The drift adjustment offsets the convexity adjustment so that the expected value relationship below is satisfied:

$$E\left[S_{t+j}\middle|S_t\right] = S_t e^{\alpha j}$$

… exactly as we wanted, with constant expected exponential growth of α.

Therefore we can express the moments of $ln(S_{t+j})$ to be consistent with this outcome:

Log-moments of the Geometric Random Walk

$$E\left[ln\left(S_{t+j}\right)\middle|S_t\right] = ln(S_t) + \left(\alpha - \frac{\sigma^2}{2}\right)j$$

$$Var\left[ln\left(S_{t+j}\right)\middle|S_t\right] = \sigma^2 j$$

$$Cov\left[ln\left(S_{t+j}\right), ln(S_{t+m})\middle|S_t\right] = \sigma^2\, min(j,m)$$

A simulated price process can therefore be computed from the GRW generator, or from these three equations --- generating the Arithmetic Random Walk first, and then computing the exponential to get the Geometric values.

The GRW process is the most popular choice for modeling discrete-time security values, and in fact forms the basis for the GBM in the continuous Black-Scholes-Merton option pricing model (1973) and for most risk models used in equity markets. However, it does have its limitations. Security returns tend to be negatively skewed and leptokurtic, meaning that sudden drops in price are more likely than sudden rises, and the probability of extreme returns is higher than would be implied by a normal distribution; the distribution has *fat tails*. More advanced security models address these limitations of the simple version of GRW or GBM when applied to security returns.

Exercise

Assume the market index is currently at 1300. If the expected annual growth rate is 10%, and the annual volatility is 18%, using monthly simulated index values. What is the probability that the index will rise over 2000 any time in the next 24 months?

Solution

Proceed using simulation, taking care to make the time step 1/12 of a year, and making each successive value of the index depend on the previous one. Tabulate the number of simulations where the maximum index level exceeds 2000. The correct answer is about 20%.

6.5 STOCHASTIC PROCESS SELECTION AND PARAMETERIZATION

There are obviously many different types of stochastic processes, and we have only covered three of the most basic models. Nevertheless, it is worthwhile to summarize the models we have discussed, explain how to parameterize them, and examine some of the most frequent adjustments to these processes.

First, of these three models, the best model for security prices is the GRW model. Even if it assumes constant expected returns and volatility, it ensures that prices can never become negative. Also, a constant proportional dividend rate can be subtracted from the drift to obtain ex-dividend simulations. However, it does not allow security values to reach zero in the event of bankruptcy or termination, and it does not allow negatively skewed or leptokurtic returns in the simple version.

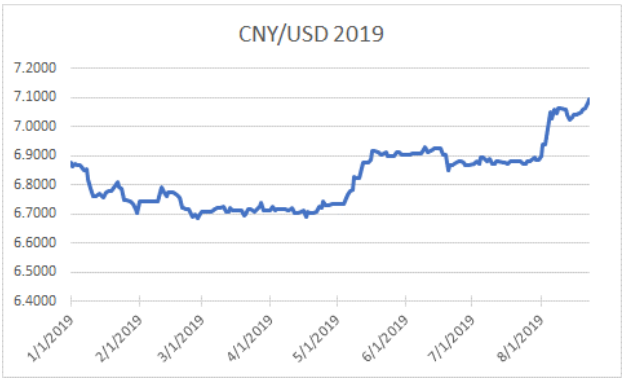
Second, the best of these three models for commodity prices and interest rates is the mean-reverting process. Importantly, commodities are not assets in the financial sense. High commodity prices stimulate production and reduce demand, causing everyone to expect lower prices in the future. If you tried to go short commodities in expectation of this, you would be disappointed to find no market for short sales of commodities! Low commodity prices have the opposite effect, causing price expectations to rise for the future. Interest rates resemble commodities in this respect since high rates encourage lending and discourage borrowing, causing future rates to fall. Finally, in many financial models, volatility itself is assumed to change over time, and an MR process can reasonably describe these movements over time.

Finally, most financial variables may be modeled short-term as ARW processes, but this generally fails long-term due to the possibility of negative values and nondecreasing forecasted volatility. The ARW process is most important as a foundation for mean-reverting and GRW processes.

Suppose we have a doubt which process to use? For some markets like foreign exchange, it is not clear if mean reversion exists. But if it does exist, we'd like to be able to estimate the parameters of the process. How can we test if mean-reversion exists, and how do we parameterize it?

Parameterizing CNY drift and volatility

The following chart shows the Chinese Yuan (CNY) exchange rate in units per dollar for about eight months in 2019.



Even though the currency may be considered partially targeted or controlled, market forces do play a role in the determination of the exchange rate. Suppose we had an income stream expected in yuan over the next few years. We might want to assess the expected appreciation or depreciation of the CNY over that time, and the volatility of the exchange rate. Not being able to see the future, we turn to the past: available data in 2019 on a daily basis, from 1/1/19 to 8/23/19.

Here is a sample from the dataset:

Annualized mean (Average x 252)	3.92%
Annualized StdDev (Stdev x Root 252)	3.72%
Adjusted mean (adding 0.5 x variance)	3.99%

	CNY	ΔIn(CNY)
1/1/2019	6.8785	
1/2/2019	6.8621	-0.0024
1/3/2019	6.8720	0.0014
1/4/2019	6.8694	-0.0004
1/5/2019	6.8694	0.0000
1/7/2019	6.8510	-0.0027
1/8/2019	6.8532	0.0003
1/9/2019	6.8180	-0.0051
1/10/2019	6.7885	-0.0043
1/11/2019	6.7627	-0.0038
1/12/2019	6.7627	0.0000
1/14/2019	6.7681	0.0008
1/15/2019	6.7610	-0.0010

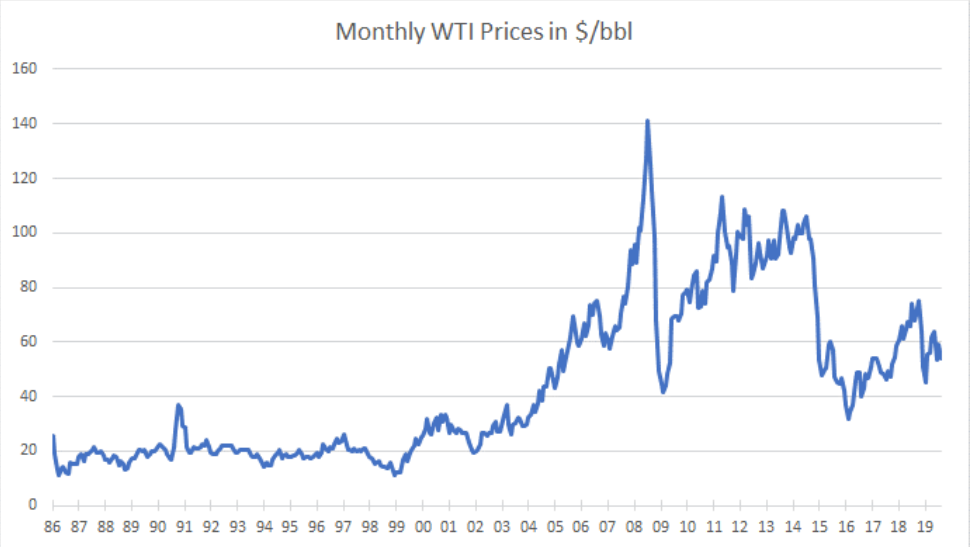
Assuming this followed a GRW process in the past, and that the first 8 months of 2019 could accurately reflect the future, both heroic assumptions indeed, we would determine the change in log prices historically. Using the convention that there are 252 trading days a year, the average log change would be multiplied by 252, and the standard deviation of the log change would be multiplied by $\sqrt{252}$.

The results are shown in the table. Based on this short history, the yuan is depreciating relative to the dollar at a continuous rate of 3.99% with a volatility of 3.72% per year. If we would like to simulate the future using these historical parameters, we may do so as long as we are clear about our assumptions.

The historical price series does not seem to show any evidence of mean-reversion, and the statistics bear that out. How can we check for mean reversion and determine the parameters?

Parameterizing a mean-reverting process

Let's consider oil prices. Using Monthly WTI prices available on MacroTrends, we obtained these data:



The x-axis denotes the beginning of each year from Jan 1986 to Aug 2019. The prices are quoted in dollars per barrel, and are not inflation-adjusted. The prices appear at times to follow a GRW process, and at other times an MR process.

To test for the presence of mean reversion, we calculate price change as the y variable, and the previous price as the x variable, and perform a simple regression. Time series statisticians may shudder at the simplicity of the test, and you should probably test for a unit root, but that is beyond the scope of this course. At least, doing a simple regression will give us an indicator. The equation is

$$(S_{t+1} - S_t) = \kappa\mu - \kappa S_t$$

So the constant is κμ, and the slope is -κ. The Excel regression output is copied below:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.085955
R Square	0.007388
Adjusted R Square	0.004913
Standard Error	4.931793
Observations	403

ANOVA

	df	*SS*	*MS*	*F*	*Signif F*
Regression	1	72.59619	72.59619	2.984724	0.084824
Residual	401	9753.355	24.32258		
Total	402	9825.951			

	Coeff	*StdErr*	*t Stat*	*P-value*
Intercept	0.7065	0.442604	1.596234	0.111224
Slope	-0.01441	0.00834	-1.72764	0.084824

The t-statistic doesn't appear to be overwhelmingly in favor of rejecting the null hypothesis, but could be considered marginal. Most interestingly, we have estimated a mean reversion speed of 1.44% per month (the slope coefficient) and a long-run mean oil price of $49 (0.7065/0.01441). Perhaps more testing is needed; mean reversion may have existed over some periods in the data and not others. The point is, if we were looking to simulate oil prices in the future, it would not be a terrible idea to use a mean-reverting process.

If we examine only the last 5 years of monthly data, a slightly different story emerges:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.404129
R Square	0.16332
Adjusted R Square	0.148895
Standard Error	5.335624
Observations	60

ANOVA

	df	*SS*	*MS*	*F*	*Signif F*
Regression	1	322.3145	322.3145	11.32164	0.001363
Residual	58	1651.195	28.46889		
Total	59	1973.51			

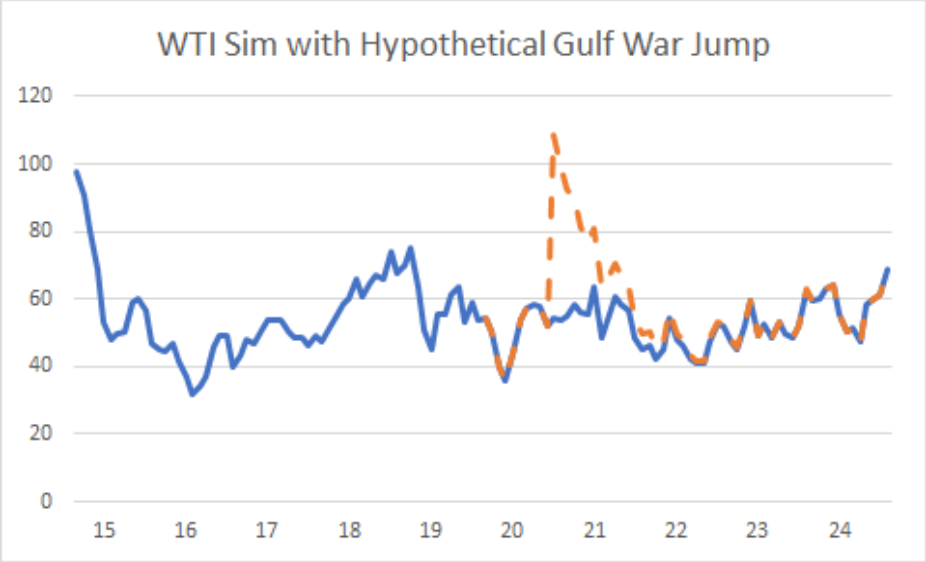
	Coeff	*StdErr*	*t Stat*	*P-value*
Intercept	8.810743	2.918485	3.018944	0.003766
Slope	-0.17098	0.050814	-3.36476	0.001363

In this period, we are seeing 17% per month mean reversion, a high degree of statistical significance, and a long-run mean value of $51 per barrel.

6.6 JUMP PROCESSES

Sometimes, prices move quickly and far. People may say "we had a 10 standard deviation move" in price on one day, but that is fairly unlikely. Most likely, they are using GRW or a MR process to compute volatility, and failed to account for the occasional jumps that occur in real-world prices outside of these model parameters. To address this concern in financial modeling, we can overlay a jump on top of any price process. The jump may represent the unlikely event that a company becomes a takeover target suddenly, or that electricity prices in the Midwest spike due to a power plant or transmission failure.

As an example, let's consider an oil shock, such as a war in the gulf. If we assume there is a 1% chance in any given month of a sudden 100% rise in the price of oil, then the future price of oil might look like this, using the 60 month parameters from the last example.



In a mean-reverting process, the impact of jumps dissipates over time, as shown in the figure. In the case of ARW or GRW processes, however, the shift is permanent.

6.7 CONCLUSIONS

In sum, there are a large number of complex and sophisticated stochastic processes used to model financial indicators such as prices and rates. This chapter explained the most basic processes understanding that in practice, you will invest significant time determining what process to use in order to model the variables that are most important to you at the time. For example, suppose you were simulating future stock prices. Your model might include any or all of the following, or even more:

- Stochastic drift or volatility (changes in expected return and risk)
- Event risks
- Changes in bid-offer spreads
- Changes in economic variables affecting the company's performance
- Changes in investor sentiments in favor or against the company's industry

… and so on

With the tools you have learned in this chapter, you will be able to design and simulate any stochastic process needed, generate cash flows from a contract, security, or activity, and value those cash flows.

With the tools we have covered in Part I of this text (through Chapter 6), you can find expected cash flows for almost any practical situation you may encounter and discount them to find the present value when discount rates are known. Chapter 7 begins Part II, which explores the determination of risk charges and discount rates used to value cash flows --- since they are never really known to you in practice.

6.8 REFERENCES AND ADDITIONAL READINGS

Black Fischer and Myron Scholes (1973),

Jarrow, Robert; Protter, Philip (2004). "A short history of stochastic integration and mathematical finance: the early years, 1880–1970". A Festschrift for Herman Rubin. Institute of Mathematical Statistics Lecture Notes - Monograph Series. pp. 75–80. CiteSeerX 10.1.1.114.632. doi:10.1214/lnms/1196285381. ISBN 978-0-940600-61-4. ISSN 0749-2170.

Markowitz

Ornstein, Uhlenbeck

6.9 EXERCISES TO BE DONE IN EXCEL

1. Use the same series of 20 independent normally distributed numbers z to generate all three of the following (annual) prices. Take initial values to be 100; volatility is 18% for GRW, and 18% of the price level in the other cases. Show your results in a graph.
    a. A GRW process with drift 10%.
    b. An ARW process with growth rate equal to a constant 10.
    c. A MR process with speed of MR of 0.35 and a long run mean of 200.

2. For the mean-reverting process in #1, what is the mean and standard deviation of the price level in periods 10 and 20? What is their covariance?

3. How would you adjust from annual means and standard deviations to monthly means and standard deviations for all three of these processes? (*Hint:* Go from monthly to annual first.)

4. How does your answer to #1 change if the processes are correlated? Implement this in your spreadsheet.